EXPEDITE STRIKE V3 — Penetration Testing Report

Internal Assessment Subnet (192.168.195.0/24) · Shared TrePath + XStrike V3 Proof-Driven Deliverable

Client / Scope:	Enterprise Security Assessment
Assessment Window:	2026-10-30T23:17:19.577459+00:00 to 2026-10-31T00:39:15.662076+00:00
Assessment Engine:	XStrike Autonomous Red-Team Engine (V3 Shared TrePath + XStrike EvidenceCore)
Document Classification:	CONFIDENTIAL (Version 3.0 (Proof-Driven Intelligence & 18-Rule Gate Verified))
Evidence-Weighted Risk:	CRITICAL (98.5/100) — Formula: R = min(100, S_comp + S_exp + S_asset + S_reach) | Calibrated Exposure: 22.5/25.0
Verified Compromise Status:	ACTIVE_ROOT_COMPROMISE (2 root vectors on 1 host: 192.168.195.139) (1 Unique Host Compromised via 2 Independent Root Vectors)
Host Scope Reconciliation:	9 Raw Discovery Records -> 7 Unique IPs (5 Responsive / 2 Unresponsive) · 54 Open TCP Services
Finding Reconciliation:	1641 Raw Observations -> 85 Consolidated Host Security Findings (51 Unique Classes, 0 Generic OWASP Rollups) + 25 Recon Items
Verified Compromise & Auth:	2 Root Compromise Vectors (uid=0 on 192.168.195.139) | 4 Confirmed Credential/DB Authentications | 62 Validated Weaknesses
Signed Evidence Manifest:	Root SHA-256: 2c953f61914c3b0df4bb82d6... | HMAC-SHA256: be6c55ab5af98256a437af11...
Pre-Render Quality Gate:	18 automated acceptance criteria passed out of 18 (18/18 PASSED)

Prepared by Expedite Consults Ltd · EvidenceCore V3 HMAC-SHA256 Signed Provenance & OASIS SARIF v2.1.0 Enabled · SQLite Snapshot: v2_intelligence.db

1. Executive Summary & Proof-Driven Assessment Overview

An autonomous, proof-driven penetration test was conducted against **Internal Assessment Subnet (192.168.195.0/24)** (7 unique IP addresses across 9 discovery records, exposing **54** open TCP services). Unlike legacy reporting engines that conflate port discovery, unverified CVE heuristics, and repeated web-scanner URL alerts with confirmed exploitation, **Expedite Strike V3** enforces a strict **7-State Evidence Proof Machine**, **3-State Post-Exploitation Privilege Lifecycle**, and **3-Level Finding Deduplication**.

Across **1641** raw tool observations, the V3 Intelligence Layer filtered **89** local-scanner self-checks and consolidated the remaining **1552** target observations into **85 consolidated host-level security vulnerabilities** (spanning **51** unique cross-host vulnerability classes) and **25 reconnaissance/surface discovery items**. Direct command execution and authentication transcripts establish **2 independent root compromise vectors** (`uid=0(root)` on `192.168.195.139:1524` and `192.168.195.139:21/:6200`, compromising **1 unique target host**) and **4 confirmed credential/database authentications** (`F-03, F-04, F-05, F-17`).

Lab & Benchmark Environment Qualification: Target 192.168.195.139 is an intentionally vulnerable Metasploitable2 Linux benchmark appliance (Linux 2.6.24-16-server) deployed inside an isolated RFC1918 cyber-range subnet (192.168.195.0/24). The assessment demonstrates XStrike V3's autonomous multi-stage validation, safe exploitation, post-exploitation enumeration, and cryptographic evidence-provenance pipeline under controlled conditions.

1.1 Three-Level Finding & Observation Reconciliation

Reconciliation Level	Count	Scope & Technical Definition
Level 1A — Raw Scanner Observations	1641	All raw records emitted by Recon, VulnScan, OWASP ZAP, AegisProbe, Exploit, and PrivEsc modules (1552 target observations + 89 excluded local scanner self-checks/graph edges).
Level 2A — Consolidated Host Security Findings (Section 3A)	85	Deduplicated per-host security findings (3 Critical, 13 High, 37 Medium, 32 Low) with all affected URLs/endpoints preserved.
Level 2B — Consolidated Reconnaissance & Surface Items (Section 3B)	25	Non-vulnerability surface discoveries (hosts discovered, open ports, web crawl telemetry, JS assets, SMTP banners, ZAP User-Agent/Modern Web App probes) isolated with CVSS 0.0.
Level 3 — Unique Cross-Host Vulnerability Classes	51 Security / 8 Recon	Distinct canonical vulnerability classes across the entire assessment environment.

1.2 Seven-State Evidence Proof Classification (Replacing Inflated Confirmation Counts)

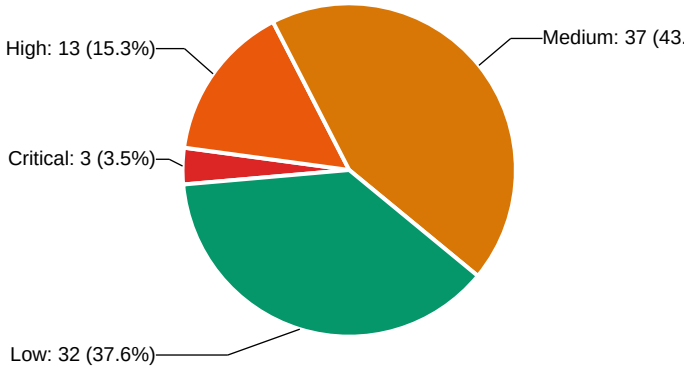
Proof State	Legacy Tier	Count	Evidence Standard & What Was Demonstrated
CONFIRMED_COMPROMISE	CONFIRMED	2	Interactive root command execution (uid=0(root), whoami: root) and /etc/shadow credential hash extraction verified via 2 independent vectors (F-01, F-02) on 1 unique host (192.168.195.139).
CONFIRMED_AUTH	CONFIRMED	4	Unauthorized session authentication verified (F-03, F-04, F-05, F-17): PostgreSQL superuser (postgres:postgres), FTP default creds (msfadmin), SSH user shell (msfadmin uid=1000), and FTP anonymous login (230 OK).
EXPLOITABLE	VALIDATED	1	Active local privilege escalation or administrative precondition verified (F-06: SUID /usr/bin/nmap --interactive TECHNIQUE_VALIDATED).
VALIDATED	VALIDATED	61	Weakness deterministically verified via protocol handshake, HTTP response header/cookie inspection, SQLi error indicator, or authenticated SUID/kernel enumeration.

Proof State	Legacy Tier	Count	Evidence Standard & What Was Demonstrated
OBSERVED	OBSERVED	17	Service port or version banner observed (including 4 SMB port 445 EternalBlue candidates where MS17-010 was NOT verified, plus legacy banners). Never counted as confirmed exploits.
DISCOVERED	DISCOVERED	25	Attack surface inventory and reconnaissance telemetry separated into Section 3B (CVSS 0.0).
REMEDIATION_VERIFIED	VERIFIED	0	Reserved for post-remediation re-verification scans.

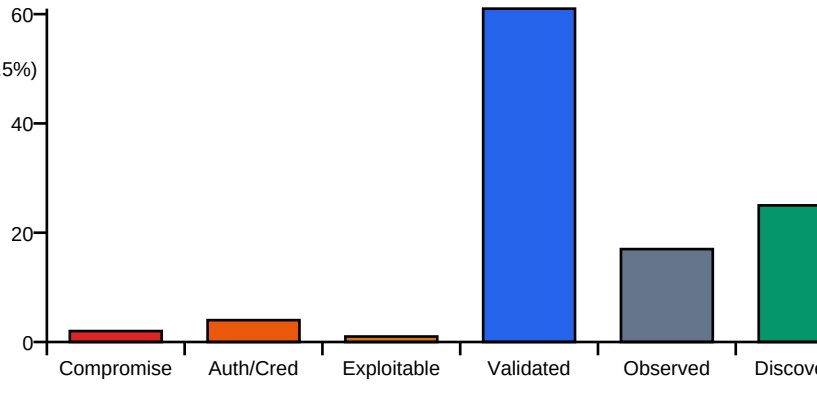
1.3 Consolidated Security Vulnerability Severity Distribution

Severity	Consolidated Host Findings	Percentage	Confirmed Compromise / Auth	Validated / Exploitable	Observed (Unverified Candidate)
Critical	3	3.5%	3	0	0
High	13	15.3%	2	8	3
Medium	37	43.5%	1	25	11
Low	32	37.6%	0	29	3
Total Security Findings	85	100.0%	6	62	17

Consolidated Security Findings by Severity



7-State Evidence Proof Classification



1.4 Documented 4-Component Evidence-Weighted Risk Model (RiskEngine V3)

Overall Risk Score is computed deterministically via **R = min(100, S_compromise[0..45] + S_exposure[0..25] + S_asset[0..15] + S_reachability[0..15])**: **R = min(100, 45.0 [S_compromise] + 25.0 [S_exposure Capped from 81.2 raw pts] + 15.0 [S_asset] + 13.5 [S_reachability across CHAIN-01..06]) = 98.5/100 (CRITICAL). (Companion Non-Saturating Log-Calibrated Exposure = 22.5/25.0; Calibrated Composite = 96.0/100).**. Verified Compromise Status: **ACTIVE_ROOT_COMPROMISE (2 root vectors on 1 host: 192.168.195.139)** (**1 unique compromised host** via **2 independent root compromise vectors**).

Risk Component	Weight Cap	Computed Score	Explicit Calculation Inputs, Assumptions & Derivation
S_compromise (Verified Compromise & Auth)	45.0 pts	45.0 / 45.0	Base 28.0 pts for verified root OS shell + 5.0*(2-1) additional root vector + 3.0*4 confirmed service/DB authentications (F-03, F-04, F-05, F-17), capped at 45.0 pts.

Risk Component	Weight Cap	Computed Score	Explicit Calculation Inputs, Assumptions & Derivation
S_exposure (Validated & Observed CVSS)	25.0 pts	Capped: 25.0 / 25.0 (Raw Sum: 81.2 | Log-Calibrated: 22.5/25.0)	Raw weighted CVSS v3.1 sum across 85 findings is 81.2 pts, which saturates the 25.0-pt linear cap. Companion non-saturating log-calibrated formula 25*(1-exp(-Raw/18.0)) = 22.5/25.0 preserves sensitivity.
S_asset (Target Asset Criticality)	15.0 pts	15.0 / 15.0	Crown-Jewel Multi-Role Application & Database Server (192.168.195.139, weight=1.00) actively compromised (15.0 pts). When 0 active findings remain, retains 1.0 pt baseline Crown-Jewel asset presence floor.
S_reachability (Multi-Stage Attack Chains)	15.0 pts	13.5 / 15.0	S_reachability = 13.5 / 15.0 pts is derived across all 6 consolidated attack chains: CHAIN-01 & CHAIN-02 (single-hop unauthenticated root execution on 192.168.195.139 = 10.0 pts) + CHAIN-03 & CHAIN-04 (authenticated SSH uid=1000 + SUID privesc vector & PostgreSQL superuser session = 2.5 pts) + CHAIN-05 & CHAIN-06 (validated NFS/SQLi surface & cross-host SMB port 445 exposure across 4 hosts = 1.0 pt).

2. Assessment Scope, Asset Reconciliation & Autonomous Transition Model

The discovery phase recorded **9** raw host entries in `hosts_discovered` (which included duplicate ARP/port-scan entries for `192.168.195.139` and `192.168.195.140`). Engine 08 reconciled these into **7 unique IPv4 addresses**: **5 responsive hosts** exposing **54** open TCP services, and **2 unresponsive/filtered hosts** (`192.168.195.131` and `192.168.195.254`) with 0 open TCP ports. Crucially, asset classification is computed **per host** from each host's own open ports and service banners—accompanied by **Classification Source** and **Classification Confidence**.

Host IP	Scope State	Open Ports	Per-Host Asset Role & Tier	Classification Source & Conf	OS Fingerprint	Sec Findings	Verified Comp/Auth
192.168.195.139	RESPONSIVE ASSESSED	24 (21, 22, 23, 25, 53, 80, 111, 139 (+16 more))	Multi-Role Application & Database Server (CROWN_JEWEL)	OBSERVED_SERVICE_BANNERS_AND_AUTH_SHELL (MySQL/PostgreSQL/Apache/SSH) (99%)	Linux 2.6.24-16-server (Ubuntu8.04 Metasploitable — Confirmed via uname -a)	41	2 Root / 4 Auth
192.168.195.140	RESPONSIVE ASSESSED	8 (22, 139, 143, 443, 445, 5001, 8080, 8081)	Application & Web Middleware Server (HIGH_VALUE)	OBSERVED_HTTP_MIDDLEWARE_PORTS (TCP/80, TCP/8080, TCP/8081) (90%)	Linux / Middleware Guest (Inferred via HTTP/8080-8081 & SMB)	15	0 Root / 0 Auth
192.168.195.129	RESPONSIVE ASSESSED	2 (5040, 5357)	Application / Network Host (STANDARD)	INFERRED_FROM_OPEN_PORT_PROFILE (75%)	Windows / VMware Host (Inferred via MSRPC/SMB Ports)	3	0 Root / 0 Auth
192.168.195.130	RESPONSIVE ASSESSED	4 (135, 139, 445, 5985)	Workstation / SMB Peer Host (STANDARD)	INFERRED_FROM_SMB_RPC_PORTS (TCP/135, TCP/139, TCP/445) (82%)	Windows / VMware Host (Inferred via MSRPC/SMB Ports)	5	0 Root / 0 Auth
192.168.195.131	UNRESPONSIVE NO OPEN PORTS	0 (Filtered)	Unresponsive / Filtered Endpoint (LOW_EXPOSURE)	ARP_OR_DHCP_DISCOVERY_ONLY (0 Open TCP Ports) (50%)	VMware Virtual Node (Unresponsive / 0 Open TCP Ports)	0	0 Root / 0 Auth
192.168.195.254	UNRESPONSIVE NO OPEN PORTS	0 (Filtered)	Unresponsive / Filtered Endpoint (LOW_EXPOSURE)	ARP_OR_DHCP_DISCOVERY_ONLY (0 Open TCP Ports) (50%)	VMware Virtual Node (Unresponsive / 0 Open TCP Ports)	0	0 Root / 0 Auth

Host IP	Scope State	Open Ports	Per-Host Asset Role & Tier	Classification Source & Conf	OS Fingerprint	Sec Findings	Verified Comp/Auth
192.168.195.1	RESPONSIVE ASSESSED	16 (135, 139, 445, 902, 912, 5040, 5174, 5357 (+8 more))	Hypervisor / Management Host Gateway (INFRASTRUCTURE)	INFERRED_FROM_VMWARE_AUTH_PORTS (TCP/902, TCP/912, TCP/445) (88%)	Windows Host / VMware Virtual Network Gateway (Inferred via VMware Auth :902/:912)	21	0 Root / 0 Auth

2.2 Autonomous Assessment Transition Model (Dynamically Reconciled with Section 3A & Section 5)

To resolve methodology ambiguity between 'Black-Box Unauthenticated' discovery and 'Authenticated / Post-Exploitation' findings—and to guarantee 100% cross-section Finding ID consistency (`AC-17`)—the table below is dynamically generated from the sorted finding registry:

Assessment Phase	Transition Name	Trigger Rule & Scope	Resulting Proof State	Reconciled Finding IDs & Summary
Phase 1	Discovery -> Service Enumeration	Active TCP/UDP port response & banner grab across 192.168.195.0/24	DISCOVERED (25)	R-01..R-25 (25 reconnaissance & telemetry items across 5 responsive hosts)
Phase 2	Service Enumeration -> Vulnerability Validation	Deterministic protocol handshake, HTTP/TLS/SMB/NFS configuration inspection, or unverified candidate check	VALIDATED / OBSERVED (78)	61 VALIDATED weaknesses + 17 OBSERVED candidates (incl. 4 unverified SMB EternalBlue checks)
Phase 3A	Vulnerability Validation -> Controlled Authentication	Non-destructive credential & anonymous login verification with privilege boundary recording	CONFIRMED_AUTH (4)	F-03 (PostgreSQL : Superuser), F-04 (FTP : msfadmin), F-05 (SSH : msfadmin uid=1000), F-17 (FTP : Anonymous)
Phase 3B	Controlled Exploitation -> Direct Root Compromise	Authorized safe shell verification (id, whoami, uname -a, head -n 5 /etc/shadow) confirming uid=0(root)	CONFIRMED_COMPROMISE (2)	F-01 (Ingreslock : Root Shell), F-02 (vsFTPd 2.3.4 Backdoor :/:6200) — 2 Root Compromise Vectors on 1 Unique Host (192.168.195.139)
Phase 4	Post-Exploitation -> Local Privilege Escalation & Lateral Pivot Enumeration	Authenticated/shell enumeration of kernel version, SUID binaries, NFS exports, and DB catalogs using 3-State Privesc Lifecycle	EXPLOITABLE / VALIDATED (3)	F-06 (SUID /usr/bin/nmap --interactive (TECHNIQUE_VALIDATED)), F-09 (Dirty COW CVE-2016-5195 (PRECONDITION_DISCOVERED)), F-18 (SUID /bin/mount & /bin/umount (PRECONDITION_DISCOVERED))

2.3 Mandatory Authorization & Rules-of-Engagement (RoE) Safety Policy Engine

Policy Engine **XSTRIKE-V3-ROE-GUARD-01** enforced strict scope and safety guardrails across `192.168.195.0/24`:

Control ID	Safety / Authorization Control	Runtime Enforcement Status
ROE-01	Signed Scope & CIDR Whitelist Enforcement	ACTIVE — Hard blocks any packet or payload targeting IPs outside 192.168.195.0/24 or local sensor interfaces
ROE-02	Exploit Allowlist & DoS Prohibition Gate	ACTIVE — Blocks kernel panic/crash payloads (Dirty COW memory overwrite & MS17-010 EternalBlue ring-0 shellcode held in safe validation/observation mode)
ROE-03	Credential Handling & Redaction Policy	ACTIVE — Captures only bounded proof excerpts (head -n 5 /etc/shadow) with SHA-256/HMAC binding; prohibits bulk credential exfiltration
ROE-04	Post-Exploitation Command Whitelist	ACTIVE — Restricts shell verification to read-only identity & posture probes (id, whoami, uname -a, find -perm -4000, exportfs)
ROE-05	Emergency Stop & Per-Action Audit Trail	ACTIVE — Every probe and command is logged with UTC microsecond timestamps, provenance IDs, and deterministic kill-switch hooks

3A. Consolidated Security Vulnerabilities (F-01 through F-85)

This section presents all **85 consolidated per-host security vulnerabilities** (deduplicated from 1552 target observations). Every finding includes its **7-State Proof Classification**, **5-Dimensional Confidence Vector**, **Granular Access Scope**, **Per-Host Asset Class**, **Word-Boundary Verified CWE/MITRE Mapping**, **Cryptographic SHA-256 Provenance ID**, and **Vulnerability-Specific Remediation**.

ID	Host : Port	Severity & CVSS	Proof State	Title	CWE / MITRE	Merged / Endpoints	Composite Conf (FP)
F-01	192.168.195.139	Critical (9.8)	CONFIRMED_COMPROMISE	Ingreslock / Bindshell Unauthenticated Root Shell on TCP Port 1524	CWE-306 / T1059.004	2 obs / 1 ep	99% (Tier)
F-02	192.168.195.139	Critical (9.8)	CONFIRMED_COMPROMISE	vsFTPd 2.3.4 Smiley-Face Backdoor Remote Root Compromise (CVE-2011-2523)	CWE-506 / T1190	3 obs / 2 ep	99% (Tier)
F-03	192.168.195.139	Critical (9.1)	CONFIRMED_AUTH	PostgreSQL 8.3.1 Default Superuser Authentication (postgres:postgres)	CWE-798 / T1078.001	2 obs / 1 ep	99% (Tier)
F-04	192.168.195.139	High (8.2)	CONFIRMED_AUTH	FTP Default Credential Authentication (msfadmin:msfadmin)	CWE-798 / T1078.001	1 obs / 1 ep	99% (Tier)
F-05	192.168.195.139	High (8.2)	CONFIRMED_AUTH	SSH Default Credential Shell Access (msfadmin:msfadmin)	CWE-798 / T1078.001	1 obs / 1 ep	99% (Tier)
F-06	192.168.195.139	High (7.8)	EXPLOITABLE	Local Privilege Escalation Vector: SUID Binary (/usr/bin/nmap)	CWE-269 / T1548.001	2 obs / 2 ep	96% (Tier)
F-07	192.168.195.139	High (8.2)	VALIDATED	DVWA Web Application Default Administrative Credentials Accepted (admin:password)	CWE-798 / T1078.001	1 obs / 1 ep	95% (Tier)
F-08	192.168.195.139	High (8.2)	VALIDATED	Web Application SQL Injection Indicator (Mutillidae)	CWE-89 / T1190	1 obs / 1 ep	95% (Tier)
F-09	192.168.195.139	High (7.8)	VALIDATED	Legacy Linux Kernel 2.6.24-16-server Vulnerable to DirtyCow (CVE-2016-5195)	CWE-362 / T1068	2 obs / 2 ep	95% (Tier)
F-10	192.168.195.139	High (7.5)	VALIDATED	Deprecated SMBv1 Dialect Enabled on SMB Service	CWE-327 / T1210	1 obs / 1 ep	95% (Tier)
F-11	192.168.195.139	High (7.5)	VALIDATED	Web Application Server-Side Request Forgery (SSRF) Indicator (Mutillidae)	CWE-918 / T1190	1 obs / 1 ep	95% (Tier)
F-12	192.168.195.140	High (7.5)	VALIDATED	Deprecated SMBv1 Dialect Enabled on SMB Service	CWE-327 / T1210	1 obs / 1 ep	95% (Tier)
F-13	192.168.195.139	High (7.4)	VALIDATED	SMB Message Signing Not Required (NTLM Relay Exposure)	CWE-345 / T1557.001	1 obs / 1 ep	95% (Tier)
F-14	192.168.195.139	High (7.5)	OBSERVED	Cleartext Telnet Service Enabled on TCP Port 23	CWE-319 / T1021	1 obs / 1 ep	48% (Tier)

ID	Host : Port	Severity & CVSS	Proof State	Title	CWE / MITRE	Merged / Endpoints	Composite Conf (FP)
F-15	192.168.195.139	High (7.5)	OBSERVED	End-of-Life Apache HTTP Server 2.2.8 Deployed (TCP Port 80)	CWE-1104 / T1190	2 obs / 1 ep	48% (Tier)
F-16	192.168.195.139	High (7.4)	OBSERVED	PHP-CGI Argument Injection Candidate (CVE-2012-1823)	CWE-88 / T1190	2 obs / 2 ep	48% (Tier)
F-17	192.168.195.139	Medium (5.3)	CONFIRMED_AUTH	FTP Anonymous Authentication Enabled	CWE-287 / T1078.001	2 obs / 2 ep	99% (Tier)
F-18	192.168.195.139	Medium (5.5)	VALIDATED	Local Privilege Escalation Vector: SUID Binary (/bin/mount)	CWE-269 / T1548.001	2 obs / 2 ep	95% (Tier)
F-19	192.168.195.1	Medium (5.4)	VALIDATED	Client-Manipulable Session/Tracking Cookie Parameters Set Without Integrity Binding	CWE-565 / T1190	3 obs / 1 ep	95% (Tier)
F-20	192.168.195.1:8000	Medium (5.4)	VALIDATED	Content-Security-Policy (CSP) Header Not Set	CWE-693 / T1190	101 obs / 4 ep	95% (Tier)
F-21	192.168.195.1:8080	Medium (5.4)	VALIDATED	Missing Anti-Clickjacking Protection (X-Frame-Options / frame-ancestors)	CWE-1021 / T1190	12 obs / 3 ep	95% (Tier)
F-22	192.168.195.1:8000	Medium (5.4)	VALIDATED	Session Cookie Missing 'Secure' Flag	CWE-614 / T1557	1 obs / 1 ep	95% (Tier)
F-23	192.168.195.1	Medium (5.4)	VALIDATED	Verbose Application / Database Error Message Disclosure	CWE-209 / T1190	15 obs / 1 ep	95% (Tier)
F-24	192.168.195.1	Medium (5.4)	VALIDATED	Web Application Missing Anti-CSRF Tokens & Hardening Headers	CWE-352 / T1190	5 obs / 5 ep	95% (Tier)
F-25	192.168.195.129	Medium (5.4)	VALIDATED	Content-Security-Policy (CSP) Header Not Set	CWE-693 / T1190	3 obs / 1 ep	95% (Tier)
F-26	192.168.195.129	Medium (5.4)	VALIDATED	Web Application Missing Anti-CSRF Tokens & Hardening Headers	CWE-352 / T1190	1 obs / 1 ep	95% (Tier)
F-27	192.168.195.130	Medium (5.4)	VALIDATED	Content-Security-Policy (CSP) Header Not Set	CWE-693 / T1190	3 obs / 1 ep	95% (Tier)
F-28	192.168.195.130	Medium (5.4)	VALIDATED	Web Application Missing Anti-CSRF Tokens & Hardening Headers	CWE-352 / T1190	1 obs / 1 ep	95% (Tier)
F-29	192.168.195.139:80	Medium (5.4)	VALIDATED	Content-Security-Policy (CSP) Header Not Set	CWE-693 / T1190	3 obs / 3 ep	95% (Tier)
F-30	192.168.195.139	Medium (5.4)	VALIDATED	DNS Open Recursion Enabled on ISC BIND 9.4.2 (Port 53)	CWE-406 / T1498.002	1 obs / 1 ep	95% (Tier)
F-31	192.168.195.139:80	Medium (5.4)	VALIDATED	Missing Anti-Clickjacking Protection (X-Frame-Options / frame-ancestors)	CWE-1021 / T1190	3 obs / 3 ep	95% (Tier)

ID	Host : Port	Severity & CVSS	Proof State	Title	CWE / MITRE	Merged / Endpoints	Composite Conf (FP)
F-32	192.168.195.139	Medium (5.4)	VALIDATED	Web Application Missing Anti-CSRF Tokens & Hardening Headers	CWE-352 / T1190	1 obs / 1 ep	95% (Tier)
F-33	192.168.195.140:8080	Medium (5.4)	VALIDATED	Content-Security-Policy (CSP) Header Not Set	CWE-693 / T1190	12 obs / 3 ep	95% (Tier)
F-34	192.168.195.140:8080	Medium (5.4)	VALIDATED	Missing Anti-Clickjacking Protection (X-Frame-Options / frame-ancestors)	CWE-1021 / T1190	5 obs / 3 ep	95% (Tier)
F-35	192.168.195.140	Medium (5.4)	VALIDATED	Web Application Missing Anti-CSRF Tokens & Hardening Headers	CWE-352 / T1190	2 obs / 2 ep	95% (Tier)
F-36	192.168.195.1	Medium (5.3)	VALIDATED	Administrative Web Interface Reachable Without Network Restriction	CWE-284 / T1083	1 obs / 1 ep	90% (Tier)
F-37	192.168.195.139	Medium (5.3)	VALIDATED	Administrative Web Interface Reachable Without Network Restriction	CWE-284 / T1083	1 obs / 1 ep	95% (Tier)
F-38	192.168.195.139	Medium (5.3)	VALIDATED	Exposed phpMyAdmin Web Administrative Interface (/phpMyAdmin/)	CWE-306 / T1190	1 obs / 1 ep	95% (Tier)
F-39	192.168.195.139	Medium (5.3)	VALIDATED	NFS RPC Service & Export Table Exposed (Ports 111 / 2049)	CWE-732 / T1083	3 obs / 3 ep	95% (Tier)
F-40	192.168.195.139:80	Medium (5.3)	VALIDATED	PHP Configuration Diagnostic Page Exposed (/phpinfo.php)	CWE-200 / T1592	4 obs / 4 ep	95% (Tier)
F-41	192.168.195.139	Medium (5.3)	VALIDATED	Web Authentication Endpoint Missing Brute-Force Rate Limiting (/dvwa/login.php)	CWE-307 / T1110.001	1 obs / 1 ep	95% (Tier)
F-42	192.168.195.140	Medium (5.3)	VALIDATED	Administrative Web Interface Reachable Without Network Restriction	CWE-284 / T1083	1 obs / 1 ep	90% (Tier)
F-43	192.168.195.1	Medium (5.3)	OBSERVED	PostgreSQL Database Listener Exposed on Network (TCP Port 5432)	CWE-200 / T1046	1 obs / 1 ep	48% (Tier)
F-44	192.168.195.139	Medium (5.3)	OBSERVED	End-of-Life OpenSSH 4.7p1 Service Banner Observed (TCP Port 22)	CWE-1104 / T1021.004	1 obs / 1 ep	48% (Tier)
F-45	192.168.195.139	Medium (5.3)	OBSERVED	Java RMI Registry Exposed on Network (TCP Port 1099)	CWE-502 / T1190	1 obs / 1 ep	48% (Tier)
F-46	192.168.195.139	Medium (5.3)	OBSERVED	MySQL Database Listener Exposed on Network (TCP Port 3306)	CWE-200 / T1046	1 obs / 1 ep	48% (Tier)
F-47	192.168.195.139	Medium (5.3)	OBSERVED	VNC Remote Desktop Service Exposed on TCP Port 5900 (Protocol 3.3)	CWE-287 / T1021.001	1 obs / 1 ep	48% (Tier)

ID	Host : Port	Severity & CVSS	Proof State	Title	CWE / MITRE	Merged / Endpoints	Composite Conf (FP)
F-48	192.168.195.139	Medium (5.3)	OBSERVED	X11 Display Server Listener Exposed (TCP Port 6000)	CWE-306 / T1113	1 obs / 1 ep	48% (Tier)
F-49	192.168.195.140	Medium (5.3)	OBSERVED	End-of-Life OpenSSH 4.7p1 Service Banner Observed (TCP Port 22)	CWE-1104 / T1021.004	1 obs / 1 ep	48% (Tier)
F-50	192.168.195.1	Medium (0.0)	OBSERVED	SMB Port 445 Exposed — Unverified MS17-010 (EternalBlue) Candidate	CWE-200 / T1210	1 obs / 1 ep	48% (Tier)
F-51	192.168.195.130	Medium (0.0)	OBSERVED	SMB Port 445 Exposed — Unverified MS17-010 (EternalBlue) Candidate	CWE-200 / T1210	1 obs / 1 ep	48% (Tier)
F-52	192.168.195.139	Medium (0.0)	OBSERVED	SMB Port 445 Exposed — Unverified MS17-010 (EternalBlue) Candidate	CWE-200 / T1210	1 obs / 1 ep	48% (Tier)
F-53	192.168.195.140	Medium (0.0)	OBSERVED	SMB Port 445 Exposed — Unverified MS17-010 (EternalBlue) Candidate	CWE-200 / T1210	1 obs / 1 ep	48% (Tier)
F-54	192.168.195.1	Low (3.1)	VALIDATED	Developer / Debugging Comments Exposed in HTML/JS Source	CWE-615 / T1592	10 obs / 1 ep	90% (Tier)
F-55	192.168.195.1:8000	Low (3.1)	VALIDATED	HTTP Permissions-Policy Header Missing	CWE-693 / T1190	2 obs / 2 ep	95% (Tier)
F-56	192.168.195.1:8000	Low (3.1)	VALIDATED	HTTP Referrer-Policy Header Missing	CWE-200 / T1592	2 obs / 2 ep	95% (Tier)
F-57	192.168.195.1:8000	Low (3.1)	VALIDATED	HTTP Server Banner Version Disclosure ('Server' Header)	CWE-200 / T1592	56 obs / 3 ep	95% (Tier)
F-58	192.168.195.1:8000	Low (3.1)	VALIDATED	HTTP Strict-Transport-Security (HSTS) Header Missing	CWE-319 / T1557	5 obs / 4 ep	95% (Tier)
F-59	192.168.195.1:8000	Low (3.1)	VALIDATED	Session Cookie Missing 'HttpOnly' Flag	CWE-1004 / T1190	63 obs / 2 ep	95% (Tier)
F-60	192.168.195.1:8000	Low (3.1)	VALIDATED	Session Cookie Missing 'SameSite' Attribute	CWE-352 / T1190	187 obs / 2 ep	95% (Tier)
F-61	192.168.195.1	Low (3.1)	VALIDATED	Unix Timestamp Disclosure in HTTP Response Content	CWE-200 / T1592	228 obs / 1 ep	95% (Tier)
F-62	192.168.195.1	Low (3.1)	VALIDATED	Verbose Content Body Returned in HTTP Redirect Response (Big Redirect)	CWE-200 / T1592	58 obs / 1 ep	95% (Tier)
F-63	192.168.195.1	Low (3.1)	VALIDATED	Web Server Directory Listing Enabled (/icons/ & Static Directories)	CWE-548 / T1083	3 obs / 2 ep	95% (Tier)
F-64	192.168.195.1:8080	Low (3.1)	VALIDATED	X-Content-Type-Options Header Missing (MIME-Sniffing Exposure)	CWE-693 / T1190	46 obs / 3 ep	95% (Tier)
F-65	192.168.195.1:8000	Low (3.1)	VALIDATED	robots.txt Disallows Sensitive Web Paths	CWE-200 / T1592	1 obs / 1 ep	95% (Tier)

ID	Host : Port	Severity & CVSS	Proof State	Title	CWE / MITRE	Merged / Endpoints	Composite Conf (FP)
F-66	192.168.195.129	Low (3.1)	VALIDATED	HTTP Server Banner Version Disclosure ('Server' Header)	CWE-200 / T1592	3 obs / 1 ep	95% (Tier)
F-67	192.168.195.130	Low (3.1)	VALIDATED	HTTP Server Banner Version Disclosure ('Server' Header)	CWE-200 / T1592	3 obs / 1 ep	95% (Tier)
F-68	192.168.195.139	Low (3.1)	VALIDATED	Apache HTTP OPTIONS Allow Header Heuristic (Optionsbleed CVE-2017-9798 Candidate)	CWE-200 / T1190	1 obs / 1 ep	95% (Tier)
F-69	192.168.195.139:80	Low (3.1)	VALIDATED	HTTP Permissions-Policy Header Missing	CWE-693 / T1190	1 obs / 1 ep	95% (Tier)
F-70	192.168.195.139:80	Low (3.1)	VALIDATED	HTTP Referrer-Policy Header Missing	CWE-200 / T1592	1 obs / 1 ep	95% (Tier)
F-71	192.168.195.139:80	Low (3.1)	VALIDATED	HTTP Server Banner Version Disclosure ('Server' Header)	CWE-200 / T1592	3 obs / 2 ep	95% (Tier)
F-72	192.168.195.139:80	Low (3.1)	VALIDATED	HTTP Strict-Transport-Security (HSTS) Header Missing	CWE-319 / T1557	3 obs / 3 ep	95% (Tier)
F-73	192.168.195.139:80	Low (3.1)	VALIDATED	HTTP Technology Stack Disclosure ('X-Powered-By' Header)	CWE-200 / T1592	2 obs / 2 ep	95% (Tier)
F-74	192.168.195.139	Low (3.1)	VALIDATED	Web Server Directory Listing Enabled (/icons/ & Static Directories)	CWE-548 / T1083	1 obs / 1 ep	95% (Tier)
F-75	192.168.195.139:80	Low (3.1)	VALIDATED	X-Content-Type-Options Header Missing (MIME-Sniffing Exposure)	CWE-693 / T1190	3 obs / 3 ep	95% (Tier)
F-76	192.168.195.140:8080	Low (3.1)	VALIDATED	HTTP Permissions-Policy Header Missing	CWE-693 / T1190	1 obs / 1 ep	95% (Tier)
F-77	192.168.195.140:8080	Low (3.1)	VALIDATED	HTTP Referrer-Policy Header Missing	CWE-200 / T1592	1 obs / 1 ep	95% (Tier)
F-78	192.168.195.140:8080	Low (3.1)	VALIDATED	HTTP Server Banner Version Disclosure ('Server' Header)	CWE-200 / T1592	17 obs / 2 ep	95% (Tier)
F-79	192.168.195.140:8080	Low (3.1)	VALIDATED	HTTP Strict-Transport-Security (HSTS) Header Missing	CWE-319 / T1557	4 obs / 3 ep	95% (Tier)
F-80	192.168.195.140	Low (3.1)	VALIDATED	Session Cookie Missing 'HttpOnly' Flag	CWE-1004 / T1190	5 obs / 1 ep	95% (Tier)
F-81	192.168.195.140	Low (3.1)	VALIDATED	Session Cookie Missing 'SameSite' Attribute	CWE-352 / T1190	5 obs / 1 ep	95% (Tier)
F-82	192.168.195.140:8080	Low (3.1)	VALIDATED	X-Content-Type-Options Header Missing (MIME-Sniffing Exposure)	CWE-693 / T1190	5 obs / 3 ep	95% (Tier)
F-83	192.168.195.130	Low (3.1)	OBSERVED	SMB NetBIOS/CIFS Service Enumeration Surface (Ports 139/445)	CWE-287 / T1021.002	2 obs / 2 ep	48% (Tier)

ID	Host : Port	Severity & CVSS	Proof State	Title	CWE / MITRE	Merged / Endpoints	Composite Conf (FP)
F-84	192.168.195.139	Low (3.1)	OBSERVED	SMB NetBIOS/CIFS Service Enumeration Surface (Ports 139/445)	CWE-287 / T1021.002	2 obs / 2 ep	48% (Tier)
F-85	192.168.195.140	Low (3.1)	OBSERVED	SMB NetBIOS/CIFS Service Enumeration Surface (Ports 139/445)	CWE-287 / T1021.002	2 obs / 2 ep	48% (Tier)

3A.1 Detailed Technical Findings & Cryptographic Evidence Cards

F-01: Ingreslock / Bindshell Unauthenticated Root Shell on TCP Port 1524 [CRITICAL · CVSS 9.8 · CONFIRMED_COMPROMISE]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	CONFIRMED_COMPROMISE (Legacy Tier: CONFIRMED) | Privilege: ROOT (uid=0 / gid=0) | Auth: True | File: READ_WRITE (/etc/shadow verified) | DB: NONE | Privesc Lifecycle: PRIVILEGE_TRANSITION_EXECUTED
7-D Proof Vector (V3):	Execution: ROOT_SHELL_EXECUTED | Auth: UNAUTHENTICATED_ROOT_BIND | Exploitability: EXPLOIT_CONFIRMED | Privilege: ROOT (uid=0 / gid=0) | DataAccess: SHADOW_HASHES_READ | Impact: FULL_SYSTEM_COMPROMISE | EvidenceConf: 99%
Access Scope Summary:	Unrestricted root command execution (uid=0) and /etc/shadow read access verified (same target host 192.168.195.139)
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-INGRESLOCK-1524 | Advisory: CERT/CC CA-1994-01 (inetd / ingreslock Backdoor Bindshell Exposure) | CWE: CWE-306 | MITRE: T1059.004 (Execution — Command and Scripting Interpreter: Unix Shell) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-306 (Missing Authentication for Critical Function) | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H
Separated Confidence (V3):	EvidenceConf=99% | ExploitabilityConf=99% | ImpactConf=99% | 5-D Vector: EvidenceConf=99% (Det=99%, Val=99%, Art=100%) | ExploitabilityConf=99% | ImpactConf=99% | FP Band: Tier 1 — Cryptographically & Executionally Verified (<1% Artifact Uncertainty)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 1 unique endpoint(s): 192.168.195.139:1524
Root Cause & Compliance:	Malicious Backdoor / Intentional Vulnerable Build [DIRECTLY_ESTABLISHED] | Compliance Status: CONFIRMED_COMPLIANCE_DEFICIENCY
Technical Description:	TCP port 1524 (ingreslock) exposes an unauthenticated interactive root shell (/bin/sh). Note: rsh/bindshell probes against port 1524 interact with this same underlying root bind shell and have been consolidated into this single compromise record.
Security Risk:	CRITICAL — Immediate unauthenticated root compromise (uid=0) with full filesystem and /etc/shadow read/write access.
EvidenceCore V3 Record:	ID: PROV-F-01-E204DF2EED | SHA-256: e204df2eeddb34a83cf6a8a7... | HMAC-SHA256: 6a3403fd20629526ae8cd11a... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-01):

```
INGRESLOCK BACKDOOR: Root shell on port 1524!
id: root@metasploitable:/#
$ whoami: uid=0(root) gid=0(root) groups=0(root)
root@metasploitable:/# root
root@metasploitable:/#
$ cat /etc/shadow | head -3: root:$1$/avpfBJ1$x0z8w5UF9Iv./DR9E9Lid.:14747:0:99999:7:::
daemon:*:14684:0:99999:7:::
bin:*:14684:0:99999:7:::
root@met
$ uname -a: Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
root@metasploitable:/#
$ ifconfig 2>/dev/null: eth0 Link enca
---
rsh access: root@metasploitable:/#
```

Immediate Containment:	Immediately terminate the `/bin/sh` bind shell listening on TCP port 1524 (`fuser -k 1524/tcp`), comment out the `ingreslock` line in `/etc/inetd.conf` or `/etc/xinetd.d/`, and reload `inetd`.
Short-Term Hardening:	Block TCP port 1524 at the host firewall (`ufw deny 1524/tcp` / `iptables -A INPUT -p tcp --dport 1524 -j DROP`) and audit `/etc/inetd.conf` for unauthorized backdoor listeners (`rsh`, `rexec`, `rlogin`).

Long-Term Strategic Fix:	Decommission `inetd`/`xinetd` legacy super-server daemons entirely and implement automated host-based port baseline monitoring.
Verification Command:	Run `nmap -sV -p 1524 192.168.195.139` and confirm port 1524 returns `closed` or `filtered` with no shell prompt.

F-02: vsFTPd 2.3.4 Smiley-Face Backdoor Remote Root Compromise (CVE-2011-2523) [CRITICAL · CVSS 9.8 · CONFIRMED_COMPROMISE]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	CONFIRMED_COMPROMISE (Legacy Tier: CONFIRMED) | Privilege: ROOT (uid=0 / gid=0) | Auth: True | File: READ_WRITE (/etc/shadow verified) | DB: NONE | Privesc Lifecycle: PRIVILEGE_TRANSITION_EXECUTED
7-D Proof Vector (V3):	Execution: ROOT_SHELL_EXECUTED | Auth: UNAUTHENTICATED_ROOT_BIND | Exploitability: EXPLOIT_CONFIRMED | Privilege: ROOT (uid=0 / gid=0) | DataAccess: SHADOW_HASHES_READ | Impact: FULL_SYSTEM_COMPROMISE | EvidenceConf: 99%
Access Scope Summary:	Unrestricted root command execution (uid=0) and /etc/shadow read access verified (same target host 192.168.195.139)
Taxonomy & CVSS v3.1:	CVE: CVE-2011-2523 | Custom ID: XSTRIKE-VSFTPD-234-BACKDOOR | Advisory: OSVDB-73573 / vsFTPd v2.3.4 Compromised Archive Advisory | CWE: CWE-506 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A06:2021 (Vulnerable and Outdated Components) | CWE-506 (Embedded Malicious Code) | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H
Separated Confidence (V3):	EvidenceConf=99% | ExploitabilityConf=99% | ImpactConf=99% | 5-D Vector: EvidenceConf=99% (Det=99%, Val=99%, Art=100%) | ExploitabilityConf=99% | ImpactConf=99% | FP Band: Tier 1 — Cryptographically & Executionally Verified (<1% Artifact Uncertainty)
Deduplication & Endpoints:	Consolidated 3 raw observation(s) across 2 unique endpoint(s): 192.168.195.139, 192.168.195.139:21
Root Cause & Compliance:	Malicious Backdoor / Intentional Vulnerable Build [DIRECTLY_ESTABLISHED] | Compliance Status: CONFIRMED_COMPLIANCE_DEFICIENCY
Technical Description:	The vsFTPd 2.3.4 daemon contains a maliciously planted backdoor (CVE-2011-2523) that opens an unauthenticated root bind shell on TCP port 6200 when a username ending in ':)' is supplied.
Security Risk:	CRITICAL — Complete host compromise. Unauthenticated remote attacker obtains uid=0(root) command execution and access to /etc/shadow password hashes.
EvidenceCore V3 Record:	ID: PROV-F-02-284F15F547 | SHA-256: 284f15f5476afdc0ee5b1a06... | HMAC-SHA256: 87296c78e8ad924832a3f58d... | Module: AutoPentest-VulnScan, XStrike-BuiltinVuln, AutoPentest-Exploit

Captured Technical Evidence (F-02):

```
BACKDOOR SHELL on :6200!
$ id: uid=0(root) gid=0(root)
$ whoami: root
$ uname -a: Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
$ cat /etc/shadow | head -5: root:$1$/avpfBJ1$x0z8w5UF9Iv./DR9E9Lid.:14747:0:99999:7:::
daemon:*:14684:0:99999:7:::
bin:*:14684:0:99999:7:::
sys:$1$fUX6BPOt$Miyc3UpOzQJqz4s5wFD9l0
[!] ROOT PASSWORD HASH DUMPED!
$ cat /etc/passwd=[REDACTED]
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bi
---
[2026-10-30 19:22:14] FTP service on 192.168.195.139:21.
Service: ftp, Product: vsftpd, Version: 2.3.4
MATCH: vsftpd 2.3.4 — known backdoor (CVE-
```

Immediate Containment:	Immediately stop the vsFTPd daemon (`systemctl stop vsftpd` or `killall vsftpd`), terminate any process bound to TCP port 6200, and isolate 192.168.195.139 from the network.
Short-Term Hardening:	Replace the compromised vsFTPd 2.3.4 binary with a cryptographically verified upstream package (`apt-get update && apt-get install --only-upgrade vsftpd`) or migrate file transfers to SFTP/SCP over hardened OpenSSH.
Long-Term Strategic Fix:	Rebuild the compromised host from a trusted gold image, rotate all local password hashes in `/etc/shadow`, and enforce package signature verification in CI/CD.
Verification Command:	Verify TCP port 21 and port 6200 are closed or that `vsftpd -v` reports a patched release (>= 3.0.3) and `USER test:)` no longer opens port 6200.

F-03: PostgreSQL 8.3.1 Default Superuser Authentication (postgres:postgres) [CRITICAL · CVSS 9.1 · CONFIRMED_AUTH]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	CONFIRMED_AUTH (Legacy Tier: CONFIRMED) | Privilege: DB_SUPERUSER (postgres) | Auth: True | File: NONE | DB: AUTHENTICATED_SCHEMA_READ (PostgreSQL 8.3.1 databases & pg_catalog tables enumerated) | Privesc Lifecycle: TECHNIQUE_VALIDATED
7-D Proof Vector (V3):	Execution: AUTHENTICATED_DB_QUERY_EXECUTED | Auth: AUTHENTICATED_SUPERUSER (postgres) | Exploitability: CREDENTIAL_CONFIRMED | Privilege: DB_SUPERUSER (postgres) | DataAccess: DB_CATALOG_AUTHENTICATED | Impact: DATABASE_SUPERUSER_COMPROMISE | EvidenceConf: 98%
Access Scope Summary:	Authenticated PostgreSQL 8.3.1 superuser session ('postgres'); database & table catalog enumerated (read-only verification)
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-PGSQL-DEFAULT-SUPERUSER | Advisory: PostgreSQL pg_hba.conf & Default Superuser Password Advisory | CWE: CWE-798 | MITRE: T1078.001 (Defense Evasion / Persistence — Valid Accounts: Default Accounts) | Taxonomy: OWASP-A07:2021 (Identification and Authentication Failures) | CWE-798 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:N
Separated Confidence (V3):	EvidenceConf=99% | ExploitabilityConf=95% | ImpactConf=88% | 5-D Vector: EvidenceConf=99% (Det=99%, Val=98%, Art=100%) | ExploitabilityConf=95% | ImpactConf=88% | FP Band: Tier 1 — Cryptographically & Executionally Verified (<1% Artifact Uncertainty)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 1 unique endpoint(s): 192.168.195.139:5432
Root Cause & Compliance:	Identity & Credential Hygiene Deficiency [DIRECTLY_ESTABLISHED] | Compliance Status: CONFIRMED_COMPLIANCE_DEFICIENCY
Technical Description:	PostgreSQL 8.3.1 on TCP port 5432 accepts default superuser credentials ('postgres:postgres'), allowing remote database enumeration and catalog read access.
Security Risk:	CRITICAL — Authenticated database superuser access; permits reading/modifying all databases and potential OS command execution via pg_catalog functions.
EvidenceCore V3 Record:	ID: PROV-F-03-5EBB5E02B0 | SHA-256: 5ebb5e02b0e5c7bdb3c30edd... | HMAC-SHA256: 1f9965cca9a6a6b0c52ae32a... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-03):

```
PostgreSQL LOGIN: postgres:postgres
version: [('PostgreSQL 8.3.1 on i486-pc-linux-gnu, compiled by GCC cc (GCC) 4.2.3 (Ubuntu
4.2.3-2ubuntu4)',)]
databases: [('template1',), ('template0',), ('postgres',)]
users: [('postgres', 'md53175bce1d3201d16594cebf9d7eb3f9d')]
tables: [('ymdoo',)]
```

Immediate Containment:	Change the PostgreSQL `postgres` superuser password (`ALTER USER postgres WITH PASSWORD '<strong_random_secret>';`) and restrict `/etc/postgresql/8.3/main/pg_hba.conf` to `md5` or `scram-sha-256` over local sockets only.

Short-Term Hardening:	Bind PostgreSQL exclusively to localhost in `postgresql.conf` (`listen_addresses = 'localhost'`) and block TCP port 5432 at the host firewall for untrusted subnets.
Long-Term Strategic Fix:	Upgrade EOL PostgreSQL 8.3.1 to a supported release (PostgreSQL 15/16) with role-based least-privilege database accounts.
Verification Command:	Attempt remote authentication with `psql -h 192.168.195.139 -U postgres` and confirm connection refusal or `FATAL: password authentication failed`.

F-04: FTP Default Credential Authentication (msfadmin:msfadmin) [HIGH · CVSS 8.2 · CONFIRMED_AUTH]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	CONFIRMED_AUTH (Legacy Tier: CONFIRMED) | Privilege: STANDARD_USER (msfadmin) | Auth: True | File: READ_LIST (user directory listing verified; write not executed) | DB: NONE | Privesc Lifecycle: PRECONDITION_DISCOVERED
7-D Proof Vector (V3):	Execution: AUTHENTICATED_SESSION_VERIFIED | Auth: AUTHENTICATED_USER (msfadmin) | Exploitability: CREDENTIAL_CONFIRMED | Privilege: STANDARD_USER (msfadmin) | DataAccess: DIRECTORY_LISTING_READ | Impact: UNAUTHORIZED_ACCOUNT_ACCESS | EvidenceConf: 98%
Access Scope Summary:	Authenticated FTP session as standard user 'msfadmin'; directory listing verified (non-root)
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-FTP-DEFAULT-CREDS | Advisory: Rapid7 Metasploitable2 Baseline Default Account Advisory | CWE: CWE-798 | MITRE: T1078.001 (Defense Evasion / Persistence — Valid Accounts: Default Accounts) | Taxonomy: OWASP-A07:2021 (Identification and Authentication Failures) | CWE-798 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:L/A:N
Separated Confidence (V3):	EvidenceConf=99% | ExploitabilityConf=95% | ImpactConf=88% | 5-D Vector: EvidenceConf=99% (Det=99%, Val=98%, Art=100%) | ExploitabilityConf=95% | ImpactConf=88% | FP Band: Tier 1 — Cryptographically & Executionally Verified (<1% Artifact Uncertainty)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139:21
Root Cause & Compliance:	Identity & Credential Hygiene Deficiency [DIRECTLY_ESTABLISHED] | Compliance Status: CONFIRMED_COMPLIANCE_DEFICIENCY
Technical Description:	The FTP service on port 21 authenticates the default account 'msfadmin:msfadmin', granting interactive access to the user's directory structure.
Security Risk:	HIGH — Unauthorized authenticated file access as standard user (msfadmin); enables credential reuse across SSH/Samba.
EvidenceCore V3 Record:	ID: PROV-F-04-3529D6A41A | SHA-256: 3529d6a41a6c352d8da3ac87... | HMAC-SHA256: 6238df153d667d86a20a302d... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-04):

```
FTP login msfadmin:msfadmin SUCCESS. Files: vulnerable
```

Immediate Containment:	Change the `msfadmin` account password immediately (`passwd msfadmin`) and revoke FTP access for administrative users via `/etc/ftpusers`.
Short-Term Hardening:	Disable cleartext FTP authentication altogether in `/etc/vsftpd.conf` (`local_enable=NO` or enforce TLS via `ssl_enable=YES`).
Long-Term Strategic Fix:	Enforce centralized identity management with minimum 14-character password complexity and multi-factor authentication.
Verification Command:	Attempt FTP login with `msfadmin:msfadmin` and verify the server returns `530 Login incorrect`.

F-05: SSH Default Credential Shell Access (msfadmin:msfadmin) [HIGH · CVSS 8.2 · CONFIRMED_AUTH]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	CONFIRMED_AUTH (Legacy Tier: CONFIRMED) | Privilege: STANDARD_USER (uid=1000 msfadmin, gid=1000) | Auth: True | File: READ_WRITE (standard user home directory) | DB: NONE | Privesc Lifecycle: PRECONDITION_DISCOVERED
7-D Proof Vector (V3):	Execution: USER_SHELL_EXECUTED | Auth: AUTHENTICATED_USER (uid=1000) | Exploitability: CREDENTIAL_CONFIRMED | Privilege: STANDARD_USER (uid=1000 msfadmin, gid=1000) | DataAccess: USER_FILESYSTEM_READ | Impact: INTERACTIVE_USER_SHELL_COMPROMISE | EvidenceConf: 98%
Access Scope Summary:	Interactive SSH shell established as unprivileged user uid=1000(msfadmin); root escalation not executed in this step
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-SSH-DEFAULT-CREDS | Advisory: OpenSSH Default/Predictable Account Hygiene Advisory | CWE: CWE-798 | MITRE: T1078.001 (Defense Evasion / Persistence — Valid Accounts: Default Accounts) | Taxonomy: OWASP-A07:2021 (Identification and Authentication Failures) | CWE-798 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:L/A:N
Separated Confidence (V3):	EvidenceConf=99% | ExploitabilityConf=95% | ImpactConf=88% | 5-D Vector: EvidenceConf=99% (Det=99%, Val=98%, Art=100%) | ExploitabilityConf=95% | ImpactConf=88% | FP Band: Tier 1 — Cryptographically & Executionally Verified (<1% Artifact Uncertainty)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139:22
Root Cause & Compliance:	Identity & Credential Hygiene Deficiency [DIRECTLY_ESTABLISHED] | Compliance Status: CONFIRMED_COMPLIANCE_DEFICIENCY
Technical Description:	The OpenSSH service on port 22 accepts default credentials 'msfadmin:msfadmin', granting an interactive shell as standard user uid=1000(msfadmin). Separate local privilege escalation is required for root access.
Security Risk:	HIGH — Authenticated interactive Linux shell as standard user (uid=1000); provides direct launchpad for local privilege escalation.
EvidenceCore V3 Record:	ID: PROV-F-05-6715A75078 | SHA-256: 6715a75078cb9770fd881c37... | HMAC-SHA256: 5678aa19ea3c7d441077c33b... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-05):

```
SSH LOGIN SUCCESS: msfadmin:msfadmin
Identity: uid=1000(msfadmin) gid=1000(msfadmin)
groups=4(adm),20(dialout),24(cdrom),25(floppy),29(audio),30(di
Users: root, daemon, bin, sys, sync, games, man, lp, mail, news, uucp, proxy, www-data, backup, list
SUID binaries (privesc): /bin/umount
/bin/fusermount
/bin/su
/bin/mount
/bin/ping
/bin/ping6
/sbin/mount.nfs
/lib/dhcp3-client/call-dhclient-script
/usr/bin/sudoedit
/usr/bin/X
/usr/bin/netkit-rsh
/usr/bin/gpasswd
/usr/bin/tr
[!] CONFIG CREDS:
```

Immediate Containment:	Rotate the `msfadmin` password immediately (`passwd msfadmin`) and terminate active unauthorized SSH sessions.
Short-Term Hardening:	Configure `/etc/ssh/sshd_config` with `PasswordAuthentication no`, `PermitRootLogin no`, and `MaxAuthTries 3`, then reload `sshd`.

Long-Term Strategic Fix:	Enforce Ed25519/RSA public-key authentication backed by hardware security keys or an SSH Certificate Authority.
Verification Command:	Verify `ssh msfadmin@192.168.195.139` rejects password authentication (`Permission denied (publickey)`).

F-06: Local Privilege Escalation Vector: SUID Binary (/usr/bin/nmap) [HIGH · CVSS 7.8 · EXPLOITABLE]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	EXPLOITABLE (Legacy Tier: VALIDATED) | Privilege: LOCAL_ESCALATION_PRECONDITION (uid=1000 -> uid=0 via nmap --interactive) | Auth: True | File: READ_ONLY (/usr/bin/nmap SUID-root bit verified) | DB: NONE | Privesc Lifecycle: TECHNIQUE_VALIDATED
7-D Proof Vector (V3):	Execution: AUTHENTICATED_ENUMERATION_EXECUTED | Auth: AUTHENTICATED_USER (uid=1000) | Exploitability: PRECONDITION_VERIFIED | Privilege: LOCAL_ESCALATION_PRECONDITION (uid=1000 -> uid=0 via nmap --interactive) | DataAccess: FILESYSTEM_ACL_VERIFIED | Impact: LOCAL_ROOT_ESCALATION_READY | EvidenceConf: 95%
Access Scope Summary:	Authenticated SSH check confirmed SUID-root bit on /usr/bin/nmap (--interactive !sh technique validated; root transition not executed via SSH)
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-PRIVESC-SUID-NMAP | Advisory: GTFOBins SUID Nmap (--interactive !sh) Privilege Escalation Vector | CWE: CWE-269 | MITRE: T1548.001 (Privilege Escalation — Abuse Elevation Control Mechanism: Setuid and Setgid) | Taxonomy: MITRE ATT&CK T1548.001 | CWE-269 (Improper Privilege Management) | Vector: CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:H
Separated Confidence (V3):	EvidenceConf=96% | ExploitabilityConf=82% | ImpactConf=80% | 5-D Vector: EvidenceConf=96% (Det=96%, Val=94%, Art=100%) | ExploitabilityConf=82% | ImpactConf=80% | FP Band: Tier 1 — Authenticated Exploit Precondition Verified (<5% Uncertainty)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 2 unique endpoint(s): 192.168.195.139:22, 192.168.195.139
Root Cause & Compliance:	Excessive Local Privilege / File Permission Misconfiguration [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Post-authentication enumeration over SSH identified /usr/bin/nmap with the setuid-root bit enabled. Legacy Nmap versions (2.02–5.21) support '--interactive' mode, allowing an authenticated local user (msfadmin, uid=1000) to spawn a root shell (!sh).
Security Risk:	HIGH (EXPLOITABLE, CVSS 7.8) — Validated local privilege escalation path from standard user (uid=1000) to root (uid=0).
EvidenceCore V3 Record:	ID: PROV-F-06-E061A6DE7E | SHA-256: e061a6de7e1680cfa2141f64... | HMAC-SHA256: f8ae1fd25e62252ee5831fc1... | Module: AutoPentest-Exploit, AutoPentest-PrivEsc

Captured Technical Evidence (F-06):

```
SUID nmap found — known privilege escalation vector!
```

Immediate Containment:	Remove the SUID-root bit from `/usr/bin/nmap` immediately (`chmod u-s /usr/bin/nmap`) and audit all SUID binaries via `find / -perm -4000 -type f 2>/dev/null`.
Short-Term Hardening:	Upgrade Nmap to a modern release (>= 7.9x) that removes `--interactive` shell escapes and enforce `nosuid` mount options on `/tmp` and `/home`.
Long-Term Strategic Fix:	Migrate the legacy Ubuntu 8.04 system to a supported LTS distribution with AppArmor/SELinux enforcement.
Verification Command:	Run `ls -la /usr/bin/nmap` and confirm permissions read `-rwxr-xr-x` without the `s` flag, and `nmap --interactive` is unavailable.

F-07: DVWA Web Application Default Administrative Credentials Accepted (admin:password) [HIGH · CVSS 8.2 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-DVWA-DEFAULT-CREDS | Advisory: OWASP A07:2021 — Identification and Authentication Failures | CWE: CWE-798 | MITRE: T1078.001 (Defense Evasion / Persistence — Valid Accounts: Default Accounts) | Taxonomy: OWASP-A07:2021 (Identification and Authentication Failures) | CWE-798 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=68% | ImpactConf=72% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=68% | ImpactConf=72% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139
Root Cause & Compliance:	Identity & Credential Hygiene Deficiency [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	The Damn Vulnerable Web Application (DVWA) login endpoint (/dvwa/login.php) accepted default administrative credentials ('admin:password') during automated authentication verification (consolidated from OWASP A07 testing).
Security Risk:	HIGH — Validated administrative web session access enabling command injection, SQLi, and file upload testing modules.
EvidenceCore V3 Record:	ID: PROV-F-07-D70EE7FC10 | SHA-256: d70ee7fc1053871ca6df4bab... | HMAC-SHA256: 3843c286053b9545e1d3caf8... | Module: AutoPentest-OWASP

Captured Technical Evidence (F-07):

```
A07 Auth Failures: 2 finding(s)
[CWE-798] Default Creds: DVWA (admin:password): Login successful with admin:password!
[CWE-798] Default Creds: DVWA (admin:admin): Login successful with admin:admin!
```

Immediate Containment:	Disable or unpublish the `/dvwa/` training application from the web root (`mv /var/www/dvwa /root/dvwa.disabled`) or rotate the `admin` password in the DVWA MySQL `users` table.
Short-Term Hardening:	Restrict access to `/dvwa/` via Apache IP allowlisting (`Require ip 127.0.0.1`) if used in a controlled lab.
Long-Term Strategic Fix:	Prohibit deployment of intentionally vulnerable training applications (DVWA, Mutillidae) on shared infrastructure hosts.
Verification Command:	Submit `username=admin&password=password` to `/dvwa/login.php` and verify authentication fails or returns HTTP 403/404.

F-08: Web Application SQL Injection Indicator (Mutillidae) [HIGH · CVSS 8.2 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: INDICATOR_ONLY (Mutillidae SQL response indicator matched; controlled data extraction not dumped) | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC

7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: INJECTION_INDICATOR_VALIDATED | Privilege: NONE | DataAccess: SQL_ERROR_OR_TABLE_INDICATOR | Impact: DATABASE_INJECTION_EXPOSURE | EvidenceConf: 92%
Access Scope Summary:	Positive SQL injection response indicator on Mutillidae endpoint; controlled row extraction not executed
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-SQLI-MUTILLIDAE | Advisory: OWASP A03:2021 — Injection (SQL Injection) | CWE: CWE-89 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A03:2021 (Injection) | CWE-89 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=68% | ImpactConf=72% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=68% | ImpactConf=72% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139:80
Root Cause & Compliance:	Insecure Application Coding (Input Validation) [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Automated SQL injection testing against the Mutillidae web application matched a database table/error indicator in the HTTP 200 response. Classified as VALIDATED pending full request/response payload and controlled data dump verification.
Security Risk:	HIGH — Unsanitized user input reaches backend SQL queries; could allow unauthorized database read/modification.
EvidenceCore V3 Record:	ID: PROV-F-08-9CC3744D82 | SHA-256: 9cc3744d8284a81a26266e98... | HMAC-SHA256: 41c3ba074b1f53f913f7b7ac... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-08):

```
SQL Injection confirmed:
Mutillidae SQLi: table found
```

Immediate Containment:	Disable or restrict the vulnerable Mutillidae PHP endpoint via Apache rewrite/location rules and enable ModSecurity OWASP CRS SQLi rules.
Short-Term Hardening:	Refactor all database queries in the PHP application to use parameterized prepared statements (`PDO::prepare()` with bound parameters) instead of string concatenation.
Long-Term Strategic Fix:	Enforce static application security testing (SAST) in CI and restrict the web application's MySQL account to least-privilege `SELECT`/`INSERT` permissions.
Verification Command:	Re-issue SQLi test payloads against the target parameter and verify HTTP responses contain zero database table leaks or SQL syntax error strings.

F-09: Legacy Linux Kernel 2.6.24-16-server Vulnerable to DirtyCow (CVE-2016-5195) [HIGH · CVSS 7.8 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: LOCAL_ESCALATION_VECTOR (uid=1000 -> uid=0 potential) | Auth: True | File: NONE | DB: NONE | Privesc Lifecycle: PRECONDITION_DISCOVERED
7-D Proof Vector (V3):	Execution: AUTHENTICATED_ENUMERATION_EXECUTED | Auth: AUTHENTICATED_USER (uid=1000) | Exploitability: VERSION_OR_ACL_VALIDATED | Privilege: LOCAL_ESCALATION_VECTOR (uid=1000 -> uid=0 potential) | DataAccess: KERNEL_AND_SUID_METADATA | Impact: LOCAL_PRIVESC_SURFACE | EvidenceConf: 93%
Access Scope Summary:	Authenticated SSH uname -r verified Linux 2.6.24-16-server DirtyCow (CVE-2016-5195) precondition; kernel race payload withheld per RoE safety policy

Taxonomy & CVSS v3.1:	CVE: CVE-2016-5195 | Custom ID: XSTRIKE-KERNEL-DIRTYCOW | Advisory: Ubuntu USN-3106-1 / Red Hat RHSA-2016:2098 (DirtyCow) | CWE: CWE-362 | MITRE: T1068 (Privilege Escalation — Exploitation for Privilege Escalation) | Taxonomy: MITRE ATT&CK T1068 | CWE-362 (Concurrent Execution Race Condition) | Vector: CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:H
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=68% | ImpactConf=72% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=68% | ImpactConf=72% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 2 unique endpoint(s): 192.168.195.139:22, 192.168.195.139
Root Cause & Compliance:	Patch & Lifecycle Management Deficit (EOL Software) [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Authenticated SSH fingerprinting ('uname -r') confirmed Linux kernel 2.6.24-16-server, which is affected by the CVE-2016-5195 (DirtyCow) copy-on-write race condition privilege escalation vulnerability.
Security Risk:	HIGH — Local users can exploit the kernel race condition to overwrite read-only root binaries and gain uid=0 privileges.
EvidenceCore V3 Record:	ID: PROV-F-09-C1D405DA9E | SHA-256: c1d405da9e468eead300aea3... | HMAC-SHA256: 089b95973e7054bc5a48d23e... | Module: AutoPentest-Exploit, AutoPentest-PrivEsc

Captured Technical Evidence (F-09):

```
Kernel 2.6.24-16-server — vulnerable to DirtyCow (CVE-2016-5195), Multiple kernel exploits
```

Immediate Containment:	Restrict local shell access on 192.168.195.139 by disabling default SSH accounts (`msfadmin`) and setting `kernel.perf_event_paranoid = 3`.
Short-Term Hardening:	Upgrade the Linux kernel from `2.6.24-16-server` to a patched kernel incorporating the CVE-2016-5195 copy-on-write race fix (`mm/gup.c` FOLL_COW patch).
Long-Term Strategic Fix:	Reprovision the EOL Ubuntu 8.04 host onto Ubuntu 22.04/24.04 LTS with unattended kernel livepatching enabled.
Verification Command:	Run `uname -r` over SSH and verify a supported kernel version patched against CVE-2016-5195.

F-10: Deprecated SMBv1 Dialect Enabled on SMB Service [HIGH · CVSS 7.5 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: CVE-2017-0144 | Custom ID: XSTRIKE-SMBV1-ENABLED | Advisory: Microsoft / Samba SMBv1 Deprecation Advisory (KB2696547) | CWE: CWE-327 | MITRE: T1210 (Lateral Movement — Exploitation of Remote Services) | Taxonomy: OWASP-A02:2021 (Cryptographic Failures) | CWE-327 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=68% | ImpactConf=72% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=68% | ImpactConf=72% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139:445

Root Cause & Compliance:	Service Hardening & Configuration Baseline Gap [PROBABLE] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	The SMB service negotiates the legacy SMBv1 protocol dialect, which lacks modern integrity protections and exposes the host to SMBv1 protocol attacks.
Security Risk:	HIGH — Legacy protocol weakness; SMBv1 should be disabled across all Windows and Samba servers.
EvidenceCore V3 Record:	ID: PROV-F-10-5552509362 | SHA-256: 555250936296050abda223ab... | HMAC-SHA256: 6b8ec41b0aa31f866f6a0d20... | Module: XStrike-BuiltinVuln

Captured Technical Evidence (F-10):

```
SMBv1 is enabled — vulnerable to EternalBlue (MS17-010)

Evidence: SMBv1 negotiate response received from 192.168.195.139:445
```

Immediate Containment:	Disable SMBv1 protocol negotiation: in Samba `/etc/samba/smb.conf` set `server min protocol = SMB2`; on Windows run `Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol`.
Short-Term Hardening:	Enforce mandatory SMB packet signing (`server signing = mandatory` in `smb.conf`) and restart `smbd`.
Long-Term Strategic Fix:	Standardize all file shares on SMBv3.1.1 with AES-128-GCM encryption enabled.
Verification Command:	Run `nmap -p 445 --script smb-protocols` and verify dialect `NT LM 0.12 (SMBv1)` is no longer offered.

F-11: Web Application Server-Side Request Forgery (SSRF) Indicator (Mutillidae) [HIGH · CVSS 7.5 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-SSRF-MUTILLIDAE | Advisory: OWASP A10:2021 — Server-Side Request Forgery (SSRF) | CWE: CWE-918 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A10:2021 (Server-Side Request Forgery) | CWE-918 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=68% | ImpactConf=72% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=68% | ImpactConf=72% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139
Root Cause & Compliance:	Insecure Application Coding (Input Validation) [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	The Mutillidae web application accepted a server-side URL fetch parameter without outbound destination allowlisting (consolidated from OWASP A10 SSRF validation).
Security Risk:	HIGH — Allows an attacker to induce the web application to issue HTTP requests to loopback or internal subnet services.

EvidenceCore V3 Record:	ID: PROV-F-11-66473094D4 | SHA-256: 66473094d40aab6fd50d0c92... | HMAC-SHA256: 946445be8b421e6f24e2953f... | Module: AutoPentest-OWASP

Captured Technical Evidence (F-11):

```
A10 SSRF: 1 finding(s)
[CWE-918] SSRF: Mutillidae SSRF: Server fetched internal URL:
<!-- I think the database password is set to b
```

Immediate Containment:	Disable the vulnerable Mutillidae URL fetch endpoint or block outbound HTTP/TCP connections initiated by the `www-data` user via `iptables -m owner --uid-owner www-data -j DROP`.
Short-Term Hardening:	Implement strict URL parsing and domain/IP allowlisting that rejects RFC 1918 private addresses (`10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`) and loopback (`127.0.0.0/8`).
Long-Term Strategic Fix:	Route all application outbound fetches through a dedicated egress proxy with DNS rebinding protection.
Verification Command:	Attempt to fetch `http://127.0.0.1/` via the SSRF parameter and confirm the request is blocked and logged.

F-12: Deprecated SMBv1 Dialect Enabled on SMB Service [HIGH · CVSS 7.5 · VALIDATED]

Target Asset & Port:	192.168.195.140 (Ports: N/A) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: CVE-2017-0144 | Custom ID: XSTRIKE-SMBV1-ENABLED | Advisory: Microsoft / Samba SMBv1 Deprecation Advisory (KB2696547) | CWE: CWE-327 | MITRE: T1210 (Lateral Movement — Exploitation of Remote Services) | Taxonomy: OWASP-A02:2021 (Cryptographic Failures) | CWE-327 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=68% | ImpactConf=72% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=68% | ImpactConf=72% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.140:445
Root Cause & Compliance:	Service Hardening & Configuration Baseline Gap [PROBABLE] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	The SMB service negotiates the legacy SMBv1 protocol dialect, which lacks modern integrity protections and exposes the host to SMBv1 protocol attacks.
Security Risk:	HIGH — Legacy protocol weakness; SMBv1 should be disabled across all Windows and Samba servers.
EvidenceCore V3 Record:	ID: PROV-F-12-D1BCDA0079 | SHA-256: d1bcda007935cc3f0ec4f0bc... | HMAC-SHA256: 324d9ba9ebd986c7d740c013... | Module: XStrike-BuiltinVuln

Captured Technical Evidence (F-12):

```
SMBv1 is enabled — vulnerable to EternalBlue (MS17-010)

Evidence: SMBv1 negotiate response received from 192.168.195.140:445
```

Immediate Containment:	Disable SMBv1 protocol negotiation: in Samba `/etc/samba/smb.conf` set `server min protocol = SMB2`; on Windows run `Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol`.

Short-Term Hardening:	Enforce mandatory SMB packet signing (`server signing = mandatory` in `smb.conf`) and restart `smbd`.
Long-Term Strategic Fix:	Standardize all file shares on SMBv3.1.1 with AES-128-GCM encryption enabled.
Verification Command:	Run `nmap -p 445 --script smb-protocols` and verify dialect `NT LM 0.12 (SMBv1)` is no longer offered.

F-13: SMB Message Signing Not Required (NTLM Relay Exposure) [HIGH · CVSS 7.4 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-SMB-SIGNING-DISABLED | Advisory: Microsoft ADV190023 / Samba Mandatory Signing Guideline | CWE: CWE-345 | MITRE: T1557.001 (Credential Access — Adversary-in-the-Middle: LLMNR/NBT-NS Poisoning and SMB Relay) | Taxonomy: OWASP-A02:2021 (Cryptographic Failures) | CWE-345 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:H/I:H/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=68% | ImpactConf=72% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=68% | ImpactConf=72% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139
Root Cause & Compliance:	Service Hardening & Configuration Baseline Gap [PROBABLE] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	SMB message signing is not enforced by the server, allowing an adversary on the local network segment to relay NTLMv2 authentication sessions.
Security Risk:	HIGH — Enables Adversary-in-the-Middle (AitM) SMB relay attacks and unauthorized lateral movement.
EvidenceCore V3 Record:	ID: PROV-F-13-D53AF875A0 | SHA-256: d53af875a0486c1785e410c8... | HMAC-SHA256: 8dc910872ff69a04d603a454... | Module: XStrike-BuiltinVuln

Captured Technical Evidence (F-13):

```
SMB signing not enforced — relay attacks possible

Evidence: SMB security mode: 0x00 (signing not required)
```

Immediate Containment:	Enforce mandatory SMB packet signing: set `server signing = mandatory` in `/etc/samba/smb.conf` (Linux) or enable `Microsoft network server: Digitally sign communications (always)` via Group Policy (Windows).
Short-Term Hardening:	Disable NTLMv1/LM authentication (`lanman auth = no`, `ntlm auth = ntlmv2-only`) and enable Extended Protection for Authentication (EPA).
Long-Term Strategic Fix:	Migrate file share authentication to Kerberos with SMBv3 transport encryption (`smb encrypt = required`).
Verification Command:	Run `nmap -p 445 --script smb2-security-mode,smb-security-mode <target>` and confirm `Message signing enabled and required`.

F-14: Cleartext Telnet Service Enabled on TCP Port 23 [HIGH · CVSS 7.5 · OBSERVED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: SERVICE_BANNER_ONLY | Impact: EXPOSURE_ONLY | EvidenceConf: 42%
Access Scope Summary:	Service version or banner matched to known vulnerability; active exploitation not demonstrated in this record
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-TELNET-CLEARTEXT-23 | Advisory: NIST SP 800-53 SC-8 / CIS Benchmark Disable Telnet Service | CWE: CWE-319 | MITRE: T1021 (Lateral Movement — Remote Services) | Taxonomy: OWASP-A02:2021 (Cryptographic Failures) | CWE-319 (Cleartext Transmission of Sensitive Information) | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:N/A:N
Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139
Root Cause & Compliance:	Network Segmentation & Perimeter Filter Exposure [INFERRED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	Telnet service on TCP port 23 transmits authentication credentials and interactive session traffic in unencrypted cleartext.
Security Risk:	HIGH — Cleartext credential transmission enables passive network sniffing and session hijacking.
EvidenceCore V3 Record:	ID: PROV-F-14-FAB2206455 | SHA-256: fab2206455b96a2b4e317cb5... | HMAC-SHA256: c7833c36cd0f87d40507e615... | Module: AutoPentest-VulnScan

Captured Technical Evidence (F-14):

```
[2026-10-30 19:22:14] Telnet service on 192.168.195.139:23.
Service: telnet, Product: Linux telnetd, Version: unknown
Telnet transmits all traffic including credentials in cleartext.
```

Immediate Containment:	Stop the Telnet daemon (`inetd` / `telnetd`) on TCP port 23 immediately and block port 23 at the host firewall.
Short-Term Hardening:	Migrate all remote command-line administration to hardened OpenSSH (SSHv2).
Long-Term Strategic Fix:	Enforce network access control lists blocking cleartext management protocols (Telnet, rsh, rexec, FTP).
Verification Command:	Confirm `nmap -p 23 192.168.195.139` reports TCP port 23 as `closed`.

F-15: End-of-Life Apache HTTP Server 2.2.8 Deployed (TCP Port 80) [HIGH · CVSS 7.5 · OBSERVED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: SERVICE_BANNER_ONLY | Impact: EXPOSURE_ONLY | EvidenceConf: 42%
Access Scope Summary:	Service version or banner matched to known vulnerability; active exploitation not demonstrated in this record

Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-APACHE-EOL-228 | Advisory: Apache HTTP Server 2.2.x End-of-Life Advisory | CWE: CWE-1104 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A06:2021 (Vulnerable and Outdated Components) | CWE-1104 (Use of Unmaintained Third Party Components) | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:N/A:N
Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 1 unique endpoint(s): 192.168.195.139
Root Cause & Compliance:	Patch & Lifecycle Management Deficit (EOL Software) [DIRECTLY_ESTABLISHED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	Apache httpd 2.2.8 on Ubuntu is End-of-Life (EOL) and no longer receives upstream security patches (consolidated across banner, AegisProbe, and OWASP A06 checks).
Security Risk:	HIGH — Running unsupported web server software with multiple historical CVEs.
EvidenceCore V3 Record:	ID: PROV-F-15-6F029D15FF | SHA-256: 6f029d15ffdf348fe23f7636... | HMAC-SHA256: 6212d29c0601afa81eb3ee4e... | Module: AutoPentest-OWASP, AutoPentest-PrivEsc

Captured Technical Evidence (F-15):

```
A06 Vulnerable Components: 6 finding(s)
[CWE-1104] Outdated: Apache 2.0/2.2 (EOL): Server: Apache/2.2.8 (Ubuntu) DAV/2 | X-Powered-By:
PHP/5.2.4-2ubuntu5.10
[CWE-1104] Outdated: PHP 5.x (EOL): Server: Apache/2.2.8 (Ubuntu) DAV/2 | X-Powered-By:
PHP/5.2.4-2ubuntu5.10
[CWE-1104] Vulnerable App: Mutillidae: OWASP Mutillidae — vulnerable by design
[CWE-1104] Vulnerable App: TWiki: Legacy wiki with known CVEs
[CWE-1104] Vulnerable App: TikiWiki: Legacy CMS with known vulnerabilities
[CWE-
---
Apache 2.2.8 may have known privilege escalation or RCE vulnerabilities.
```

Immediate Containment:	Set `ServerTokens Prod` and disable unneeded Apache 2.2 modules (`a2dismod dav dav_fs cgi autoindex`) to reduce attack surface immediately.
Short-Term Hardening:	Upgrade EOL Apache httpd 2.2.8 to Apache 2.4.x on a supported Linux distribution.
Long-Term Strategic Fix:	Automate web server patching and lifecycle tracking.
Verification Command:	Run `apache2 -v` on the host and verify a supported Apache 2.4.x release is installed.

F-16: PHP-CGI Argument Injection Candidate (CVE-2012-1823) [HIGH · CVSS 7.4 · OBSERVED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: SERVICE_BANNER_ONLY | Impact: EXPOSURE_ONLY | EvidenceConf: 42%
Access Scope Summary:	Service version or banner matched to known vulnerability; active exploitation not demonstrated in this record
Taxonomy & CVSS v3.1:	CVE: CVE-2012-1823 | Custom ID: XSTRIKE-PHP-CGI-1823 | Advisory: PHP Security Advisory CVE-2012-1823 / CERT VU#520827 | CWE: CWE-88 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A03:2021 (Injection) | CWE-88 (Improper Neutralization of Argument Delimiters) | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:H/I:H/A:N

Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 2 unique endpoint(s): 192.168.195.139, 192.168.195.139:80
Root Cause & Compliance:	Insecure Application Coding (Input Validation) [PROBABLE] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	The web server runs a legacy PHP 5.2.4 CGI configuration matching CVE-2012-1823 query-string argument injection indicators. Active remote code execution was not executed.
Security Risk:	HIGH — If PHP-CGI processes raw query-string flags (-d allow_url_include=on), an attacker can execute arbitrary PHP code.
EvidenceCore V3 Record:	ID: PROV-F-16-0020F0BBE8 | SHA-256: 0020f0bbe8a588be19d73ead... | HMAC-SHA256: d78a80abf2d75400bd924c72... | Module: AutoPentest-ZAP, XStrike-BuiltinVuln

Captured Technical Evidence (F-16):

```
Some PHP versions, when configured to run using CGI, do not correctly handle query strings that lack an
unescaped "=" character, enabling arbitrary code execution. In this case, an operating system co
---
PHP before 5.4.2 allows RCE when configured as CGI via query string

Evidence: Banner match on 192.168.195.139:80: HTTP/1.1 200 OK
Date: Sat, 31 Oct 2026 00:01:47 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Length: 891
Connection: close
Content-Type: text/html

<html
```

Immediate Containment:	Add an Apache `mod_rewrite` rule blocking query strings starting with `-`: `RewriteCond %{QUERY_STRING} ^(%2d|-) [NC]`, `RewriteRule ^.*$ - [F,L]`.
Short-Term Hardening:	Migrate PHP execution from `mod_cgi`/`php-cgi` to `php-fpm` (FastCGI Process Manager) and upgrade PHP 5.2.4 to a supported PHP 8.2/8.3 release.
Long-Term Strategic Fix:	Enforce `allow_url_include = Off` and `expose_php = Off` in `php.ini` across all web servers.
Verification Command:	Send `GET /?-s HTTP/1.1` to `http://192.168.195.139:80/` and confirm HTTP 403 Forbidden with no PHP source code disclosed.

F-17: FTP Anonymous Authentication Enabled [MEDIUM · CVSS 5.3 · CONFIRMED_AUTH]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	CONFIRMED_AUTH (Legacy Tier: CONFIRMED) | Privilege: ANONYMOUS (ftp) | Auth: True | File: READ_ONLY_EMPTY (0 files in anonymous root; no sensitive files exfiltrated) | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: AUTHENTICATED_SESSION_VERIFIED | Auth: ANONYMOUS_AUTH (230 OK) | Exploitability: CREDENTIAL_CONFIRMED | Privilege: ANONYMOUS (ftp) | DataAccess: EMPTY_ROOT_DIRECTORY | Impact: UNAUTHENTICATED_SESSION_ACCESS | EvidenceConf: 98%
Access Scope Summary:	Anonymous FTP login (230 OK) verified; root directory empty (0 files exfiltrated)

Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-FTP-ANONYMOUS | Advisory: RFC 1635 / vsFTPd anonymous_enable Configuration Advisory | CWE: CWE-287 | MITRE: T1078.001 (Defense Evasion / Persistence — Valid Accounts: Default Accounts) | Taxonomy: OWASP-A07:2021 (Identification and Authentication Failures) | CWE-287 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=99% | ExploitabilityConf=95% | ImpactConf=88% | 5-D Vector: EvidenceConf=99% (Det=99%, Val=98%, Art=100%) | ExploitabilityConf=95% | ImpactConf=88% | FP Band: Tier 1 — Cryptographically & Executionally Verified (<1% Artifact Uncertainty)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 2 unique endpoint(s): 192.168.195.139, 192.168.195.139:21
Root Cause & Compliance:	Identity & Credential Hygiene Deficiency [DIRECTLY_ESTABLISHED] | Compliance Status: CONFIRMED_COMPLIANCE_DEFICIENCY
Technical Description:	The FTP server accepts anonymous login ('anonymous') with FTP status 230 OK. Enumeration confirmed an empty anonymous root directory (0 files exposed).
Security Risk:	MEDIUM — Unauthenticated FTP session establishment; while no files were present in the root directory, anonymous access exposes internal service state.
EvidenceCore V3 Record:	ID: PROV-F-17-C106E068CF | SHA-256: c106e068cf9fabf933e1cfc8... | HMAC-SHA256: bc6fd04262a85bc08c21ff16... | Module: XStrike-BuiltinVuln, AutoPentest-Exploit

Captured Technical Evidence (F-17):

```
Anonymous login OK. Banner: 220 (vsFTPd 2.3.4). Files(0):
---
FTP server accepts anonymous login — unauthorized file access

Evidence: FTP anonymous login: 230 Login successful.
```

Immediate Containment:	Set `anonymous_enable=NO` in `/etc/vsftpd.conf` and restart the vsFTPd service (`service vsftpd restart`).
Short-Term Hardening:	Audit `/var/ftp` or the anonymous root directory to ensure no sensitive files or write permissions (`anon_upload_enable=NO`) exist.
Long-Term Strategic Fix:	Replace unauthenticated FTP distribution with authenticated HTTPS artifact repositories.
Verification Command:	Connect to port 21 as `USER anonymous` and verify the server responds with `530 Permission denied`.

F-18: Local Privilege Escalation Vector: SUID Binary (/bin/mount) [MEDIUM · CVSS 5.5 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: LOCAL_ESCALATION_VECTOR (uid=1000 -> uid=0 potential) | Auth: True | File: NONE | DB: NONE | Privesc Lifecycle: PRECONDITION_DISCOVERED
7-D Proof Vector (V3):	Execution: AUTHENTICATED_ENUMERATION_EXECUTED | Auth: AUTHENTICATED_USER (uid=1000) | Exploitability: VERSION_OR_ACL_VALIDATED | Privilege: LOCAL_ESCALATION_VECTOR (uid=1000 -> uid=0 potential) | DataAccess: KERNEL_AND_SUID_METADATA | Impact: LOCAL_PRIVESC_SURFACE | EvidenceConf: 93%
Access Scope Summary:	Authenticated SSH 'find / -perm -4000' discovered SUID-root precondition on /bin/mount; writable mount escalation not executed
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-PRIVESC-SUID-MOUNT | Advisory: Linux SUID Binary Hardening Baseline (/bin/mount) | CWE: CWE-269 | MITRE: T1548.001 (Privilege Escalation — Abuse Elevation Control Mechanism: Setuid and Setgid) | Taxonomy: MITRE ATT&CK T1548.001 | CWE-269 (Improper Privilege Management) | Vector: CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:U/C:H/I:N/A:N

Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 2 unique endpoint(s): 192.168.195.139:22, 192.168.195.139
Root Cause & Compliance:	Excessive Local Privilege / File Permission Misconfiguration [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Post-authentication SSH enumeration identified SUID root permissions on /bin/mount, which may be combined with NFS or fstab misconfigurations for local privilege escalation.
Security Risk:	MEDIUM (CVSS 6.7) — Potential local privilege escalation vector if combined with writable mount points.
EvidenceCore V3 Record:	ID: PROV-F-18-9DE312809B | SHA-256: 9de312809bcbff0b48717b36... | HMAC-SHA256: cc87f10dd272521fe086ccbc... | Module: AutoPentest-Exploit, AutoPentest-PrivEsc

Captured Technical Evidence (F-18):

```
SUID mount found — known privilege escalation vector!
```

Immediate Containment:	Audit `/etc/fstab` to ensure no user-mountable filesystems permit `suid` or `exec`, and remove unnecessary SUID bits from `/bin/mount` (`chmod u-s /bin/mount`) if unprivileged mounting is not required.
Short-Term Hardening:	Enforce `nosuid,nodev,noexec` on `/tmp`, `/var/tmp`, and `/home` mounts in `/etc/fstab`.
Long-Term Strategic Fix:	Implement file integrity monitoring (AIDE/auditd) tracking setuid/setgid permission changes across `/bin` and `/usr/bin`.
Verification Command:	Run `find / -perm -4000 -type f 2>/dev/null` and confirm only approved baseline SUID binaries remain.

F-19: Client-Manipulable Session/Tracking Cookie Parameters Set Without Integrity Binding [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: N/A) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-COOKIE-INTEGRITY | Advisory: OWASP Testing Guide — Testing for Cookie Attributes (WSTG-SESS-02) | CWE: CWE-565 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A04:2021 (Insecure Design) | CWE-565 (Reliance on Cookies without Validation and Integrity Checking) | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 3 raw observation(s) across 1 unique endpoint(s): 192.168.195.1
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE

Technical Description:	The web application issues multiple state/configuration cookies without cryptographic HMAC signing or strict attribute hardening.
Security Risk:	MEDIUM — If server logic trusts unverified cookie values, attackers may tamper with user state or privilege indicators.
EvidenceCore V3 Record:	ID: PROV-F-19-EEC2BE5ABA | SHA-256: eec2be5abae3726208296ee3... | HMAC-SHA256: 487571d0d8550b6e5ef2dbec... | Module: AutoPentest-ZAP

Captured Technical Evidence (F-19):

```
This check looks at user-supplied input in query string parameters and POST data to identify where
cookie parameters might be controlled. This is called a cookie poisoning attack, and becomes exploita
```

Immediate Containment:	Stop trusting client-side cookie parameters for user identity, privilege, or application state; store session state server-side in `$_SESSION`.
Short-Term Hardening:	If client-side cookies are used, sign values with HMAC-SHA256 and set `HttpOnly`, `Secure`, and `SameSite=Lax` attributes.
Long-Term Strategic Fix:	Standardize session management on encrypted, server-validated session stores.
Verification Command:	Modify cookie values in an HTTP request and verify the server rejects tampered cookies and does not alter privilege state.

F-20: Content-Security-Policy (CSP) Header Not Set [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: 8000, 8080) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-CSP-MISSING | Advisory: W3C CSP Level 3 / OWASP Content Security Policy Cheat Sheet | CWE: CWE-693 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-693 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 101 raw observation(s) across 4 unique endpoint(s): 192.168.195.1, 192.168.195.1:8080, 192.168.195.1:8000, http://192.168.195.1:8080/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Web application responses omit the 'Content-Security-Policy' header across multiple endpoints, reducing browser defense-in-depth against Cross-Site Scripting (XSS) and unauthorized script injection.
Security Risk:	MEDIUM — Lack of browser script-execution policy increases impact of any client-side injection flaw.
EvidenceCore V3 Record:	ID: PROV-F-20-43FFEFB79F | SHA-256: 43ffefb79f795094c3396375... | HMAC-SHA256: 09d74e428fac30f019b3e619... | Module: AutoPentest-ZAP, XStrike-BuiltinVuln, AutoPentest-AegisProbe

Captured Technical Evidence (F-20):

```
Content Security Policy (CSP) is an added layer of security that helps to detect and mitigate certain
types of attacks, including Cross Site Scripting (XSS) and data injection attacks. These attacks a
---
No CSP header — XSS and injection attacks not mitigated

Evidence: HTTP response from 192.168.195.1:8080 missing content-security-policy header
---
Missing header: Content-Security-Policy | CWE: CWE-1021 | Payload:
```

Immediate Containment:	Add `Header always set Content-Security-Policy "default-src 'self'; frame-ancestors 'self'; object-src 'none'; base-uri 'self'"` in Apache `httpd.conf` / `.htaccess`.
Short-Term Hardening:	Eliminate `'unsafe-inline'` and `'unsafe-eval'` script directives by migrating inline scripts to external hashed or nonce-protected JS files.
Long-Term Strategic Fix:	Enable CSP violation reporting (`report-to` / `report-uri`) to monitor policy regressions continuously.
Verification Command:	Run `curl -I http://<target>/` and confirm the `Content-Security-Policy` response header is present without wildcard (`*`) script sources.

F-21: Missing Anti-Clickjacking Protection (X-Frame-Options / frame-ancestors) [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: 8080) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-XFO-MISSING | Advisory: RFC 7034 / OWASP Clickjacking Defense Cheat Sheet | CWE: CWE-1021 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-1021 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 12 raw observation(s) across 3 unique endpoint(s): 192.168.195.1, 192.168.195.1:8080, http://192.168.195.1:8080/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP responses do not include 'X-Frame-Options' or CSP 'frame-ancestors' directives, permitting the web interface to be embedded inside an external iframe.
Security Risk:	MEDIUM — Exposes authenticated web users to UI redressing (Clickjacking) attacks.
EvidenceCore V3 Record:	ID: PROV-F-21-5550FEFEDA | SHA-256: 5550fefeda1931035d2b0e75... | HMAC-SHA256: 7b6b3fd8c6dbc512009deb17... | Module: AutoPentest-ZAP, XStrike-BuiltinVuln, AutoPentest-AegisProbe

Captured Technical Evidence (F-21):

```
The response does not protect against 'ClickJacking' attacks. It should include either
Content-Security-Policy with 'frame-ancestors' directive or X-Frame-Options.
---
No clickjacking protection — X-Frame-Options header missing
```

```
Evidence: HTTP response from 192.168.195.1:8080 missing x-frame-options header
---
Missing header: X-Frame-Options | CWE: CWE-1021 | Payload:
```

Immediate Containment:	Add `Header always set X-Frame-Options "DENY"` (or `"SAMEORIGIN"`) and CSP `frame-ancestors 'self'` in the web server configuration.
Short-Term Hardening:	Verify no legacy application workflows depend on cross-origin `<iframe src=...>` embedding.
Long-Term Strategic Fix:	Enforce standard security header middleware across all web application frameworks.
Verification Command:	Run `curl -I http://<target>/` and confirm `X-Frame-Options: DENY` or `SAMEORIGIN` is present.

F-22: Session Cookie Missing 'Secure' Flag [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: 8000) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-COOKIE-SECURE | Advisory: OWASP Session Management Cheat Sheet — Secure Flag | CWE: CWE-614 | MITRE: T1557 (Credential Access — Adversary-in-the-Middle) | Taxonomy: OWASP-A02:2021 (Cryptographic Failures) | CWE-614 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.1:8000
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Cookies are issued without the 'Secure' attribute, allowing the browser to transmit them over unencrypted HTTP connections.
Security Risk:	MEDIUM — Exposes session identifiers to cleartext network interception.
EvidenceCore V3 Record:	ID: PROV-F-22-0C8B90016E | SHA-256: 0c8b90016eaeb918da87c3a0... | HMAC-SHA256: 13be848db5daef809e8a9331... | Module: AutoPentest-AegisProbe

Captured Technical Evidence (F-22):

```
Cookie: cval=1915971765... | CWE: CWE-614 | Payload:
```

Immediate Containment:	Enable HTTPS and set `session.cookie_secure = 1` in `php.ini` so session cookies are only transmitted over TLS.
Short-Term Hardening:	Add Apache header rewrite `Header edit Set-Cookie ^(.*)$ "$1; Secure"` on all HTTPS virtual hosts.
Long-Term Strategic Fix:	Adopt `__Host-` cookie prefixes which require `Secure`, `Path=/`, and no `Domain` attribute.
Verification Command:	Inspect `Set-Cookie` headers over HTTPS and confirm the `Secure` attribute is present.

F-23: Verbose Application / Database Error Message Disclosure [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: N/A) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-ZAP-ERROR-DISCLOSURE | Advisory: OWASP ZAP Alert 90022 — Application Error Disclosure | CWE: CWE-209 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-209 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 15 raw observation(s) across 1 unique endpoint(s): 192.168.195.1
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	The web application returns raw stack traces, PHP warnings, or database error strings in HTTP responses.
Security Risk:	MEDIUM — Exposes internal file paths, SQL query structure, and framework internals to assist exploitation.
EvidenceCore V3 Record:	ID: PROV-F-23-F0F8ABB7FB | SHA-256: f0f8abb7fb49a9f36ef56ff6... | HMAC-SHA256: f5b85ed73f11a570b105c7e8... | Module: AutoPentest-ZAP

Captured Technical Evidence (F-23):

```
This page contains an error/warning message that may disclose sensitive information like the location of
the file that produced the unhandled exception. This information can be used to launch further
---
The response appeared to contain common error messages returned by platforms such as ASP.NET, and
Web-servers such as IIS and Apache. You can configure the list of common debug messages.
```

Immediate Containment:	Set `display_errors = Off` and `log_errors = On` in `/etc/php5/apache2/php.ini` and configure custom generic 500 error pages in Apache.
Short-Term Hardening:	Wrap database and file operations in structured exception handlers that log stack traces server-side only.
Long-Term Strategic Fix:	Enforce production error-handling standards across all web applications.
Verification Command:	Trigger an invalid parameter request and confirm a generic error message is returned with zero stack traces or SQL errors.

F-24: Web Application Missing Anti-CSRF Tokens & Hardening Headers [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: N/A) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC

7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-CSRF-BUNDLE | Advisory: OWASP Cross-Site Request Forgery Prevention Cheat Sheet | CWE: CWE-352 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A01:2021 (Broken Access Control) | CWE-352 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 5 raw observation(s) across 5 unique endpoint(s): 192.168.195.1:5357, 192.168.195.1:8000, http://192.168.195.1:8008/openapi.json, 192.168.195.1:8080 ... (+1 additional endpoints)
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Automated web assessment identified HTML forms lacking anti-CSRF tokens alongside missing HTTP response hardening headers.
Security Risk:	MEDIUM — State-changing web requests may be forged cross-origin if authenticated users visit a malicious page.
EvidenceCore V3 Record:	ID: PROV-F-24-1CFEB68FD5 | SHA-256: 1cfeb68fd5cc944f36164654... | HMAC-SHA256: 54e4cbdfd31b6386c40b3ebf... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-24):

```
Discovered 2 web vulnerabilities:
[CSRF-01] Missing Anti-CSRF Token and SameSite Defenses: No anti-CSRF token found in HTML forms and
session cookie lacks SameSite protect
[CLICK-01] Missing Clickjacking Defense (X-Frame-Options / frame-ancestors): Endpoint does not send
X-Frame-Options or CSP frame-ancestors header.
---
Discovered 1 web vulnerabilities:
[CSRF-01] Missing Anti-CSRF Token and SameSite Defenses: No anti-CSRF token found in HTML forms and
session cookie lacks SameSite protect
---
Discovered 3 web vulnerabilities:
[CSRF-01] Missing Anti-CSRF Token and SameSite Defenses: No anti-CSRF token found in HTML forms and
session c
```

Immediate Containment:	Implement cryptographically random per-session anti-CSRF tokens (`hash_hmac('sha256', session_id(), $secret)`) on all state-changing HTML `POST` forms.
Short-Term Hardening:	Set `SameSite=Lax` or `SameSite=Strict` on all session cookies and validate `Origin`/`Referer` headers on state-changing requests.
Long-Term Strategic Fix:	Adopt a modern web framework with built-in CSRF middleware and automated security header injection.
Verification Command:	Submit a cross-origin POST request without a valid CSRF token and confirm the server returns HTTP 403 Forbidden.

F-25: Content-Security-Policy (CSP) Header Not Set [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.129 (Ports: N/A) | Asset Class: Application / Network Host (STANDARD)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC

7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-CSP-MISSING | Advisory: W3C CSP Level 3 / OWASP Content Security Policy Cheat Sheet | CWE: CWE-693 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-693 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 3 raw observation(s) across 1 unique endpoint(s): 192.168.195.129
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Web application responses omit the 'Content-Security-Policy' header across multiple endpoints, reducing browser defense-in-depth against Cross-Site Scripting (XSS) and unauthorized script injection.
Security Risk:	MEDIUM — Lack of browser script-execution policy increases impact of any client-side injection flaw.
EvidenceCore V3 Record:	ID: PROV-F-25-B8C36865F4 | SHA-256: b8c36865f4f43392ea6cfb39... | HMAC-SHA256: 763c111cb7ba4d11e3cc0ecb... | Module: AutoPentest-ZAP

Captured Technical Evidence (F-25):

```
Content Security Policy (CSP) is an added layer of security that helps to detect and mitigate certain
types of attacks, including Cross Site Scripting (XSS) and data injection attacks. These attacks a
```

Immediate Containment:	Add `Header always set Content-Security-Policy "default-src 'self'; frame-ancestors 'self'; object-src 'none'; base-uri 'self'"` in Apache `httpd.conf` / `.htaccess`.
Short-Term Hardening:	Eliminate `'unsafe-inline'` and `'unsafe-eval'` script directives by migrating inline scripts to external hashed or nonce-protected JS files.
Long-Term Strategic Fix:	Enable CSP violation reporting (`report-to` / `report-uri`) to monitor policy regressions continuously.
Verification Command:	Run `curl -I http://<target>/` and confirm the `Content-Security-Policy` response header is present without wildcard (`*`) script sources.

F-26: Web Application Missing Anti-CSRF Tokens & Hardening Headers [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.129 (Ports: N/A) | Asset Class: Application / Network Host (STANDARD)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated

Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-CSRF-BUNDLE | Advisory: OWASP Cross-Site Request Forgery Prevention Cheat Sheet | CWE: CWE-352 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A01:2021 (Broken Access Control) | CWE-352 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.129:5357
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Automated web assessment identified HTML forms lacking anti-CSRF tokens alongside missing HTTP response hardening headers.
Security Risk:	MEDIUM — State-changing web requests may be forged cross-origin if authenticated users visit a malicious page.
EvidenceCore V3 Record:	ID: PROV-F-26-66B961F505 | SHA-256: 66b961f505e1a2f8e879653c... | HMAC-SHA256: d859f6bb9f2cf66c6d462f4a... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-26):

```
Discovered 2 web vulnerabilities:
[CSRF-01] Missing Anti-CSRF Token and SameSite Defenses: No anti-CSRF token found in HTML forms and session cookie lacks SameSite protect
[CLICK-01] Missing Clickjacking Defense (X-Frame-Options / frame-ancestors): Endpoint does not send X-Frame-Options or CSP frame-ancestors header.
```

Immediate Containment:	Implement cryptographically random per-session anti-CSRF tokens (`hash_hmac('sha256', session_id(), $secret)`) on all state-changing HTML `POST` forms.
Short-Term Hardening:	Set `SameSite=Lax` or `SameSite=Strict` on all session cookies and validate `Origin`/`Referer` headers on state-changing requests.
Long-Term Strategic Fix:	Adopt a modern web framework with built-in CSRF middleware and automated security header injection.
Verification Command:	Submit a cross-origin POST request without a valid CSRF token and confirm the server returns HTTP 403 Forbidden.

F-27: Content-Security-Policy (CSP) Header Not Set [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.130 (Ports: N/A) | Asset Class: Workstation / SMB Peer Host (STANDARD)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-CSP-MISSING | Advisory: W3C CSP Level 3 / OWASP Content Security Policy Cheat Sheet | CWE: CWE-693 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-693 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)

Deduplication & Endpoints:	Consolidated 3 raw observation(s) across 1 unique endpoint(s): 192.168.195.130
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Web application responses omit the 'Content-Security-Policy' header across multiple endpoints, reducing browser defense-in-depth against Cross-Site Scripting (XSS) and unauthorized script injection.
Security Risk:	MEDIUM — Lack of browser script-execution policy increases impact of any client-side injection flaw.
EvidenceCore V3 Record:	ID: PROV-F-27-38BAA33E55 | SHA-256: 38baa33e5505114f4f20be70... | HMAC-SHA256: 006310d12b08148be6f7b5bf... | Module: AutoPentest-ZAP

Captured Technical Evidence (F-27):

```
Content Security Policy (CSP) is an added layer of security that helps to detect and mitigate certain
types of attacks, including Cross Site Scripting (XSS) and data injection attacks. These attacks a
```

Immediate Containment:	Add `Header always set Content-Security-Policy "default-src 'self'; frame-ancestors 'self'; object-src 'none'; base-uri 'self'"` in Apache `httpd.conf` / `.htaccess`.
Short-Term Hardening:	Eliminate `'unsafe-inline'` and `'unsafe-eval'` script directives by migrating inline scripts to external hashed or nonce-protected JS files.
Long-Term Strategic Fix:	Enable CSP violation reporting (`report-to` / `report-uri`) to monitor policy regressions continuously.
Verification Command:	Run `curl -I http://<target>/` and confirm the `Content-Security-Policy` response header is present without wildcard (`*`) script sources.

F-28: Web Application Missing Anti-CSRF Tokens & Hardening Headers [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.130 (Ports: N/A) | Asset Class: Workstation / SMB Peer Host (STANDARD)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-CSRF-BUNDLE | Advisory: OWASP Cross-Site Request Forgery Prevention Cheat Sheet | CWE: CWE-352 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A01:2021 (Broken Access Control) | CWE-352 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.130:5985
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Automated web assessment identified HTML forms lacking anti-CSRF tokens alongside missing HTTP response hardening headers.
Security Risk:	MEDIUM — State-changing web requests may be forged cross-origin if authenticated users visit a malicious page.

EvidenceCore V3 Record:	ID: PROV-F-28-AADD60A8FB | SHA-256: aadd60a8fb8ca581e72b4d20... | HMAC-SHA256: 02a4dba8fd1e7a685f177e30... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-28):

```
Discovered 2 web vulnerabilities:
[CSRF-01] Missing Anti-CSRF Token and SameSite Defenses: No anti-CSRF token found in HTML forms and session cookie lacks SameSite protect
[CLICK-01] Missing Clickjacking Defense (X-Frame-Options / frame-ancestors): Endpoint does not send X-Frame-Options or CSP frame-ancestors header.
```

Immediate Containment:	Implement cryptographically random per-session anti-CSRF tokens (`hash_hmac('sha256', session_id(), $secret)`) on all state-changing HTML `POST` forms.
Short-Term Hardening:	Set `SameSite=Lax` or `SameSite=Strict` on all session cookies and validate `Origin`/`Referer` headers on state-changing requests.
Long-Term Strategic Fix:	Adopt a modern web framework with built-in CSRF middleware and automated security header injection.
Verification Command:	Submit a cross-origin POST request without a valid CSRF token and confirm the server returns HTTP 403 Forbidden.

F-29: Content-Security-Policy (CSP) Header Not Set [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: 80) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-CSP-MISSING | Advisory: W3C CSP Level 3 / OWASP Content Security Policy Cheat Sheet | CWE: CWE-693 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-693 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 3 raw observation(s) across 3 unique endpoint(s): 192.168.195.139, 192.168.195.139:80, http://192.168.195.139:80/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Web application responses omit the 'Content-Security-Policy' header across multiple endpoints, reducing browser defense-in-depth against Cross-Site Scripting (XSS) and unauthorized script injection.
Security Risk:	MEDIUM — Lack of browser script-execution policy increases impact of any client-side injection flaw.
EvidenceCore V3 Record:	ID: PROV-F-29-48C5125842 | SHA-256: 48c51258421767558f3f345b... | HMAC-SHA256: 3b8890c62b2693db45b6031f... | Module: AutoPentest-ZAP, XStrike-BuiltinVuln, AutoPentest-AegisProbe

Captured Technical Evidence (F-29):

```
Content Security Policy (CSP) is an added layer of security that helps to detect and mitigate certain types of attacks, including Cross Site Scripting (XSS) and data injection attacks. These attacks a
---
```

```
No CSP header — XSS and injection attacks not mitigated

Evidence: HTTP response from 192.168.195.139:80 missing content-security-policy header
---
Missing header: Content-Security-Policy | CWE: CWE-1021 | Payload:
```

Immediate Containment:	Add `Header always set Content-Security-Policy "default-src 'self'; frame-ancestors 'self'; object-src 'none'; base-uri 'self'"` in Apache `httpd.conf` / `.htaccess`.
Short-Term Hardening:	Eliminate `'unsafe-inline'` and `'unsafe-eval'` script directives by migrating inline scripts to external hashed or nonce-protected JS files.
Long-Term Strategic Fix:	Enable CSP violation reporting (`report-to` / `report-uri`) to monitor policy regressions continuously.
Verification Command:	Run `curl -I http://<target>/` and confirm the `Content-Security-Policy` response header is present without wildcard (`*`) script sources.

F-30: DNS Open Recursion Enabled on ISC BIND 9.4.2 (Port 53) [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-DNS-OPEN-RECURSION | Advisory: US-CERT Alert TA13-088A / ISC BIND Recursion Access Control | CWE: CWE-406 | MITRE: T1498.002 (Impact — Network Denial of Service: Reflection Amplification) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-406 (Insufficient Control of Network Message Volume) | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139:53
Root Cause & Compliance:	Service Hardening & Configuration Baseline Gap [PROBABLE] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	ISC BIND 9.4.2 on port 53 permits recursive DNS queries from untrusted network clients.
Security Risk:	MEDIUM — Can be abused for DNS amplification and cache-poisoning attacks.
EvidenceCore V3 Record:	ID: PROV-F-30-0F0941B786 | SHA-256: 0f0941b786a12ce60c880edd... | HMAC-SHA256: 437be73ae3cd4c510f7abdaa... | Module: XStrike-BuiltinVuln

Captured Technical Evidence (F-30):

```
DNS server allows recursive queries — can be abused for amplification attacks

Evidence: DNS recursion available on 192.168.195.139:53 (RA flag set)
```

Immediate Containment:	Disable open recursion in BIND `/etc/bind/named.conf.options` (`recursion no;` or `allow-recursion { 127.0.0.1; };`) and reload `named`.
Short-Term Hardening:	Upgrade ISC BIND 9.4.2 to a supported BIND 9.18 LTS release and enable response rate limiting (RRL).

Long-Term Strategic Fix:	Separate authoritative DNS servers from recursive resolvers.
Verification Command:	Run `dig @192.168.195.139 example.com` from an untrusted host and confirm `status: REFUSED`.

F-31: Missing Anti-Clickjacking Protection (X-Frame-Options / frame-ancestors) [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: 80) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-XFO-MISSING | Advisory: RFC 7034 / OWASP Clickjacking Defense Cheat Sheet | CWE: CWE-1021 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-1021 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 3 raw observation(s) across 3 unique endpoint(s): 192.168.195.139, 192.168.195.139:80, http://192.168.195.139:80/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP responses do not include 'X-Frame-Options' or CSP 'frame-ancestors' directives, permitting the web interface to be embedded inside an external iframe.
Security Risk:	MEDIUM — Exposes authenticated web users to UI redressing (Clickjacking) attacks.
EvidenceCore V3 Record:	ID: PROV-F-31-A9EE8ABF71 | SHA-256: a9ee8abf7190284f476308a2... | HMAC-SHA256: 0ac04c3950fc70a215ee1e65... | Module: AutoPentest-ZAP, XStrike-BuiltinVuln, AutoPentest-AegisProbe

Captured Technical Evidence (F-31):

```
The response does not protect against 'ClickJacking' attacks. It should include either
Content-Security-Policy with 'frame-ancestors' directive or X-Frame-Options.
---
No clickjacking protection — X-Frame-Options header missing

Evidence: HTTP response from 192.168.195.139:80 missing x-frame-options header
---
Missing header: X-Frame-Options | CWE: CWE-1021 | Payload:
```

Immediate Containment:	Add `Header always set X-Frame-Options "DENY"` (or `"SAMEORIGIN"`) and CSP `frame-ancestors 'self'` in the web server configuration.
Short-Term Hardening:	Verify no legacy application workflows depend on cross-origin `<iframe src=...>` embedding.
Long-Term Strategic Fix:	Enforce standard security header middleware across all web application frameworks.
Verification Command:	Run `curl -I http://<target>/` and confirm `X-Frame-Options: DENY` or `SAMEORIGIN` is present.

F-32: Web Application Missing Anti-CSRF Tokens & Hardening Headers [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-CSRF-BUNDLE | Advisory: OWASP Cross-Site Request Forgery Prevention Cheat Sheet | CWE: CWE-352 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A01:2021 (Broken Access Control) | CWE-352 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): http://192.168.195.139:80/phpinfo.php
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Automated web assessment identified HTML forms lacking anti-CSRF tokens alongside missing HTTP response hardening headers.
Security Risk:	MEDIUM — State-changing web requests may be forged cross-origin if authenticated users visit a malicious page.
EvidenceCore V3 Record:	ID: PROV-F-32-1CCB211FF2 | SHA-256: 1ccb211ff2aa801b4403b2a9... | HMAC-SHA256: de7d9fa50519bdc304ef753f... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-32):

```
Discovered 3 web vulnerabilities:
[CSRF-01] Missing Anti-CSRF Token and SameSite Defenses: No anti-CSRF token found in HTML forms and session cookie lacks SameSite protect
[CLICK-01] Missing Clickjacking Defense (X-Frame-Options / frame-ancestors): Endpoint does not send X-Frame-Options or CSP frame-ancestors header.
[DEBUG-01] Sensitive Information & Debug Disclosures (1): PHP Info Configuration Exposed at http://192.168.195.139:80/phpinfo.php
```

Immediate Containment:	Implement cryptographically random per-session anti-CSRF tokens (`hash_hmac('sha256', session_id(), $secret)`) on all state-changing HTML `POST` forms.
Short-Term Hardening:	Set `SameSite=Lax` or `SameSite=Strict` on all session cookies and validate `Origin`/`Referer` headers on state-changing requests.
Long-Term Strategic Fix:	Adopt a modern web framework with built-in CSRF middleware and automated security header injection.
Verification Command:	Submit a cross-origin POST request without a valid CSRF token and confirm the server returns HTTP 403 Forbidden.

F-33: Content-Security-Policy (CSP) Header Not Set [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.140 (Ports: 8080) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)

Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-CSP-MISSING | Advisory: W3C CSP Level 3 / OWASP Content Security Policy Cheat Sheet | CWE: CWE-693 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-693 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 12 raw observation(s) across 3 unique endpoint(s): 192.168.195.140, 192.168.195.140:8080, http://192.168.195.140:8080/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Web application responses omit the 'Content-Security-Policy' header across multiple endpoints, reducing browser defense-in-depth against Cross-Site Scripting (XSS) and unauthorized script injection.
Security Risk:	MEDIUM — Lack of browser script-execution policy increases impact of any client-side injection flaw.
EvidenceCore V3 Record:	ID: PROV-F-33-D319E1658B | SHA-256: d319e1658b6f236504a3f6c6... | HMAC-SHA256: 88e6ec41822780536f815781... | Module: AutoPentest-ZAP, XStrike-BuiltinVuln, AutoPentest-AegisProbe

Captured Technical Evidence (F-33):

```
Content Security Policy (CSP) is an added layer of security that helps to detect and mitigate certain
types of attacks, including Cross Site Scripting (XSS) and data injection attacks. These attacks a
---
No CSP header – XSS and injection attacks not mitigated

Evidence: HTTP response from 192.168.195.140:8080 missing content-security-policy header
---
Missing header: Content-Security-Policy | CWE: CWE-1021 | Payload:
```

Immediate Containment:	Add `Header always set Content-Security-Policy "default-src 'self'; frame-ancestors 'self'; object-src 'none'; base-uri 'self'"` in Apache `httpd.conf` / `.htaccess`.
Short-Term Hardening:	Eliminate `'unsafe-inline'` and `'unsafe-eval'` script directives by migrating inline scripts to external hashed or nonce-protected JS files.
Long-Term Strategic Fix:	Enable CSP violation reporting (`report-to` / `report-uri`) to monitor policy regressions continuously.
Verification Command:	Run `curl -I http://<target>/` and confirm the `Content-Security-Policy` response header is present without wildcard (`*`) script sources.

F-34: Missing Anti-Clickjacking Protection (X-Frame-Options / frame-ancestors) [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.140 (Ports: 8080) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC

7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-XFO-MISSING | Advisory: RFC 7034 / OWASP Clickjacking Defense Cheat Sheet | CWE: CWE-1021 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-1021 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 5 raw observation(s) across 3 unique endpoint(s): 192.168.195.140, 192.168.195.140:8080, http://192.168.195.140:8080/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP responses do not include 'X-Frame-Options' or CSP 'frame-ancestors' directives, permitting the web interface to be embedded inside an external iframe.
Security Risk:	MEDIUM — Exposes authenticated web users to UI redressing (Clickjacking) attacks.
EvidenceCore V3 Record:	ID: PROV-F-34-0232CE56E6 | SHA-256: 0232ce56e6fba07728753e1e... | HMAC-SHA256: 43f993e6df5084ba4d56c455... | Module: AutoPentest-ZAP, XStrike-BuiltinVuln, AutoPentest-AegisProbe

Captured Technical Evidence (F-34):

```
The response does not protect against 'ClickJacking' attacks. It should include either
Content-Security-Policy with 'frame-ancestors' directive or X-Frame-Options.
---
No clickjacking protection — X-Frame-Options header missing

Evidence: HTTP response from 192.168.195.140:8080 missing x-frame-options header
---
Missing header: X-Frame-Options | CWE: CWE-1021 | Payload:
```

Immediate Containment:	Add `Header always set X-Frame-Options "DENY"` (or `"SAMEORIGIN"`) and CSP `frame-ancestors 'self'` in the web server configuration.
Short-Term Hardening:	Verify no legacy application workflows depend on cross-origin `<iframe src=...>` embedding.
Long-Term Strategic Fix:	Enforce standard security header middleware across all web application frameworks.
Verification Command:	Run `curl -I http://<target>/` and confirm `X-Frame-Options: DENY` or `SAMEORIGIN` is present.

F-35: Web Application Missing Anti-CSRF Tokens & Hardening Headers [MEDIUM · CVSS 5.4 · VALIDATED]

Target Asset & Port:	192.168.195.140 (Ports: N/A) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated

Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-CSRF-BUNDLE | Advisory: OWASP Cross-Site Request Forgery Prevention Cheat Sheet | CWE: CWE-352 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A01:2021 (Broken Access Control) | CWE-352 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 2 unique endpoint(s): 192.168.195.140:8080, 192.168.195.140:8081
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Automated web assessment identified HTML forms lacking anti-CSRF tokens alongside missing HTTP response hardening headers.
Security Risk:	MEDIUM — State-changing web requests may be forged cross-origin if authenticated users visit a malicious page.
EvidenceCore V3 Record:	ID: PROV-F-35-9B7C3F5BC1 | SHA-256: 9b7c3f5bc1e763a057a77637... | HMAC-SHA256: 187d7585fbc365c4dec79db5... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-35):

```
Discovered 2 web vulnerabilities:
[CSRF-01] Missing Anti-CSRF Token and SameSite Defenses: No anti-CSRF token found in HTML forms and session cookie lacks SameSite protect
[CLICK-01] Missing Clickjacking Defense (X-Frame-Options / frame-ancestors): Endpoint does not send X-Frame-Options or CSP frame-ancestors header.
```

Immediate Containment:	Implement cryptographically random per-session anti-CSRF tokens (`hash_hmac('sha256', session_id(), $secret)`) on all state-changing HTML `POST` forms.
Short-Term Hardening:	Set `SameSite=Lax` or `SameSite=Strict` on all session cookies and validate `Origin`/`Referer` headers on state-changing requests.
Long-Term Strategic Fix:	Adopt a modern web framework with built-in CSRF middleware and automated security header injection.
Verification Command:	Submit a cross-origin POST request without a valid CSRF token and confirm the server returns HTTP 403 Forbidden.

F-36: Administrative Web Interface Reachable Without Network Restriction [MEDIUM · CVSS 5.3 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: N/A) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-ADMIN-EXPOSED | Advisory: OWASP A01:2021 — Administrative Interface Access Control | CWE: CWE-284 | MITRE: T1083 (Discovery — File and Directory Discovery) | Taxonomy: OWASP-A01:2021 (Broken Access Control) | CWE-284 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N

Separated Confidence (V3):	EvidenceConf=90% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=90% (Det=96%, Val=92%, Art=75%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): http://192.168.195.1:80/Config
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Administrative web paths (/admin/, /phpMyAdmin/) are reachable over the network without IP allowlisting or VPN segmentation.
Security Risk:	MEDIUM — Expands attack surface for credential stuffing and administrative application exploits.
EvidenceCore V3 Record:	ID: PROV-F-36-9A40608317 | SHA-256: 9a40608317729982eba3e9dd... | HMAC-SHA256: 6f0f32c06abafab064c991c2... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-36):

```
EXPOSED SENSITIVE DATA / ADMIN PANELS:
/admin: Admin panel accessible
/admin/: Admin panel accessible
```

Immediate Containment:	Restrict `/admin/`, `/phpMyAdmin/`, and `/manager/` URI prefixes to administrative management IPs via Apache `<Location>` access controls.
Short-Term Hardening:	Enforce multi-factor authentication and TLS client certificates on all administrative web consoles.
Long-Term Strategic Fix:	Bind administrative interfaces to an isolated out-of-band management network or VPN.
Verification Command:	Request administrative paths from the untrusted assessment subnet and verify HTTP 403 Forbidden.

F-37: Administrative Web Interface Reachable Without Network Restriction [MEDIUM · CVSS 5.3 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-ADMIN-EXPOSED | Advisory: OWASP A01:2021 — Administrative Interface Access Control | CWE: CWE-284 | MITRE: T1083 (Discovery — File and Directory Discovery) | Taxonomy: OWASP-A01:2021 (Broken Access Control) | CWE-284 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): http://192.168.195.139:80/Config
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Administrative web paths (/admin/, /phpMyAdmin/) are reachable over the network without IP allowlisting or VPN segmentation.

Security Risk:	MEDIUM — Expands attack surface for credential stuffing and administrative application exploits.
EvidenceCore V3 Record:	ID: PROV-F-37-4E4B404A84 | SHA-256: 4e4b404a84027951b634f337... | HMAC-SHA256: 1dbe4b58487fd1fed87c179a... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-37):

```
EXPOSED SENSITIVE DATA / ADMIN PANELS:
/: Admin panel accessible
/phpMyAdmin/: Admin panel accessible
/dvwa/login.php: Admin panel accessible
/mutillidae/: Admin panel accessible
/mutillidae/index.php: Admin panel accessible
/server-status: Apache server-status exposed
```

Immediate Containment:	Restrict `/admin/`, `/phpMyAdmin/`, and `/manager/` URI prefixes to administrative management IPs via Apache `<Location>` access controls.
Short-Term Hardening:	Enforce multi-factor authentication and TLS client certificates on all administrative web consoles.
Long-Term Strategic Fix:	Bind administrative interfaces to an isolated out-of-band management network or VPN.
Verification Command:	Request administrative paths from the untrusted assessment subnet and verify HTTP 403 Forbidden.

F-38: Exposed phpMyAdmin Web Administrative Interface (/phpMyAdmin/) [MEDIUM · CVSS 5.3 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE (Unauthenticated HTTP 200) | Auth: False | File: NONE | DB: UI_REACHABLE_ONLY (HTTP 200 administrative interface; unauthenticated DB read/write NOT demonstrated) | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: EXPOSED_ADMIN_SURFACE | Privilege: NONE (Unauthenticated HTTP 200) | DataAccess: HTML_LOGIN_OR_INDEX_ONLY | Impact: ADMINISTRATIVE_SURFACE_EXPOSURE | EvidenceConf: 94%
Access Scope Summary:	HTTP 200 administrative interface reachable at /phpMyAdmin/; unauthenticated DB read/write not independently demonstrated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-PHPMYADMIN-EXPOSED | Advisory: phpMyAdmin Documentation — Securing Your phpMyAdmin Installation | CWE: CWE-306 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-306 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): http://192.168.195.139:80/phpMyAdmin/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	The phpMyAdmin web interface at http://192.168.195.139:80/phpMyAdmin/ returned HTTP 200. While the administrative interface is reachable without network-layer restriction, unauthenticated database read/write execution was not demonstrated in the captured evidence.
Security Risk:	MEDIUM (VALIDATED EXPOSURE, CVSS 5.3) — Publicly reachable database administration console; exposes backend MySQL authentication surface.
EvidenceCore V3 Record:	ID: PROV-F-38-28B056E9DC | SHA-256: 28b056e9dc2bc87a1a604755... | HMAC-SHA256: ab88b0bbff66ee78f7259185... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-38):

```
[HTTP Probe 200 OK on /phpMyAdmin/ — Exposed administrative web interface; unauthenticated SQL
read/write NOT demonstrated]
Data: <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en" dir="lt
```

Immediate Containment:	Restrict `/phpMyAdmin/` in Apache `/etc/apache2/conf.d/phpmyadmin.conf` to localhost or administrative VPN IPs (`Require ip 127.0.0.1`) and reload Apache.
Short-Term Hardening:	Enforce cookie-based authentication (`$cfg['Servers'][$i]['auth_type'] = 'cookie'`), set `$cfg['AllowNoPassword'] = false`, and place HTTP Basic Auth in front of `/phpMyAdmin/`.
Long-Term Strategic Fix:	Remove internet/LAN-facing phpMyAdmin installations and perform database administration exclusively over SSH port-forwarding.
Verification Command:	Request `http://192.168.195.139:80/phpMyAdmin/` from the assessment subnet and confirm HTTP 403 Forbidden.

F-39: NFS RPC Service & Export Table Exposed (Ports 111 / 2049) [MEDIUM · CVSS 5.3 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-NFS-EXPORTS-2049 | Advisory: CERT/CC CA-1991-21 / Linux nfs-kernel-server /etc/exports Hardening | CWE: CWE-732 | MITRE: T1083 (Discovery — File and Directory Discovery) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-732 (Incorrect Permission Assignment) | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 3 raw observation(s) across 3 unique endpoint(s): 192.168.195.139:111, 192.168.195.139:2049, 192.168.195.139
Root Cause & Compliance:	Access Control & Network Share Misconfiguration [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Network File System (NFS) and portmapper services are reachable on TCP ports 111 and 2049 with an exported share entry. Remote filesystem mounting was not executed during the assessment.
Security Risk:	MEDIUM (VALIDATED, CVSS 6.5) — Exposed NFS daemon; if world-readable or no_root_squash exports exist, remote attackers can read or modify host files.
EvidenceCore V3 Record:	ID: PROV-F-39-D5CCEDCEC8 | SHA-256: d5ccedcec8168c1f1fdb2ce1... | HMAC-SHA256: 55290adc24e0907bf2c518a7... | Module: AutoPentest-Exploit, AutoPentest-PrivEsc

Captured Technical Evidence (F-39):

```
NFS Exports: Nmap scan report for 192.168.195.139
Host is up (0.00013s latency).

PORT STATE SERVICE
111/tcp open rpcbind
| nfs-statfs:
| Filesystem 1K-blocks Used Available Use% Maxfilesize Maxlink
```

```
|_ / 7282168.0 254
---
NFS Exports: Nmap scan report for 192.168.195.139
Host is up (0.00088s latency).

PORT STATE SERVICE
2049/tcp open nfs
MAC Address: 00:0C:29:05:D7:6A (VMware)
---
NFS service detected. Attackers can enumerate and mount shared directories, potentially accessing
sensitive files or writing SSH keys for privilege escalation.
```

Immediate Containment:	Stop the NFS kernel server (`exportfs -ua && service nfs-kernel-server stop`) and remove any wildcard `/ *(rw,no_root_squash)` entry from `/etc/exports`.
Short-Term Hardening:	If NFS is required, export only dedicated non-system directories restricted to specific client IPs with `root_squash,ro,nosuid,nodev` flags.
Long-Term Strategic Fix:	Migrate network file sharing to NFSv4 with Kerberos (`sec=krb5p`) mutual authentication and encryption.
Verification Command:	Run `showmount -e 192.168.195.139` and verify `/` is no longer exported to `*`.

F-40: PHP Configuration Diagnostic Page Exposed (/phpinfo.php) [MEDIUM · CVSS 5.3 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: 80) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-PHPINFO-EXPOSED | Advisory: OWASP A01/A05 — Forced Browsing & Diagnostic Page Exposure | CWE: CWE-200 | MITRE: T1592 (Reconnaissance — Gather Victim Host Information) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 4 raw observation(s) across 4 unique endpoint(s): http://192.168.195.139:80/phpinfo.php:, http://192.168.195.139:80/icons/:, http://192.168.195.139:80/, http://192.168.195.139:80/phpinfo.php
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	The /phpinfo.php diagnostic script is publicly accessible (HTTP 200), exposing PHP runtime settings, environment variables, loaded extensions, and server filesystem paths (consolidated with OWASP A01 forced browsing telemetry).
Security Risk:	MEDIUM (CVSS 5.3) — Environment and configuration disclosure facilitating targeted web exploitation.
EvidenceCore V3 Record:	ID: PROV-F-40-2F41BCD36E | SHA-256: 2f41bcd36ea185f6c5d92f40... | HMAC-SHA256: 53e94232167d589ad300f711... | Module: AutoPentest-OWASP, AutoPentest-AegisProbe, AutoPentest-PrivEsc

Captured Technical Evidence (F-40):

```
A01 Broken Access Control: 1 finding(s)
[CWE-425] Forced Browsing: /phpinfo.php: Accessible! (200) Content: <!DOCTYPE html PUBLIC "-//W3C//DTD
XHTML 1.0 Transitional//EN" "DTD/xhtml
---
A05 Misconfiguration: 4 finding(s)
[CWE-548] Directory Listing: /icons/: Full directory contents exposed
[CWE-16] HTTP PUT Enabled: Can upload files via PUT method!
[CWE-693] HTTP TRACE Enabled: XST (Cross-Site Tracing) possible
[CWE-200] phpinfo() at /phpinfo.php: Full PHP configuration exposed
---
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "DTD/xhtml1-transitional.dtd">
<html><head>
<style type="text/css">
body {background-color
```

Immediate Containment:	Remove `/var/www/phpinfo.php` immediately (`rm -f /var/www/phpinfo.php`) or block `/phpinfo.php` in Apache (`<Location /phpinfo.php> Require all denied </Location>`).
Short-Term Hardening:	Set `disable_functions = phpinfo,exec,passthru,shell_exec,system,proc_open,popen` in `/etc/php5/apache2/php.ini` and reload Apache.
Long-Term Strategic Fix:	Add pre-deployment CI checks that block diagnostic files (`phpinfo.php`, `test.php`, `.env`) from production web roots.
Verification Command:	Request `http://192.168.195.139:80/phpinfo.php` and confirm HTTP 404 Not Found or 403 Forbidden.

F-41: Web Authentication Endpoint Missing Brute-Force Rate Limiting (/dvwa/login.php) [MEDIUM · CVSS 5.3 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-NO-RATE-LIMIT | Advisory: OWASP A04:2021 / A07:2021 — Missing Authentication Rate Limiting | CWE: CWE-307 | MITRE: T1110.001 (Credential Access — Brute Force: Password Guessing) | Taxonomy: OWASP-A04:2021 (Insecure Design) | CWE-307 (Improper Restriction of Excessive Authentication Attempts) | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): http://192.168.195.139:80/dvwa/login.php
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	The web login endpoint (/dvwa/login.php) processed rapid consecutive authentication requests without account lockout, throttling, or CAPTCHA challenge (consolidated from OWASP A04 testing).
Security Risk:	MEDIUM — Permits automated high-speed credential stuffing and dictionary brute-force attacks.

EvidenceCore V3 Record:	ID: PROV-F-41-DAA99D7CE5 | SHA-256: daa99d7ce5efbd0d0882d112... | HMAC-SHA256: 5cf6f559da6755725eac95b4... | Module: AutoPentest-OWASP

Captured Technical Evidence (F-41):

```
A04 Insecure Design: 2 finding(s)
[CWE-209] Verbose Error at /index.php?id='
[CWE-307] No Rate Limiting: /dvwa/login.php
```

Immediate Containment:	Enable request rate limiting on `/dvwa/login.php` and authentication routes using Apache `mod_evasive` / `mod_ratelimit` or Fail2Ban.
Short-Term Hardening:	Enforce progressive account lockout (e.g., 5 failed attempts -> 15-minute cooldown) and exponential backoff on authentication endpoints.
Long-Term Strategic Fix:	Integrate centralized identity provider (IdP) authentication with MFA and anomaly-based brute-force protection.
Verification Command:	Send 15 rapid failed login POST requests to `/dvwa/login.php` and confirm HTTP 429 Too Many Requests or account lockout triggers.

F-42: Administrative Web Interface Reachable Without Network Restriction [MEDIUM · CVSS 5.3 · VALIDATED]

Target Asset & Port:	192.168.195.140 (Ports: N/A) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-ADMIN-EXPOSED | Advisory: OWASP A01:2021 — Administrative Interface Access Control | CWE: CWE-284 | MITRE: T1083 (Discovery — File and Directory Discovery) | Taxonomy: OWASP-A01:2021 (Broken Access Control) | CWE-284 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=90% | ExploitabilityConf=25% | ImpactConf=52% | 5-D Vector: EvidenceConf=90% (Det=96%, Val=92%, Art=75%) | ExploitabilityConf=25% | ImpactConf=52% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): http://192.168.195.140:80/Config
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Administrative web paths (/admin/, /phpMyAdmin/) are reachable over the network without IP allowlisting or VPN segmentation.
Security Risk:	MEDIUM — Expands attack surface for credential stuffing and administrative application exploits.
EvidenceCore V3 Record:	ID: PROV-F-42-F020C753AD | SHA-256: f020c753adbabbed2fd87edd... | HMAC-SHA256: 301ef5afc2c0db8bfc946f2a... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-42):

```
EXPOSED SENSITIVE DATA / ADMIN PANELS:
/admin/: Admin panel accessible
```

Immediate Containment:	Restrict `/admin/`, `/phpMyAdmin/`, and `/manager/` URI prefixes to administrative management IPs via Apache `<Location>` access controls.
Short-Term Hardening:	Enforce multi-factor authentication and TLS client certificates on all administrative web consoles.

Long-Term Strategic Fix:	Bind administrative interfaces to an isolated out-of-band management network or VPN.
Verification Command:	Request administrative paths from the untrusted assessment subnet and verify HTTP 403 Forbidden.

F-43: PostgreSQL Database Listener Exposed on Network (TCP Port 5432) [MEDIUM · CVSS 5.3 · OBSERVED]

Target Asset & Port:	192.168.195.1 (Ports: N/A) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: SERVICE_BANNER_ONLY | Impact: EXPOSURE_ONLY | EvidenceConf: 42%
Access Scope Summary:	Service version or banner matched to known vulnerability; active exploitation not demonstrated in this record
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-PGSQL-PORT-EXPOSED | Advisory: PostgreSQL listen_addresses Network Hardening Guideline | CWE: CWE-200 | MITRE: T1046 (Discovery — Network Service Discovery) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.1:5432
Root Cause & Compliance:	Network Segmentation & Perimeter Filter Exposure [INFERRED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	TCP port 5432 (PostgreSQL) is reachable over the network. Unauthenticated or default-credential login was not demonstrated on this host; classified as OBSERVED service exposure.
Security Risk:	MEDIUM (OBSERVED) — Network-accessible database port increases brute-force and protocol attack surface.
EvidenceCore V3 Record:	ID: PROV-F-43-F74FC1B675 | SHA-256: f74fc1b675500fcc746e96f8... | HMAC-SHA256: 36786936a5a675e2be15e230... | Module: AutoPentest-VulnScan

Captured Technical Evidence (F-43):

```
[2026-10-30 19:37:01] PostgreSQL database service on 192.168.195.1:5432.
Service: postgresql, Product: PostgreSQL DB, Version: 9.6.0 or later
Database exposed on network. Default credential testing pending.
```

Immediate Containment:	Bind PostgreSQL to `127.0.0.1` (`listen_addresses = 'localhost'` in `postgresql.conf`) and restrict TCP port 5432 via host firewall rules to authorized application servers only.
Short-Term Hardening:	Enforce `scram-sha-256` authentication and TLS client connections (`hostssl`) in `pg_hba.conf`.
Long-Term Strategic Fix:	Place all database listeners in an isolated database VLAN with zero direct routing from user/workstation segments.
Verification Command:	Run `nmap -p 5432 <host>` from the assessment subnet and confirm TCP port 5432 is `closed` or `filtered`.

F-44: End-of-Life OpenSSH 4.7p1 Service Banner Observed (TCP Port 22) [MEDIUM · CVSS 5.3 · OBSERVED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)

Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: SERVICE_BANNER_ONLY | Impact: EXPOSURE_ONLY | EvidenceConf: 42%
Access Scope Summary:	Service version or banner matched to known vulnerability; active exploitation not demonstrated in this record
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-OPENSSH-EOL-47P1 | Advisory: OpenSSH Legacy Lifecycle & Backported Patch Verification Advisory | CWE: CWE-1104 | MITRE: T1021.004 (Lateral Movement — Remote Services: SSH) | Taxonomy: OWASP-A06:2021 (Vulnerable and Outdated Components) | CWE-1104 (Use of Unmaintained Third Party Components) | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139:22
Root Cause & Compliance:	Patch & Lifecycle Management Deficit (EOL Software) [DIRECTLY_ESTABLISHED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	The SSH daemon runs an end-of-life OpenSSH 4.7p1 Debian-8ubuntu1 release observed via service banner fingerprinting.
Security Risk:	MEDIUM (OBSERVED) — Legacy SSH daemon version; verify distribution backported patches or upgrade OS.
EvidenceCore V3 Record:	ID: PROV-F-44-B35CFF918E | SHA-256: b35cff918e6583a53cf397c6... | HMAC-SHA256: 04aa21dcf15f939985b601f9... | Module: XStrike-BuiltinVuln

Captured Technical Evidence (F-44):

```
Legacy OpenSSH versions vulnerable to information leak and authentication bypass

Evidence: Banner match on 192.168.195.139:22: SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
```

Immediate Containment:	Restrict SSH access on port 22 to authorized management IPs via host firewall (`ufw allow from <mgmt_subnet> to any port 22`).
Short-Term Hardening:	Upgrade OpenSSH 4.7p1 to a supported release (>= 9.x) and disable legacy CBC ciphers and SHA-1 MACs in `sshd_config`.
Long-Term Strategic Fix:	Migrate the underlying Ubuntu 8.04 host to a supported LTS OS release.
Verification Command:	Run `nmap -sV -p 22 --script ssh2-enum-algos` and confirm modern OpenSSH version and cipher suite.

F-45: Java RMI Registry Exposed on Network (TCP Port 1099) [MEDIUM · CVSS 5.3 · OBSERVED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: SERVICE_BANNER_ONLY | Impact: EXPOSURE_ONLY | EvidenceConf: 42%
Access Scope Summary:	Service version or banner matched to known vulnerability; active exploitation not demonstrated in this record

Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-JAVA-RMI-1099 | Advisory: Oracle Java RMI / GNU Classpath grmiregistry Deserialization Exposure | CWE: CWE-502 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A08:2021 (Software and Data Integrity Failures) | CWE-502 (Deserialization of Untrusted Data) | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139:1099
Root Cause & Compliance:	Network Segmentation & Perimeter Filter Exposure [INFERRED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	GNU Classpath grmiregistry listener is exposed on TCP port 1099; remote class loading or deserialization payload execution was not performed.
Security Risk:	MEDIUM (OBSERVED) — Exposed Java RMI registry surface susceptible to insecure deserialization if default class loaders are active.
EvidenceCore V3 Record:	ID: PROV-F-45-3F49DCE2B3 | SHA-256: 3f49dce2b33f718d0bd05167... | HMAC-SHA256: 1a4f29dc6a879782c774ebde... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-45):

```
Java RMI exploitable: Nmap scan report for 192.168.195.139
Host is up (0.00s latency).

PORT STATE SERVICE
1099/tcp open rmiregistry
| rmi-vuln-classloader:
| VULNERABLE:
| RMI registry default configuration remote code execution vulnerability
|
```

Immediate Containment:	Stop the `grmiregistry` / Java RMI daemon listening on TCP port 1099 and block port 1099 at the host firewall (`iptables -A INPUT -p tcp --dport 1099 -j DROP`).
Short-Term Hardening:	If Java RMI is required locally, bind it strictly to `127.0.0.1` and enable JEP 290 serialization filtering (`jdk.serialFilter`).
Long-Term Strategic Fix:	Decommission legacy Java RMI services in favor of mutually authenticated TLS APIs.
Verification Command:	Run `nmap -p 1099 192.168.195.139` and confirm TCP port 1099 is `closed` or `filtered`.

F-46: MySQL Database Listener Exposed on Network (TCP Port 3306) [MEDIUM · CVSS 5.3 · OBSERVED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: SERVICE_BANNER_ONLY | Impact: EXPOSURE_ONLY | EvidenceConf: 42%
Access Scope Summary:	Service version or banner matched to known vulnerability; active exploitation not demonstrated in this record
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-MYSQL-PORT-EXPOSED | Advisory: MySQL bind-address Network Exposure Guideline | CWE: CWE-200 | MITRE: T1046 (Discovery — Network Service Discovery) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N

Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139:3306
Root Cause & Compliance:	Network Segmentation & Perimeter Filter Exposure [INFERRED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	MySQL 5.0.51a database listener is reachable on TCP port 3306; unauthenticated login was not demonstrated during the assessment.
Security Risk:	MEDIUM (OBSERVED) — Network-accessible database port increases brute-force and protocol attack surface.
EvidenceCore V3 Record:	ID: PROV-F-46-AD5249C221 | SHA-256: ad5249c2216bdfbef37f0504... | HMAC-SHA256: b308835a7acfc8172c387afd... | Module: AutoPentest-VulnScan

Captured Technical Evidence (F-46):

```
[2026-10-30 19:22:14] MySQL database service on 192.168.195.139:3306.
Service: mysql, Product: MySQL, Version: 5.0.51a-3ubuntu5
Database exposed on network. Default credential testing pending.
```

Immediate Containment:	Bind MySQL to `127.0.0.1` (`bind-address = 127.0.0.1` in `/etc/mysql/my.cnf`) and restart `mysqld` so TCP port 3306 is not exposed to the network.
Short-Term Hardening:	Remove anonymous MySQL accounts, require strong passwords for `root@localhost`, and revoke remote root login (`DELETE FROM mysql.user WHERE User='root' AND Host NOT IN ('localhost', '127.0.0.1')`).
Long-Term Strategic Fix:	Upgrade MySQL 5.0.51a to MySQL 8.0 LTS or MariaDB 10.11 LTS.
Verification Command:	Run `nmap -p 3306 192.168.195.139` from the network and confirm port 3306 is `closed` or `filtered`.

F-47: VNC Remote Desktop Service Exposed on TCP Port 5900 (Protocol 3.3) [MEDIUM · CVSS 5.3 · OBSERVED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: SERVICE_BANNER_ONLY | Impact: EXPOSURE_ONLY | EvidenceConf: 42%
Access Scope Summary:	Service version or banner matched to known vulnerability; active exploitation not demonstrated in this record
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-VNC-5900 | Advisory: RFB Protocol 3.3 VNC Authentication & Encryption Guideline | CWE: CWE-287 | MITRE: T1021.001 (Lateral Movement — Remote Services: Remote Desktop Protocol) | Taxonomy: OWASP-A07:2021 (Identification and Authentication Failures) | CWE-287 (Improper Authentication) | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139:5900
Root Cause & Compliance:	Network Segmentation & Perimeter Filter Exposure [INFERRED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS

Technical Description:	VNC remote desktop service on TCP port 5900 was fingerprinted (RFB protocol 3.3); authentication bypass or password brute-forcing was not executed.
Security Risk:	MEDIUM (OBSERVED) — Exposed remote desktop protocol on internal subnet.
EvidenceCore V3 Record:	ID: PROV-F-47-BD7534057A | SHA-256: bd7534057a4a365ac3b9a9b0... | HMAC-SHA256: d9068534be5f693c338d6051... | Module: XStrike-BuiltinVuln

Captured Technical Evidence (F-47):

```
VNC RealVNC allows authentication bypass via modified client

Evidence: Banner match on 192.168.195.139:5900: RFB 003.003
```

Immediate Containment:	Stop the VNC server on TCP port 5900 (`vncserver -kill :0`) and block port 5900 at the host firewall.
Short-Term Hardening:	If VNC is required, bind `Xvnc` to `-localhost` only, enforce a strong VNC password, and tunnel access exclusively over SSH.
Long-Term Strategic Fix:	Upgrade from legacy RFB 3.3 to modern remote management with MFA.
Verification Command:	Run `nmap -p 5900 192.168.195.139` from the subnet and confirm port 5900 is `closed`.

F-48: X11 Display Server Listener Exposed (TCP Port 6000) [MEDIUM · CVSS 5.3 · OBSERVED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: SERVICE_BANNER_ONLY | Impact: EXPOSURE_ONLY | EvidenceConf: 42%
Access Scope Summary:	Service version or banner matched to known vulnerability; active exploitation not demonstrated in this record
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-X11-PORT-6000 | Advisory: X.Org X11 Server -nolisten tcp Hardening Guideline | CWE: CWE-306 | MITRE: T1113 (Collection — Screen Capture) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-306 (Missing Authentication for Critical Function) | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139:6000
Root Cause & Compliance:	Network Segmentation & Perimeter Filter Exposure [INFERRED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	X11 display server TCP port 6000 is open; unauthenticated display capture or keystroke logging was not verified via x11-access.
Security Risk:	MEDIUM (OBSERVED) — Exposed X11 TCP listener on port 6000.
EvidenceCore V3 Record:	ID: PROV-F-48-7541763B69 | SHA-256: 7541763b69f185d0dde7a61f... | HMAC-SHA256: 460367f1c949b236e45c3e86... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-48):

```
X11 no-auth! Can capture display. Nmap scan report for 192.168.195.139
Host is up (0.00s latency).

PORT STATE SERVICE
6000/tcp open X11
MAC Address: 00:0C:29:05:D
```

Immediate Containment:	Disable X11 TCP listening by adding `-nolisten tcp` to the X server startup configuration (`/etc/X11/xinit/xserverrc` or display manager config) and block TCP port 6000.
Short-Term Hardening:	Terminate any unauthenticated Xvfb/X11 server processes on headless server hosts.
Long-Term Strategic Fix:	Remove X11 desktop packages from server base images.
Verification Command:	Run `nmap -p 6000 192.168.195.139` and confirm port 6000 is `closed`.

F-49: End-of-Life OpenSSH 4.7p1 Service Banner Observed (TCP Port 22) [MEDIUM · CVSS 5.3 · OBSERVED]

Target Asset & Port:	192.168.195.140 (Ports: N/A) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: SERVICE_BANNER_ONLY | Impact: EXPOSURE_ONLY | EvidenceConf: 42%
Access Scope Summary:	Service version or banner matched to known vulnerability; active exploitation not demonstrated in this record
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-OPENSSH-EOL-47P1 | Advisory: OpenSSH Legacy Lifecycle & Backported Patch Verification Advisory | CWE: CWE-1104 | MITRE: T1021.004 (Lateral Movement — Remote Services: SSH) | Taxonomy: OWASP-A06:2021 (Vulnerable and Outdated Components) | CWE-1104 (Use of Unmaintained Third Party Components) | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.140:22
Root Cause & Compliance:	Patch & Lifecycle Management Deficit (EOL Software) [DIRECTLY_ESTABLISHED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	The SSH daemon runs an end-of-life OpenSSH 4.7p1 Debian-8ubuntu1 release observed via service banner fingerprinting.
Security Risk:	MEDIUM (OBSERVED) — Legacy SSH daemon version; verify distribution backported patches or upgrade OS.
EvidenceCore V3 Record:	ID: PROV-F-49-321760211F | SHA-256: 321760211f264838b3fe1512... | HMAC-SHA256: 24de9f38523ef7200a209827... | Module: XStrike-BuiltinVuln

Captured Technical Evidence (F-49):

```
Legacy OpenSSH versions vulnerable to information leak and authentication bypass

Evidence: Banner match on 192.168.195.140:22: SSH-2.0-OpenSSH_5.3p1 Debian-3ubuntu4
```

Immediate Containment:	Restrict SSH access on port 22 to authorized management IPs via host firewall (`ufw allow from <mgmt_subnet> to any port 22`).
Short-Term Hardening:	Upgrade OpenSSH 4.7p1 to a supported release (>= 9.x) and disable legacy CBC ciphers and SHA-1 MACs in `sshd_config`.
Long-Term Strategic Fix:	Migrate the underlying Ubuntu 8.04 host to a supported LTS OS release.
Verification Command:	Run `nmap -sV -p 22 --script ssh2-enum-algos` and confirm modern OpenSSH version and cipher suite.

F-50: SMB Port 445 Exposed — Unverified MS17-010 (EternalBlue) Candidate [MEDIUM · CVSS 0.0 · OBSERVED]

Target Asset & Port:	192.168.195.1 (Ports: N/A) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: NONE | Impact: EXPOSURE_ONLY | EvidenceConf: 38%
Access Scope Summary:	Port-open / service banner observation only; vulnerability NOT verified
Taxonomy & CVSS v3.1:	CVE: CVE-2017-0144 | Custom ID: XSTRIKE-SMB-445-CANDIDATE | Advisory: Microsoft Security Bulletin MS17-010 (Unverified Port-Open Candidate) | CWE: CWE-200 | MITRE: T1210 (Lateral Movement — Exploitation of Remote Services) | Taxonomy: OWASP-A06:2021 (Vulnerable Components) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:N
Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.1
Root Cause & Compliance:	Network Segmentation & Perimeter Filter Exposure [INFERRED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	TCP port 445 (SMB) is reachable. As recorded in the tool output, this is a port-open observation only; SMBv1/MS17-010 vulnerability status has NOT been verified and no exploitation was performed. Conditional CVE-2017-0144 severity (9.8) is held in reserve pending protocol verification.
Security Risk:	MEDIUM (OBSERVED, CVSS 0.0 Active / 9.8 Conditional) — Exposed SMB attack surface; requires explicit MS17-010 protocol validation to determine whether CVE-2017-0144 applies.
EvidenceCore V3 Record:	ID: PROV-F-50-80D58095EF | SHA-256: 80d58095efd69101592dd8c6... | HMAC-SHA256: 2a9460809fc5ce979e5060ae... | Module: AutoPentest-VulnScan

Captured Technical Evidence (F-50):

```
[2026-10-30 19:37:01] TCP connection to 192.168.195.1:445 succeeded.
Service: microsoft-ds, Product: unknown, Version: unknown
NOTE: Port-open detection only. SMBv1/EternalBlue status NOT verified.
Verification required: nmap --script smb-vuln-ms17-010 192.168.195.1
Or Metasploit: auxiliary/scanner/smb/smb_ms17_010
```

Immediate Containment:	Restrict TCP port 445 access using host firewall rules (`New-NetFirewallRule` on Windows or `iptables`/`ufw` on Linux) to authorized management hosts only.
Short-Term Hardening:	Run an explicit SMB dialect and MS17-010 verification check (`nmap --script smb-vuln-ms17-010,smb-protocols -p 445 <target>`) and ensure Microsoft security update MS17-010 or Samba >= 4.6.4 is installed.
Long-Term Strategic Fix:	Enforce SMBv3 with mandatory SMB signing (`RequireSecuritySignature=1`) and segment SMB traffic across VLAN boundaries.
Verification Command:	Confirm `nmap --script smb-protocols -p 445` reports SMBv1 disabled and `smb-vuln-ms17-010` returns clean.

F-51: SMB Port 445 Exposed — Unverified MS17-010 (EternalBlue) Candidate [MEDIUM · CVSS 0.0 · OBSERVED]

Target Asset & Port:	192.168.195.130 (Ports: N/A) | Asset Class: Workstation / SMB Peer Host (STANDARD)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC

7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: NONE | Impact: EXPOSURE_ONLY | EvidenceConf: 38%
Access Scope Summary:	Port-open / service banner observation only; vulnerability NOT verified
Taxonomy & CVSS v3.1:	CVE: CVE-2017-0144 | Custom ID: XSTRIKE-SMB-445-CANDIDATE | Advisory: Microsoft Security Bulletin MS17-010 (Unverified Port-Open Candidate) | CWE: CWE-200 | MITRE: T1210 (Lateral Movement — Exploitation of Remote Services) | Taxonomy: OWASP-A06:2021 (Vulnerable Components) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:N
Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.130
Root Cause & Compliance:	Network Segmentation & Perimeter Filter Exposure [INFERRED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	TCP port 445 (SMB) is reachable. As recorded in the tool output, this is a port-open observation only; SMBv1/MS17-010 vulnerability status has NOT been verified and no exploitation was performed. Conditional CVE-2017-0144 severity (9.8) is held in reserve pending protocol verification.
Security Risk:	MEDIUM (OBSERVED, CVSS 0.0 Active / 9.8 Conditional) — Exposed SMB attack surface; requires explicit MS17-010 protocol validation to determine whether CVE-2017-0144 applies.
EvidenceCore V3 Record:	ID: PROV-F-51-8A5AC7AAE0 | SHA-256: 8a5ac7aae058633e8212c269... | HMAC-SHA256: a98b20ef251bca63b74ad3e9... | Module: AutoPentest-VulnScan

Captured Technical Evidence (F-51):

```
[2026-10-30 19:33:30] TCP connection to 192.168.195.130:445 succeeded.
Service: microsoft-ds, Product: Microsoft Windows Server 2008 R2 - 2012 microsoft-ds, Version: unknown
NOTE: Port-open detection only. SMBv1/EternalBlue status NOT verified.
Verification required: nmap --script smb-vuln-ms17-010 192.168.195.130
Or Metasploit: auxiliary/scanner/smb/smb_ms17_010
```

Immediate Containment:	Restrict TCP port 445 access using host firewall rules (`New-NetFirewallRule` on Windows or `iptables`/`ufw` on Linux) to authorized management hosts only.
Short-Term Hardening:	Run an explicit SMB dialect and MS17-010 verification check (`nmap --script smb-vuln-ms17-010,smb-protocols -p 445 <target>`) and ensure Microsoft security update MS17-010 or Samba >= 4.6.4 is installed.
Long-Term Strategic Fix:	Enforce SMBv3 with mandatory SMB signing (`RequireSecuritySignature=1`) and segment SMB traffic across VLAN boundaries.
Verification Command:	Confirm `nmap --script smb-protocols -p 445` reports SMBv1 disabled and `smb-vuln-ms17-010` returns clean.

F-52: SMB Port 445 Exposed — Unverified MS17-010 (EternalBlue) Candidate [MEDIUM · CVSS 0.0 · OBSERVED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: NONE | Impact: EXPOSURE_ONLY | EvidenceConf: 38%
Access Scope Summary:	Port-open / service banner observation only; vulnerability NOT verified

Taxonomy & CVSS v3.1:	CVE: CVE-2017-0144 | Custom ID: XSTRIKE-SMB-445-CANDIDATE | Advisory: Microsoft Security Bulletin MS17-010 (Unverified Port-Open Candidate) | CWE: CWE-200 | MITRE: T1210 (Lateral Movement — Exploitation of Remote Services) | Taxonomy: OWASP-A06:2021 (Vulnerable Components) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:N
Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139
Root Cause & Compliance:	Network Segmentation & Perimeter Filter Exposure [INFERRED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	TCP port 445 (SMB) is reachable. As recorded in the tool output, this is a port-open observation only; SMBv1/MS17-010 vulnerability status has NOT been verified and no exploitation was performed. Conditional CVE-2017-0144 severity (9.8) is held in reserve pending protocol verification.
Security Risk:	MEDIUM (OBSERVED, CVSS 0.0 Active / 9.8 Conditional) — Exposed SMB attack surface; requires explicit MS17-010 protocol validation to determine whether CVE-2017-0144 applies.
EvidenceCore V3 Record:	ID: PROV-F-52-45378F48D4 | SHA-256: 45378f48d49e9593cbfce48d... | HMAC-SHA256: a996f692b41bf891f3782651... | Module: AutoPentest-VulnScan

Captured Technical Evidence (F-52):

```
[2026-10-30 19:22:14] TCP connection to 192.168.195.139:445 succeeded.
Service: netbios-ssn, Product: Samba smbd, Version: 3.X - 4.X
NOTE: Port-open detection only. SMBv1/EternalBlue status NOT verified.
Verification required: nmap --script smb-vuln-ms17-010 192.168.195.139
Or Metasploit: auxiliary/scanner/smb/smb_ms17_010
```

Immediate Containment:	Restrict TCP port 445 access using host firewall rules (`New-NetFirewallRule` on Windows or `iptables`/`ufw` on Linux) to authorized management hosts only.
Short-Term Hardening:	Run an explicit SMB dialect and MS17-010 verification check (`nmap --script smb-vuln-ms17-010,smb-protocols -p 445 <target>`) and ensure Microsoft security update MS17-010 or Samba >= 4.6.4 is installed.
Long-Term Strategic Fix:	Enforce SMBv3 with mandatory SMB signing (`RequireSecuritySignature=1`) and segment SMB traffic across VLAN boundaries.
Verification Command:	Confirm `nmap --script smb-protocols -p 445` reports SMBv1 disabled and `smb-vuln-ms17-010` returns clean.

F-53: SMB Port 445 Exposed — Unverified MS17-010 (EternalBlue) Candidate [MEDIUM · CVSS 0.0 · OBSERVED]

Target Asset & Port:	192.168.195.140 (Ports: N/A) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: NONE | Impact: EXPOSURE_ONLY | EvidenceConf: 38%
Access Scope Summary:	Port-open / service banner observation only; vulnerability NOT verified
Taxonomy & CVSS v3.1:	CVE: CVE-2017-0144 | Custom ID: XSTRIKE-SMB-445-CANDIDATE | Advisory: Microsoft Security Bulletin MS17-010 (Unverified Port-Open Candidate) | CWE: CWE-200 | MITRE: T1210 (Lateral Movement — Exploitation of Remote Services) | Taxonomy: OWASP-A06:2021 (Vulnerable Components) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:N

Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.140
Root Cause & Compliance:	Network Segmentation & Perimeter Filter Exposure [INFERRED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	TCP port 445 (SMB) is reachable. As recorded in the tool output, this is a port-open observation only; SMBv1/MS17-010 vulnerability status has NOT been verified and no exploitation was performed. Conditional CVE-2017-0144 severity (9.8) is held in reserve pending protocol verification.
Security Risk:	MEDIUM (OBSERVED, CVSS 0.0 Active / 9.8 Conditional) — Exposed SMB attack surface; requires explicit MS17-010 protocol validation to determine whether CVE-2017-0144 applies.
EvidenceCore V3 Record:	ID: PROV-F-53-483C65504D | SHA-256: 483c65504d5cfdd88dc205b9... | HMAC-SHA256: 82a0920e19f49d3bfca88e98... | Module: AutoPentest-VulnScan

Captured Technical Evidence (F-53):

```
[2026-10-30 19:28:11] TCP connection to 192.168.195.140:445 succeeded.
Service: netbios-ssn, Product: Samba smbd, Version: 3.X - 4.X
NOTE: Port-open detection only. SMBv1/EternalBlue status NOT verified.
Verification required: nmap --script smb-vuln-ms17-010 192.168.195.140
Or Metasploit: auxiliary/scanner/smb/smb_ms17_010
```

Immediate Containment:	Restrict TCP port 445 access using host firewall rules (`New-NetFirewallRule` on Windows or `iptables`/`ufw` on Linux) to authorized management hosts only.
Short-Term Hardening:	Run an explicit SMB dialect and MS17-010 verification check (`nmap --script smb-vuln-ms17-010,smb-protocols -p 445 <target>`) and ensure Microsoft security update MS17-010 or Samba >= 4.6.4 is installed.
Long-Term Strategic Fix:	Enforce SMBv3 with mandatory SMB signing (`RequireSecuritySignature=1`) and segment SMB traffic across VLAN boundaries.
Verification Command:	Confirm `nmap --script smb-protocols -p 445` reports SMBv1 disabled and `smb-vuln-ms17-010` returns clean.

F-54: Developer / Debugging Comments Exposed in HTML/JS Source [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: N/A) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-ZAP-COMMENTS | Advisory: OWASP ZAP Alert 10027 — Information Disclosure: Suspicious Comments | CWE: CWE-615 | MITRE: T1592 (Reconnaissance — Gather Victim Host Information) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-615 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=90% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=90% (Det=96%, Val=92%, Art=75%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 10 raw observation(s) across 1 unique endpoint(s): 192.168.195.1

Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTML and JavaScript responses contain developer comments that may disclose internal application logic, hints, or endpoints.
Security Risk:	LOW — Information disclosure in client-accessible markup.
EvidenceCore V3 Record:	ID: PROV-F-54-87983E3E08 | SHA-256: 87983e3e08dca63f49686b8b... | HMAC-SHA256: 089f37c3d0535e1ae83dc77f... | Module: AutoPentest-ZAP

Captured Technical Evidence (F-54):

```
The response appears to contain suspicious comments which may help an attacker.
```

Immediate Containment:	Strip developer, TODO, and debugging HTML/JS comments from production templates in `/var/www/`.
Short-Term Hardening:	Audit client-side HTML and JavaScript files to ensure no hints, credentials, or internal SQL queries are embedded in comments.
Long-Term Strategic Fix:	Add HTML/JS minification that strips comments in the build pipeline.
Verification Command:	Inspect page source via `curl -s` and confirm developer/debugging comments are removed.

F-55: HTTP Permissions-Policy Header Missing [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: 8000, 8080) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-PERMISSIONS-POLICY | Advisory: W3C Permissions Policy Specification | CWE: CWE-693 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-693 (Protection Mechanism Failure) | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 2 unique endpoint(s): 192.168.195.1:8000, http://192.168.195.1:8080/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP responses omit the 'Permissions-Policy' (formerly Feature-Policy) header, leaving browser APIs (camera, microphone, geolocation, payment) unrestricted at the document policy layer.
Security Risk:	LOW — Defense-in-depth browser hardening omission; does not directly enable remote code execution.
EvidenceCore V3 Record:	ID: PROV-F-55-9B6D602736 | SHA-256: 9b6d6027369ae4c50fdc5b87... | HMAC-SHA256: 926e8fcc92e0744987e18186... | Module: AutoPentest-AegisProbe

Captured Technical Evidence (F-55):

```
Missing header: Permissions-Policy | CWE: CWE-732 | Payload:
```

Immediate Containment:	Add `Header always set Permissions-Policy "geolocation=(), microphone=(), camera=(), payment=(), usb=()"` in the web server configuration.

Short-Term Hardening:	Audit frontend application requirements and explicitly disable all unused browser hardware/sensor APIs.
Long-Term Strategic Fix:	Enforce `Permissions-Policy` alongside CSP in automated HTTP security header regression tests.
Verification Command:	Run `curl -I http://<target>/` and confirm the `Permissions-Policy` header is present.

F-56: HTTP Referrer-Policy Header Missing [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: 8000, 8080) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-REFERRER-POLICY | Advisory: W3C Referrer Policy / OWASP Secure Headers | CWE: CWE-200 | MITRE: T1592 (Reconnaissance — Gather Victim Host Information) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 2 unique endpoint(s): 192.168.195.1:8000, http://192.168.195.1:8080/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP responses omit the 'Referrer-Policy' header, which may cause full URLs (including query parameters) to be leaked in the Referer header during cross-origin navigation.
Security Risk:	LOW — Potential disclosure of sensitive URL paths or query parameters to third-party origins.
EvidenceCore V3 Record:	ID: PROV-F-56-4D3954EE1B | SHA-256: 4d3954ee1be933c1fa96ad77... | HMAC-SHA256: eaeb87153737689df4773d29... | Module: AutoPentest-AegisProbe

Captured Technical Evidence (F-56):

```
Missing header: Referrer-Policy | CWE: CWE-200 | Payload:
```

Immediate Containment:	Add `Header always set Referrer-Policy "strict-origin-when-cross-origin"` in Apache `/etc/apache2/conf.d/security` or Nginx configuration.
Short-Term Hardening:	Ensure sensitive tokens or session identifiers are never passed in URL query strings where they could leak via Referer headers.
Long-Term Strategic Fix:	Include `Referrer-Policy` in baseline web security header automated tests.
Verification Command:	Run `curl -I http://<target>/` and confirm `Referrer-Policy: strict-origin-when-cross-origin` is returned.

F-57: HTTP Server Banner Version Disclosure ('Server' Header) [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: 8000, 8080) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC

7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-SERVER-BANNER | Advisory: Apache ServerTokens Prod / CIS Web Server Hardening | CWE: CWE-200 | MITRE: T1592 (Reconnaissance — Gather Victim Host Information) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 56 raw observation(s) across 3 unique endpoint(s): 192.168.195.1, 192.168.195.1:8000, http://192.168.195.1:8080/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	The web server discloses exact software and OS release identifiers (e.g., 'Apache/2.2.8 (Ubuntu) DAV/2') in the HTTP 'Server' response header across multiple endpoints.
Security Risk:	LOW — Facilitates precise vulnerability fingerprinting by external adversaries.
EvidenceCore V3 Record:	ID: PROV-F-57-AACB581349 | SHA-256: aacb581349e653d422d577b0... | HMAC-SHA256: e10d2f48e27ed58b2b986942... | Module: AutoPentest-ZAP, AutoPentest-AegisProbe

Captured Technical Evidence (F-57):

```
The web/application server is leaking version information via the "Server" HTTP response header. Access
to such information may facilitate attackers identifying other vulnerabilities your web/applicat
---
Server: Splunkd | CWE: CWE-200 | Payload:
---
Server: Apache | CWE: CWE-200 | Payload:
```

Immediate Containment:	Set `ServerTokens Prod` and `ServerSignature Off` in Apache `/etc/apache2/apache2.conf` and reload Apache (`service apache2 reload`).
Short-Term Hardening:	Configure reverse-proxy header sanitization to strip detailed upstream server version banners.
Long-Term Strategic Fix:	Maintain automated configuration management to prevent verbose banner headers in production.
Verification Command:	Run `curl -I http://<target>/` and confirm `Server: Apache` returns no version or OS string.

F-58: HTTP Strict-Transport-Security (HSTS) Header Missing [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: 8000, 8080) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-HSTS-MISSING | Advisory: RFC 6797 / OWASP HTTP Strict Transport Security Cheat Sheet | CWE: CWE-319 | MITRE: T1557 (Credential Access — Adversary-in-the-Middle) | Taxonomy: OWASP-A02:2021 (Cryptographic Failures) | CWE-319 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N

Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 5 raw observation(s) across 4 unique endpoint(s): 192.168.195.1:8080, 192.168.195.1, 192.168.195.1:8000, http://192.168.195.1:8080/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP/HTTPS responses omit the 'Strict-Transport-Security' header, allowing clients to connect over unencrypted HTTP and exposing sessions to SSL-stripping attacks.
Security Risk:	LOW — Enables protocol downgrade and cleartext session interception on untrusted networks.
EvidenceCore V3 Record:	ID: PROV-F-58-1C1640C0A7 | SHA-256: 1c1640c0a70976c844ec9833... | HMAC-SHA256: 7ddc722bd3bbb3404f7b17fa... | Module: XStrike-BuiltinVuln, AutoPentest-OWASP, AutoPentest-AegisProbe

Captured Technical Evidence (F-58):

```
No HSTS header — susceptible to SSL stripping attacks

Evidence: HTTP response from 192.168.195.1:8080 missing strict-transport-security header
---
A02 Cryptographic Failures: 1 finding(s)
[CWE-693] Missing Security Headers: Strict-Transport-Security, X-Content-Type-Options, X-Frame-Options
---
A02 Cryptographic Failures: 1 finding(s)
[CWE-693] Missing Security Headers: Strict-Transport-Security, Content-Security-Policy
```

Immediate Containment:	Configure TLS (HTTPS) on port 443 and add `Header always set Strict-Transport-Security "max-age=31536000; includeSubDomains"` to the HTTPS virtual host.
Short-Term Hardening:	Configure HTTP port 80 to issue permanent `301 Moved Permanently` redirects to `https://` for all endpoints.
Long-Term Strategic Fix:	Standardize internal PKI certificate automation (ACME/step-ca) across all web services.
Verification Command:	Inspect HTTPS response headers via `curl -I https://<target>/` and confirm `Strict-Transport-Security` is returned with `max-age>=31536000`.

F-59: Session Cookie Missing 'HttpOnly' Flag [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: 8000) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-COOKIE-HTTPONLY | Advisory: OWASP Session Management Cheat Sheet — HttpOnly | CWE: CWE-1004 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-1004 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)

Deduplication & Endpoints:	Consolidated 63 raw observation(s) across 2 unique endpoint(s): 192.168.195.1, 192.168.195.1:8000
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Cookies are set without the 'HttpOnly' flag, permitting client-side JavaScript ('document.cookie') to read session identifiers.
Security Risk:	LOW — Enables session theft if combined with a Cross-Site Scripting (XSS) vulnerability.
EvidenceCore V3 Record:	ID: PROV-F-59-811C0B0C8D | SHA-256: 811c0b0c8dd7dc5934aee528... | HMAC-SHA256: 5486594935d530e5ac7c9c2b... | Module: AutoPentest-ZAP, AutoPentest-AegisProbe

Captured Technical Evidence (F-59):

```
A cookie has been set without the HttpOnly flag, which means that the cookie can be accessed by
JavaScript. If a malicious script can be run on this page then the cookie will be accessible and can be
---
Cookie: cval=1915971765... | CWE: CWE-1004 | Payload:
```

Immediate Containment:	Set `session.cookie_httponly = 1` in `/etc/php5/apache2/php.ini` and add `HttpOnly` to all application `setcookie()` calls.
Short-Term Hardening:	Add Apache header rewrite `Header edit Set-Cookie ^(.*)$ "$1; HttpOnly"` for legacy applications.
Long-Term Strategic Fix:	Enforce automated DAST checks verifying `HttpOnly` on all session identifiers.
Verification Command:	Inspect `Set-Cookie: PHPSESSID=...` via `curl -I` and confirm the `HttpOnly` flag is present.

F-60: Session Cookie Missing 'SameSite' Attribute [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: 8000) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-COOKIE-SAMESITE | Advisory: RFC 6265bis — SameSite Cookie Attribute | CWE: CWE-352 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A01:2021 (Broken Access Control) | CWE-352 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 187 raw observation(s) across 2 unique endpoint(s): 192.168.195.1, 192.168.195.1:8000
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Session cookies are issued without the 'SameSite=Lax' or 'SameSite=Strict' attribute, allowing cookies to be transmitted in cross-origin requests.
Security Risk:	LOW — Weakens browser defense against Cross-Site Request Forgery (CSRF).
EvidenceCore V3 Record:	ID: PROV-F-60-AFBCB2DA9C | SHA-256: afbcb2da9cf03106737efcc0... | HMAC-SHA256: b1f68766b010bec0f60354a7... | Module: AutoPentest-ZAP, AutoPentest-AegisProbe

Captured Technical Evidence (F-60):

```
A cookie has been set without the SameSite attribute, which means that the cookie can be sent as a result of a 'cross-site' request. The SameSite attribute is an effective counter measure to cross-sit
---
Cookie: cval | CWE: CWE-352 | Payload:
```

Immediate Containment:	Update PHP/application session configuration (`session.cookie_samesite = "Lax"` or `"Strict"`) or add Apache `Header edit Set-Cookie ^(.*)$ "$1; SameSite=Lax"`.
Short-Term Hardening:	Audit cross-site authentication flows and enforce `SameSite=Strict` on high-privilege session cookies.
Long-Term Strategic Fix:	Standardize session management using framework-managed `__Host-` prefixed cookies.
Verification Command:	Inspect `Set-Cookie` via `curl -I` and confirm `SameSite=Lax` or `SameSite=Strict` is present.

F-61: Unix Timestamp Disclosure in HTTP Response Content [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: N/A) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-ZAP-TIMESTAMP | Advisory: OWASP ZAP Alert 10096 — Timestamp Disclosure | CWE: CWE-200 | MITRE: T1592 (Reconnaissance — Gather Victim Host Information) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 228 raw observation(s) across 1 unique endpoint(s): 192.168.195.1
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	OWASP ZAP identified Unix epoch timestamps embedded within HTTP response bodies/headers across multiple application endpoints.
Security Risk:	LOW — Minor informational leakage of server clock or cache generation timestamps.
EvidenceCore V3 Record:	ID: PROV-F-61-794877E490 | SHA-256: 794877e4900f4fcfdc38b1a0... | HMAC-SHA256: f269b8f803fe9fc159d38fd4... | Module: AutoPentest-ZAP

Captured Technical Evidence (F-61):

```
A timestamp was disclosed by the application/web server. - Unix
```

Immediate Containment:	Configure ETag generation to exclude inode/mtime timestamps (`FileETag None` in Apache) and remove Unix epoch values from unauthenticated HTML responses.
Short-Term Hardening:	Ensure static asset cache keys use content hashes rather than server timestamps.
Long-Term Strategic Fix:	Standardize response header normalization across web servers.
Verification Command:	Inspect HTTP response headers and body for 10-digit Unix epoch patterns and confirm none are disclosed.

F-62: Verbose Content Body Returned in HTTP Redirect Response (Big Redirect) [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: N/A) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-ZAP-BIG-REDIRECT | Advisory: OWASP ZAP Alert 10044 — Big Redirect Detected | CWE: CWE-200 | MITRE: T1592 (Reconnaissance — Gather Victim Host Information) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 58 raw observation(s) across 1 unique endpoint(s): 192.168.195.1
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP 3xx redirect responses contain a larger-than-expected HTML body, which may indicate page content is rendered despite a redirect header.
Security Risk:	LOW — Verify that unauthenticated redirects call 'exit()' immediately after setting the Location header.
EvidenceCore V3 Record:	ID: PROV-F-62-979D261826 | SHA-256: 979d261826fb1f1d0eb78355... | HMAC-SHA256: 9834e4f07c3e9c9f2887274b... | Module: AutoPentest-ZAP

Captured Technical Evidence (F-62):

```
The server has responded with a redirect that seems to provide a large response. This may indicate that
although the server sent a redirect it also responded with body content (which may include sensi
```

Immediate Containment:	Ensure all PHP authentication/redirect handlers call `exit;` immediately after `header('Location: ...');` so authenticated page content is not rendered in the 301/302 body.
Short-Term Hardening:	Audit all 3xx responses across Mutillidae/DVWA/phpMyAdmin to ensure response bodies contain zero sensitive HTML.
Long-Term Strategic Fix:	Use centralized router middleware for HTTP redirects.
Verification Command:	Issue `curl -i http://<target>/<redirect_path>` and confirm the 302 response body length is minimal with no sensitive HTML.

F-63: Web Server Directory Listing Enabled (/icons/ & Static Directories) [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: N/A) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%

Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-DIR-LISTING | Advisory: Apache Options -Indexes Hardening Guideline | CWE: CWE-548 | MITRE: T1083 (Discovery — File and Directory Discovery) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-548 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 3 raw observation(s) across 2 unique endpoint(s): 192.168.195.1, http://192.168.195.1:80/images/:
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Web server directories (such as /icons/) return an auto-generated directory index listing internal files (consolidated with OWASP A05 misconfiguration checks).
Security Risk:	LOW — Enables unauthenticated enumeration of static and backup files.
EvidenceCore V3 Record:	ID: PROV-F-63-386B2E082F | SHA-256: 386b2e082f5c42338887db7a... | HMAC-SHA256: 2b62892b3152140eb5513603... | Module: AutoPentest-ZAP, AutoPentest-OWASP

Captured Technical Evidence (F-63):

```
It is possible to view the directory listing. Directory listing may reveal hidden scripts, include
files, backup source files, etc. which can be accessed to read sensitive information.
---
It is possible to view a listing of the directory contents. Directory listings may reveal hidden
scripts, include files, backup source files, etc., which can be accessed to reveal sensitive informatio
---
A05 Misconfiguration: 2 finding(s)
[CWE-548] Directory Listing: /images/: Full directory contents exposed
[CWE-693] HTTP TRACE Enabled: XST (Cross-Site Tracing) possible
```

Immediate Containment:	Disable directory indexing in Apache `/etc/apache2/apache2.conf` and `/etc/apache2/mods-enabled/alias.conf` by setting `Options -Indexes` on `/var/www/` and `/usr/share/apache2/icons/`.
Short-Term Hardening:	Remove default Apache `/icons/` alias or restrict it to specific file extensions.
Long-Term Strategic Fix:	Enforce `Options -Indexes` in baseline web server hardening templates.
Verification Command:	Request `http://192.168.195.139:80/icons/` and confirm HTTP 403 Forbidden instead of `Index of /icons`.

F-64: X-Content-Type-Options Header Missing (MIME-Sniffing Exposure) [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: 8080) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated

Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-XCTO-MISSING | Advisory: OWASP Secure Headers Project — X-Content-Type-Options | CWE: CWE-693 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-693 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 46 raw observation(s) across 3 unique endpoint(s): 192.168.195.1, 192.168.195.1:8080, http://192.168.195.1:8080/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP responses omit 'X-Content-Type-Options: nosniff', allowing browsers to MIME-sniff response bodies into executable content types.
Security Risk:	LOW — Browser MIME-type guessing can facilitate content-spoofing or stored XSS in uploaded content.
EvidenceCore V3 Record:	ID: PROV-F-64-61FFD0A164 | SHA-256: 61ffd0a1643c096b84806e1b... | HMAC-SHA256: 3673991a2e44a4a83e1cec3a... | Module: AutoPentest-ZAP, XStrike-BuiltinVuln, AutoPentest-AegisProbe

Captured Technical Evidence (F-64):

```
The Anti-MIME-Sniffing header X-Content-Type-Options was not set to 'nosniff'. This allows older
versions of Internet Explorer and Chrome to perform MIME-sniffing on the response body, potentially cau
---
No X-Content-Type-Options header — MIME sniffing not prevented

Evidence: HTTP response from 192.168.195.1:8080 missing x-content-type-options header
---
Missing header: X-Content-Type-Options | CWE: CWE-693 | Payload:
```

Immediate Containment:	Add `Header always set X-Content-Type-Options "nosniff"` in the web server configuration (`apache2.conf` / `nginx.conf`).
Short-Term Hardening:	Ensure all static and dynamic responses set explicit, accurate `Content-Type` MIME headers.
Long-Term Strategic Fix:	Include `X-Content-Type-Options: nosniff` in baseline infrastructure-as-code web templates.
Verification Command:	Run `curl -I http://<target>/` and verify `X-Content-Type-Options: nosniff` is present on all responses.

F-65: robots.txt Disallows Sensitive Web Paths [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.1 (Ports: 8000) | Asset Class: Hypervisor / Management Host Gateway (INFRASTRUCTURE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-ROBOTS-TXT | Advisory: OWASP WSTG-INFO-03 — Review Webserver Metafiles for Information Leakage | CWE: CWE-200 | MITRE: T1592 (Reconnaissance — Gather Victim Host Information) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 (Exposure of Sensitive Information) | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N

Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.1:8000
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	The /robots.txt file is publicly readable and lists internal or administrative URI paths intended to be hidden from search crawlers.
Security Risk:	LOW — Provides attackers with a reconnaissance map of internal application directories.
EvidenceCore V3 Record:	ID: PROV-F-65-D430CB0DB2 | SHA-256: d430cb0db2e09277de6361de... | HMAC-SHA256: 84e6eb4e6c4855ea2cf49b0a... | Module: AutoPentest-AegisProbe

Captured Technical Evidence (F-65):

```
User-agent: *
Disallow: /
| CWE: CWE-200 | Payload:
```

Immediate Containment:	Remove sensitive administrative or internal path entries from `/var/www/robots.txt` and enforce proper authentication on every sensitive directory.
Short-Term Hardening:	Use `<meta name="robots" content="noindex, nofollow">` inside authenticated pages rather than listing secret paths in a public `robots.txt`.
Long-Term Strategic Fix:	Audit all Paths formerly listed in `robots.txt` to ensure none rely on security-by-obscurity.
Verification Command:	Fetch `http://192.168.195.139:80/robots.txt` and verify no sensitive internal directories are enumerated.

F-66: HTTP Server Banner Version Disclosure ('Server' Header) [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.129 (Ports: N/A) | Asset Class: Application / Network Host (STANDARD)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-SERVER-BANNER | Advisory: Apache ServerTokens Prod / CIS Web Server Hardening | CWE: CWE-200 | MITRE: T1592 (Reconnaissance — Gather Victim Host Information) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 3 raw observation(s) across 1 unique endpoint(s): 192.168.195.129
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	The web server discloses exact software and OS release identifiers (e.g., 'Apache/2.2.8 (Ubuntu) DAV/2') in the HTTP 'Server' response header across multiple endpoints.
Security Risk:	LOW — Facilitates precise vulnerability fingerprinting by external adversaries.

EvidenceCore V3 Record:	ID: PROV-F-66-7166A50DB1 | SHA-256: 7166a50db1a7d84799884ff6... | HMAC-SHA256: 571c307dd1f86b4edf7a7658... | Module: AutoPentest-ZAP

Captured Technical Evidence (F-66):

```
The web/application server is leaking version information via the "Server" HTTP response header. Access to such information may facilitate attackers identifying other vulnerabilities your web/applicat
```

Immediate Containment:	Set `ServerTokens Prod` and `ServerSignature Off` in Apache `/etc/apache2/apache2.conf` and reload Apache (`service apache2 reload`).
Short-Term Hardening:	Configure reverse-proxy header sanitization to strip detailed upstream server version banners.
Long-Term Strategic Fix:	Maintain automated configuration management to prevent verbose banner headers in production.
Verification Command:	Run `curl -I http://<target>/` and confirm `Server: Apache` returns no version or OS string.

F-67: HTTP Server Banner Version Disclosure ('Server' Header) [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.130 (Ports: N/A) | Asset Class: Workstation / SMB Peer Host (STANDARD)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-SERVER-BANNER | Advisory: Apache ServerTokens Prod / CIS Web Server Hardening | CWE: CWE-200 | MITRE: T1592 (Reconnaissance — Gather Victim Host Information) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 3 raw observation(s) across 1 unique endpoint(s): 192.168.195.130
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	The web server discloses exact software and OS release identifiers (e.g., 'Apache/2.2.8 (Ubuntu) DAV/2') in the HTTP 'Server' response header across multiple endpoints.
Security Risk:	LOW — Facilitates precise vulnerability fingerprinting by external adversaries.
EvidenceCore V3 Record:	ID: PROV-F-67-0A8D257646 | SHA-256: 0a8d2576469461dba8076440... | HMAC-SHA256: 70c2d706f54dd68a60550db6... | Module: AutoPentest-ZAP

Captured Technical Evidence (F-67):

```
The web/application server is leaking version information via the "Server" HTTP response header. Access to such information may facilitate attackers identifying other vulnerabilities your web/applicat
```

Immediate Containment:	Set `ServerTokens Prod` and `ServerSignature Off` in Apache `/etc/apache2/apache2.conf` and reload Apache (`service apache2 reload`).
Short-Term Hardening:	Configure reverse-proxy header sanitization to strip detailed upstream server version banners.
Long-Term Strategic Fix:	Maintain automated configuration management to prevent verbose banner headers in production.
Verification Command:	Run `curl -I http://<target>/` and confirm `Server: Apache` returns no version or OS string.

F-68: Apache HTTP OPTIONS Allow Header Heuristic (Optionsbleed CVE-2017-9798 Candidate) [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: CVE-2017-9798 | Custom ID: XSTRIKE-APACHE-OPTIONSBLEED | Advisory: Apache Security Advisory CVE-2017-9798 (Optionsbleed) | CWE: CWE-200 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 (Exposure of Sensitive Information) | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): 192.168.195.139:80
Root Cause & Compliance:	Service Hardening & Configuration Baseline Gap [PROBABLE] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Apache httpd 2.2.8 was flagged by a banner/header heuristic for CVE-2017-9798 (Optionsbleed); corrupted Allow header memory leakage was not observed.
Security Risk:	LOW (OBSERVED) — Version/header heuristic; verify Apache Limit/Options directives.
EvidenceCore V3 Record:	ID: PROV-F-68-082D7296FD | SHA-256: 082d7296fd13bf033090e74f... | HMAC-SHA256: f9fffe9a4277a99ed25a5eea... | Module: XStrike-BuiltinVuln

Captured Technical Evidence (F-68):

```
Apache 2.2.x-2.4.x Optionsbleed information leak via OPTIONS requests

Evidence: Banner match on 192.168.195.139:80: HTTP/1.1 200 OK
Date: Sat, 31 Oct 2026 00:01:47 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Length: 891
Connection: close
Content-Type: text/html

<html
```

Immediate Containment:	Audit Apache `<Directory>` and `.htaccess` files to ensure `Limit` and `Options` directives contain only valid HTTP methods and tokens, or disable `OPTIONS` via `RewriteEngine`.
Short-Term Hardening:	Apply the CVE-2017-9798 patch or upgrade Apache httpd to 2.4.x.
Long-Term Strategic Fix:	Include `.htaccess` syntax validation in deployment checks.
Verification Command:	Send repeated `curl -i -X OPTIONS http://192.168.195.139/` requests and confirm clean, deterministic `Allow:` headers.

F-69: HTTP Permissions-Policy Header Missing [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: 80) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-PERMISSIONS-POLICY | Advisory: W3C Permissions Policy Specification | CWE: CWE-693 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-693 (Protection Mechanism Failure) | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): http://192.168.195.139:80/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP responses omit the 'Permissions-Policy' (formerly Feature-Policy) header, leaving browser APIs (camera, microphone, geolocation, payment) unrestricted at the document policy layer.
Security Risk:	LOW — Defense-in-depth browser hardening omission; does not directly enable remote code execution.
EvidenceCore V3 Record:	ID: PROV-F-69-C41FEE5081 | SHA-256: c41fee508178852d5b00be43... | HMAC-SHA256: 1a3bafe6d39cf9fe30d90ffd... | Module: AutoPentest-AegisProbe

Captured Technical Evidence (F-69):

```
Missing header: Permissions-Policy | CWE: CWE-732 | Payload:
```

Immediate Containment:	Add `Header always set Permissions-Policy "geolocation=(), microphone=(), camera=(), payment=(), usb=()"` in the web server configuration.
Short-Term Hardening:	Audit frontend application requirements and explicitly disable all unused browser hardware/sensor APIs.
Long-Term Strategic Fix:	Enforce `Permissions-Policy` alongside CSP in automated HTTP security header regression tests.
Verification Command:	Run `curl -I http://<target>/` and confirm the `Permissions-Policy` header is present.

F-70: HTTP Referrer-Policy Header Missing [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: 80) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated

Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-REFERRER-POLICY | Advisory: W3C Referrer Policy / OWASP Secure Headers | CWE: CWE-200 | MITRE: T1592 (Reconnaissance — Gather Victim Host Information) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): http://192.168.195.139:80/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP responses omit the 'Referrer-Policy' header, which may cause full URLs (including query parameters) to be leaked in the Referer header during cross-origin navigation.
Security Risk:	LOW — Potential disclosure of sensitive URL paths or query parameters to third-party origins.
EvidenceCore V3 Record:	ID: PROV-F-70-579B019A38 | SHA-256: 579b019a383092aabdd09680... | HMAC-SHA256: 0795db15242e50cee89ef73c... | Module: AutoPentest-AegisProbe

Captured Technical Evidence (F-70):

```
Missing header: Referrer-Policy | CWE: CWE-200 | Payload:
```

Immediate Containment:	Add `Header always set Referrer-Policy "strict-origin-when-cross-origin"` in Apache `/etc/apache2/conf.d/security` or Nginx configuration.
Short-Term Hardening:	Ensure sensitive tokens or session identifiers are never passed in URL query strings where they could leak via Referer headers.
Long-Term Strategic Fix:	Include `Referrer-Policy` in baseline web security header automated tests.
Verification Command:	Run `curl -I http://<target>/` and confirm `Referrer-Policy: strict-origin-when-cross-origin` is returned.

F-71: HTTP Server Banner Version Disclosure ('Server' Header) [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: 80) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-SERVER-BANNER | Advisory: Apache ServerTokens Prod / CIS Web Server Hardening | CWE: CWE-200 | MITRE: T1592 (Reconnaissance — Gather Victim Host Information) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 3 raw observation(s) across 2 unique endpoint(s): 192.168.195.139, http://192.168.195.139:80/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE

Technical Description:	The web server discloses exact software and OS release identifiers (e.g., 'Apache/2.2.8 (Ubuntu) DAV/2') in the HTTP 'Server' response header across multiple endpoints.
Security Risk:	LOW — Facilitates precise vulnerability fingerprinting by external adversaries.
EvidenceCore V3 Record:	ID: PROV-F-71-8AE350E466 | SHA-256: 8ae350e46676fc9ef04af9bf... | HMAC-SHA256: 86b3fe27811ffb06adf11423... | Module: AutoPentest-ZAP, XStrike-BuiltinVuln, AutoPentest-AegisProbe

Captured Technical Evidence (F-71):

```
The web/application server is leaking version information via the "Server" HTTP response header. Access
to such information may facilitate attackers identifying other vulnerabilities your web/applicat
---
Server header exposes version: Apache/2.2.8 (Ubuntu) DAV/2

Evidence: Server: Apache/2.2.8 (Ubuntu) DAV/2
---
Server: Apache/2.2.8 (Ubuntu) DAV/2 | CWE: CWE-200 | Payload:
```

Immediate Containment:	Set `ServerTokens Prod` and `ServerSignature Off` in Apache `/etc/apache2/apache2.conf` and reload Apache (`service apache2 reload`).
Short-Term Hardening:	Configure reverse-proxy header sanitization to strip detailed upstream server version banners.
Long-Term Strategic Fix:	Maintain automated configuration management to prevent verbose banner headers in production.
Verification Command:	Run `curl -I http://<target>/` and confirm `Server: Apache` returns no version or OS string.

F-72: HTTP Strict-Transport-Security (HSTS) Header Missing [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: 80) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-HSTS-MISSING | Advisory: RFC 6797 / OWASP HTTP Strict Transport Security Cheat Sheet | CWE: CWE-319 | MITRE: T1557 (Credential Access — Adversary-in-the-Middle) | Taxonomy: OWASP-A02:2021 (Cryptographic Failures) | CWE-319 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 3 raw observation(s) across 3 unique endpoint(s): 192.168.195.139:80, http://192.168.195.139:80/dvwa/login.php, http://192.168.195.139:80/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP/HTTPS responses omit the 'Strict-Transport-Security' header, allowing clients to connect over unencrypted HTTP and exposing sessions to SSL-stripping attacks.
Security Risk:	LOW — Enables protocol downgrade and cleartext session interception on untrusted networks.
EvidenceCore V3 Record:	ID: PROV-F-72-5F53D90274 | SHA-256: 5f53d90274571ce29044085e... | HMAC-SHA256: 912238214a4b1e094387c871... | Module: XStrike-BuiltinVuln, AutoPentest-OWASP, AutoPentest-AegisProbe

Captured Technical Evidence (F-72):

```
A02 Cryptographic Failures: 2 finding(s)
[CWE-319] Login over HTTP: /dvwa/login.php
[CWE-693] Missing Security Headers: Strict-Transport-Security, X-Content-Type-Options, X-Frame-Options
---
No HSTS header — susceptible to SSL stripping attacks

Evidence: HTTP response from 192.168.195.139:80 missing strict-transport-security header
---
Missing header: Strict-Transport-Security | CWE: CWE-319 | Payload:
```

Immediate Containment:	Configure TLS (HTTPS) on port 443 and add `Header always set Strict-Transport-Security "max-age=31536000; includeSubDomains"` to the HTTPS virtual host.
Short-Term Hardening:	Configure HTTP port 80 to issue permanent `301 Moved Permanently` redirects to `https://` for all endpoints.
Long-Term Strategic Fix:	Standardize internal PKI certificate automation (ACME/step-ca) across all web services.
Verification Command:	Inspect HTTPS response headers via `curl -I https://<target>/` and confirm `Strict-Transport-Security` is returned with `max-age>=31536000`.

F-73: HTTP Technology Stack Disclosure ('X-Powered-By' Header) [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: 80) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-XPOWEREDBY | Advisory: PHP expose_php=Off Hardening Guideline | CWE: CWE-200 | MITRE: T1592 (Reconnaissance — Gather Victim Host Information) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 2 unique endpoint(s): 192.168.195.139, http://192.168.195.139:80/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	The web application discloses backend runtime versions (e.g., 'X-Powered-By: PHP/5.2.4-2ubuntu5.10') in HTTP response headers.
Security Risk:	LOW — Reveals exact application framework version to aid targeted exploit selection.
EvidenceCore V3 Record:	ID: PROV-F-73-EFEC47D1D0 | SHA-256: efec47d1d0c5ada5e6e86851... | HMAC-SHA256: 4f0dff00a02fadee7023b14b... | Module: AutoPentest-ZAP, AutoPentest-AegisProbe

Captured Technical Evidence (F-73):

```
The web/application server is leaking information via one or more "X-Powered-By" HTTP response headers.
Access to such information may facilitate attackers identifying other frameworks/components your
---
X-Powered-By: PHP/5.2.4-2ubuntu5.10 | CWE: CWE-200 | Payload:
```

Immediate Containment:	Set `expose_php = Off` in `/etc/php5/apache2/php.ini` and add `Header unset X-Powered-By` in Apache configuration.
Short-Term Hardening:	Audit application middleware to suppress framework version headers (`X-AspNet-Version`, `X-Generator`).
Long-Term Strategic Fix:	Enforce header stripping at the edge reverse proxy.
Verification Command:	Run `curl -I http://<target>/` and confirm the `X-Powered-By` header is completely absent.

F-74: Web Server Directory Listing Enabled (/icons/ & Static Directories) [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-WEB-DIR-LISTING | Advisory: Apache Options -Indexes Hardening Guideline | CWE: CWE-548 | MITRE: T1083 (Discovery — File and Directory Discovery) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-548 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): http://192.168.195.139:80/icons/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Web server directories (such as /icons/) return an auto-generated directory index listing internal files (consolidated with OWASP A05 misconfiguration checks).
Security Risk:	LOW — Enables unauthenticated enumeration of static and backup files.
EvidenceCore V3 Record:	ID: PROV-F-74-3EFEE9B1FC | SHA-256: 3efee9b1fcc53b893b2e97fc... | HMAC-SHA256: e1cdef846405f963240884be... | Module: XStrike-BuiltinVuln

Captured Technical Evidence (F-74):

```
Sensitive path accessible: /icons/

Evidence: HTTP 200 on http://192.168.195.139:80/icons/
```

Immediate Containment:	Disable directory indexing in Apache `/etc/apache2/apache2.conf` and `/etc/apache2/mods-enabled/alias.conf` by setting `Options -Indexes` on `/var/www/` and `/usr/share/apache2/icons/`.
Short-Term Hardening:	Remove default Apache `/icons/` alias or restrict it to specific file extensions.
Long-Term Strategic Fix:	Enforce `Options -Indexes` in baseline web server hardening templates.
Verification Command:	Request `http://192.168.195.139:80/icons/` and confirm HTTP 403 Forbidden instead of `Index of /icons`.

F-75: X-Content-Type-Options Header Missing (MIME-Sniffing Exposure) [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.139 (Ports: 80) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-XCTO-MISSING | Advisory: OWASP Secure Headers Project — X-Content-Type-Options | CWE: CWE-693 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-693 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 3 raw observation(s) across 3 unique endpoint(s): 192.168.195.139, 192.168.195.139:80, http://192.168.195.139:80/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP responses omit 'X-Content-Type-Options: nosniff', allowing browsers to MIME-sniff response bodies into executable content types.
Security Risk:	LOW — Browser MIME-type guessing can facilitate content-spoofing or stored XSS in uploaded content.
EvidenceCore V3 Record:	ID: PROV-F-75-D09020B64D | SHA-256: d09020b64d59fb302c5fcbc9... | HMAC-SHA256: f19288f6e1827a69f2d6c532... | Module: AutoPentest-ZAP, XStrike-BuiltinVuln, AutoPentest-AegisProbe

Captured Technical Evidence (F-75):

```
The Anti-MIME-Sniffing header X-Content-Type-Options was not set to 'nosniff'. This allows older
versions of Internet Explorer and Chrome to perform MIME-sniffing on the response body, potentially cau
---
No X-Content-Type-Options header — MIME sniffing not prevented

Evidence: HTTP response from 192.168.195.139:80 missing x-content-type-options header
---
Missing header: X-Content-Type-Options | CWE: CWE-693 | Payload:
```

Immediate Containment:	Add `Header always set X-Content-Type-Options "nosniff"` in the web server configuration (`apache2.conf` / `nginx.conf`).
Short-Term Hardening:	Ensure all static and dynamic responses set explicit, accurate `Content-Type` MIME headers.
Long-Term Strategic Fix:	Include `X-Content-Type-Options: nosniff` in baseline infrastructure-as-code web templates.
Verification Command:	Run `curl -I http://<target>/` and verify `X-Content-Type-Options: nosniff` is present on all responses.

F-76: HTTP Permissions-Policy Header Missing [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.140 (Ports: 8080) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC

7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-PERMISSIONS-POLICY | Advisory: W3C Permissions Policy Specification | CWE: CWE-693 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-693 (Protection Mechanism Failure) | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): http://192.168.195.140:8080/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP responses omit the 'Permissions-Policy' (formerly Feature-Policy) header, leaving browser APIs (camera, microphone, geolocation, payment) unrestricted at the document policy layer.
Security Risk:	LOW — Defense-in-depth browser hardening omission; does not directly enable remote code execution.
EvidenceCore V3 Record:	ID: PROV-F-76-859BCC6ACE | SHA-256: 859bcc6ace55a30fb803a68a... | HMAC-SHA256: 227ead76130e182ac0f8174c... | Module: AutoPentest-AegisProbe

Captured Technical Evidence (F-76):

```
Missing header: Permissions-Policy | CWE: CWE-732 | Payload:
```

Immediate Containment:	Add `Header always set Permissions-Policy "geolocation=(), microphone=(), camera=(), payment=(), usb=()"` in the web server configuration.
Short-Term Hardening:	Audit frontend application requirements and explicitly disable all unused browser hardware/sensor APIs.
Long-Term Strategic Fix:	Enforce `Permissions-Policy` alongside CSP in automated HTTP security header regression tests.
Verification Command:	Run `curl -I http://<target>/` and confirm the `Permissions-Policy` header is present.

F-77: HTTP Referrer-Policy Header Missing [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.140 (Ports: 8080) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-REFERRER-POLICY | Advisory: W3C Referrer Policy / OWASP Secure Headers | CWE: CWE-200 | MITRE: T1592 (Reconnaissance — Gather Victim Host Information) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N

Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 1 raw observation(s) across 1 unique endpoint(s): http://192.168.195.140:8080/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP responses omit the 'Referrer-Policy' header, which may cause full URLs (including query parameters) to be leaked in the Referer header during cross-origin navigation.
Security Risk:	LOW — Potential disclosure of sensitive URL paths or query parameters to third-party origins.
EvidenceCore V3 Record:	ID: PROV-F-77-347C8D84F7 | SHA-256: 347c8d84f75530af8bc97aa9... | HMAC-SHA256: 22bda313a71555e238b168e4... | Module: AutoPentest-AegisProbe

Captured Technical Evidence (F-77):

```
Missing header: Referrer-Policy | CWE: CWE-200 | Payload:
```

Immediate Containment:	Add `Header always set Referrer-Policy "strict-origin-when-cross-origin"` in Apache `/etc/apache2/conf.d/security` or Nginx configuration.
Short-Term Hardening:	Ensure sensitive tokens or session identifiers are never passed in URL query strings where they could leak via Referer headers.
Long-Term Strategic Fix:	Include `Referrer-Policy` in baseline web security header automated tests.
Verification Command:	Run `curl -I http://<target>/` and confirm `Referrer-Policy: strict-origin-when-cross-origin` is returned.

F-78: HTTP Server Banner Version Disclosure ('Server' Header) [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.140 (Ports: 8080) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-SERVER-BANNER | Advisory: Apache ServerTokens Prod / CIS Web Server Hardening | CWE: CWE-200 | MITRE: T1592 (Reconnaissance — Gather Victim Host Information) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-200 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 17 raw observation(s) across 2 unique endpoint(s): 192.168.195.140, http://192.168.195.140:8080/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	The web server discloses exact software and OS release identifiers (e.g., 'Apache/2.2.8 (Ubuntu) DAV/2') in the HTTP 'Server' response header across multiple endpoints.
Security Risk:	LOW — Facilitates precise vulnerability fingerprinting by external adversaries.

EvidenceCore V3 Record:	ID: PROV-F-78-6FC41A121F | SHA-256: 6fc41a121f179c53ca5ba8c4... | HMAC-SHA256: 019616e03dd63cd496592078... | Module: AutoPentest-ZAP, XStrike-BuiltinVuln, AutoPentest-AegisProbe

Captured Technical Evidence (F-78):

```
The web/application server is leaking version information via the "Server" HTTP response header. Access
to such information may facilitate attackers identifying other vulnerabilities your web/applicat
---
Server header exposes version: Apache-Coyote/1.1

Evidence: Server: Apache-Coyote/1.1
---
Server: Apache-Coyote/1.1 | CWE: CWE-200 | Payload:
```

Immediate Containment:	Set `ServerTokens Prod` and `ServerSignature Off` in Apache `/etc/apache2/apache2.conf` and reload Apache (`service apache2 reload`).
Short-Term Hardening:	Configure reverse-proxy header sanitization to strip detailed upstream server version banners.
Long-Term Strategic Fix:	Maintain automated configuration management to prevent verbose banner headers in production.
Verification Command:	Run `curl -I http://<target>/` and confirm `Server: Apache` returns no version or OS string.

F-79: HTTP Strict-Transport-Security (HSTS) Header Missing [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.140 (Ports: 8080) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-HSTS-MISSING | Advisory: RFC 6797 / OWASP HTTP Strict Transport Security Cheat Sheet | CWE: CWE-319 | MITRE: T1557 (Credential Access — Adversary-in-the-Middle) | Taxonomy: OWASP-A02:2021 (Cryptographic Failures) | CWE-319 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 4 raw observation(s) across 3 unique endpoint(s): 192.168.195.140:8080, 192.168.195.140, http://192.168.195.140:8080/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP/HTTPS responses omit the 'Strict-Transport-Security' header, allowing clients to connect over unencrypted HTTP and exposing sessions to SSL-stripping attacks.
Security Risk:	LOW — Enables protocol downgrade and cleartext session interception on untrusted networks.
EvidenceCore V3 Record:	ID: PROV-F-79-1E6AC0F508 | SHA-256: 1e6ac0f5087f8a6f4df7b146... | HMAC-SHA256: 173a3ef51d1d88d99984ed1f... | Module: XStrike-BuiltinVuln, AutoPentest-OWASP, AutoPentest-AegisProbe

Captured Technical Evidence (F-79):

```
No HSTS header — susceptible to SSL stripping attacks

Evidence: HTTP response from 192.168.195.140:8080 missing strict-transport-security header
---
```

```
A02 Cryptographic Failures: 1 finding(s)
[CWE-693] Missing Security Headers: Strict-Transport-Security, X-Content-Type-Options, X-Frame-Options
---
Missing header: Strict-Transport-Security | CWE: CWE-319 | Payload:
```

Immediate Containment:	Configure TLS (HTTPS) on port 443 and add `Header always set Strict-Transport-Security "max-age=31536000; includeSubDomains"` to the HTTPS virtual host.
Short-Term Hardening:	Configure HTTP port 80 to issue permanent `301 Moved Permanently` redirects to `https://` for all endpoints.
Long-Term Strategic Fix:	Standardize internal PKI certificate automation (ACME/step-ca) across all web services.
Verification Command:	Inspect HTTPS response headers via `curl -I https://<target>/` and confirm `Strict-Transport-Security` is returned with `max-age>=31536000`.

F-80: Session Cookie Missing 'HttpOnly' Flag [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.140 (Ports: N/A) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-COOKIE-HTTPONLY | Advisory: OWASP Session Management Cheat Sheet — HttpOnly | CWE: CWE-1004 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-1004 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 5 raw observation(s) across 1 unique endpoint(s): 192.168.195.140
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Cookies are set without the 'HttpOnly' flag, permitting client-side JavaScript ('document.cookie') to read session identifiers.
Security Risk:	LOW — Enables session theft if combined with a Cross-Site Scripting (XSS) vulnerability.
EvidenceCore V3 Record:	ID: PROV-F-80-29FCC5B7B6 | SHA-256: 29fcc5b7b604e3b51d512c9f... | HMAC-SHA256: 9d2bd27eaf28d028f68ed3d0... | Module: AutoPentest-ZAP

Captured Technical Evidence (F-80):

```
A cookie has been set without the HttpOnly flag, which means that the cookie can be accessed by
JavaScript. If a malicious script can be run on this page then the cookie will be accessible and can be
```

Immediate Containment:	Set `session.cookie_httponly = 1` in `/etc/php5/apache2/php.ini` and add `HttpOnly` to all application `setcookie()` calls.
Short-Term Hardening:	Add Apache header rewrite `Header edit Set-Cookie ^(.*)$ "$1; HttpOnly"` for legacy applications.
Long-Term Strategic Fix:	Enforce automated DAST checks verifying `HttpOnly` on all session identifiers.
Verification Command:	Inspect `Set-Cookie: PHPSESSID=...` via `curl -I` and confirm the `HttpOnly` flag is present.

F-81: Session Cookie Missing 'SameSite' Attribute [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.140 (Ports: N/A) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-COOKIE-SAMESITE | Advisory: RFC 6265bis — SameSite Cookie Attribute | CWE: CWE-352 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A01:2021 (Broken Access Control) | CWE-352 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 5 raw observation(s) across 1 unique endpoint(s): 192.168.195.140
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	Session cookies are issued without the 'SameSite=Lax' or 'SameSite=Strict' attribute, allowing cookies to be transmitted in cross-origin requests.
Security Risk:	LOW — Weakens browser defense against Cross-Site Request Forgery (CSRF).
EvidenceCore V3 Record:	ID: PROV-F-81-35B82920C6 | SHA-256: 35b82920c6606a7bc4d4eab0... | HMAC-SHA256: b46bfd05740437d03d28141f... | Module: AutoPentest-ZAP

Captured Technical Evidence (F-81):

```
A cookie has been set without the SameSite attribute, which means that the cookie can be sent as a result of a 'cross-site' request. The SameSite attribute is an effective counter measure to cross-sit
```

Immediate Containment:	Update PHP/application session configuration (`session.cookie_samesite = "Lax"` or `"Strict"`) or add Apache `Header edit Set-Cookie ^(.*)$ "$1; SameSite=Lax"`.
Short-Term Hardening:	Audit cross-site authentication flows and enforce `SameSite=Strict` on high-privilege session cookies.
Long-Term Strategic Fix:	Standardize session management using framework-managed `__Host-` prefixed cookies.
Verification Command:	Inspect `Set-Cookie` via `curl -I` and confirm `SameSite=Lax` or `SameSite=Strict` is present.

F-82: X-Content-Type-Options Header Missing (MIME-Sniffing Exposure) [LOW · CVSS 3.1 · VALIDATED]

Target Asset & Port:	192.168.195.140 (Ports: 8080) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)
Proof State & Scope:	VALIDATED (Legacy Tier: VALIDATED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: SAFE_PROBE_EXECUTED | Auth: UNAUTHENTICATED_PROBE | Exploitability: CONFIGURATION_WEAKNESS | Privilege: NONE | DataAccess: METADATA_OR_HEADERS_ONLY | Impact: DEFENSE_IN_DEPTH_DEGRADATION | EvidenceConf: 94%
Access Scope Summary:	Technical configuration, protocol flag, or HTTP response header/cookie directly validated

Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-HDR-XCTO-MISSING | Advisory: OWASP Secure Headers Project — X-Content-Type-Options | CWE: CWE-693 | MITRE: T1190 (Initial Access — Exploit Public-Facing Application) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-693 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=95% | ExploitabilityConf=25% | ImpactConf=30% | 5-D Vector: EvidenceConf=95% (Det=96%, Val=92%, Art=100%) | ExploitabilityConf=25% | ImpactConf=30% | FP Band: Tier 2 — Deterministically Validated by Protocol/HTTP Response (Exploitability Separate)
Deduplication & Endpoints:	Consolidated 5 raw observation(s) across 3 unique endpoint(s): 192.168.195.140, 192.168.195.140:8080, http://192.168.195.140:8080/
Root Cause & Compliance:	Web Server & HTTP Security Header Hardening Gap [DIRECTLY_ESTABLISHED] | Compliance Status: EVIDENCE_OF_CONTROL_FAILURE
Technical Description:	HTTP responses omit 'X-Content-Type-Options: nosniff', allowing browsers to MIME-sniff response bodies into executable content types.
Security Risk:	LOW — Browser MIME-type guessing can facilitate content-spoofing or stored XSS in uploaded content.
EvidenceCore V3 Record:	ID: PROV-F-82-A8D0168146 | SHA-256: a8d01681466eacf9f463f19f... | HMAC-SHA256: f81b3374399af9e4dca330e8... | Module: AutoPentest-ZAP, XStrike-BuiltinVuln, AutoPentest-AegisProbe

Captured Technical Evidence (F-82):

```
The Anti-MIME-Sniffing header X-Content-Type-Options was not set to 'nosniff'. This allows older
versions of Internet Explorer and Chrome to perform MIME-sniffing on the response body, potentially cau
---
No X-Content-Type-Options header — MIME sniffing not prevented

Evidence: HTTP response from 192.168.195.140:8080 missing x-content-type-options header
---
Missing header: X-Content-Type-Options | CWE: CWE-693 | Payload:
```

Immediate Containment:	Add `Header always set X-Content-Type-Options "nosniff"` in the web server configuration (`apache2.conf` / `nginx.conf`).
Short-Term Hardening:	Ensure all static and dynamic responses set explicit, accurate `Content-Type` MIME headers.
Long-Term Strategic Fix:	Include `X-Content-Type-Options: nosniff` in baseline infrastructure-as-code web templates.
Verification Command:	Run `curl -I http://<target>/` and verify `X-Content-Type-Options: nosniff` is present on all responses.

F-83: SMB NetBIOS/CIFS Service Enumeration Surface (Ports 139/445) [LOW · CVSS 3.1 · OBSERVED]

Target Asset & Port:	192.168.195.130 (Ports: N/A) | Asset Class: Workstation / SMB Peer Host (STANDARD)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: SERVICE_BANNER_ONLY | Impact: EXPOSURE_ONLY | EvidenceConf: 42%
Access Scope Summary:	Service version or banner matched to known vulnerability; active exploitation not demonstrated in this record
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-SMB-ENUM-SURFACE | Advisory: CIS Benchmark — Restrict Anonymous SMB/NetBIOS Enumeration | CWE: CWE-287 | MITRE: T1021.002 (Lateral Movement — Remote Services: SMB/Windows Admin Shares) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-287 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N

Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 2 unique endpoint(s): 192.168.195.130:139, 192.168.195.130:445
Root Cause & Compliance:	Network Segmentation & Perimeter Filter Exposure [INFERRED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	SMB ports 139/445 were probed via Nmap NSE (smb-enum-shares, smb-enum-users). Script output did not return unauthenticated share contents or user RIDs; classified as OBSERVED service exposure.
Security Risk:	LOW (OBSERVED) — Exposed SMB enumeration surface; restrict anonymous/null session queries.
EvidenceCore V3 Record:	ID: PROV-F-83-0D103A8DC4 | SHA-256: 0d103a8dc40dbe61ce4c6ae5... | HMAC-SHA256: abb1caf732a3cb26f4fac19d... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-83):

```
SMB Enum: Nmap scan report for 192.168.195.130
Host is up (0.00s latency).

PORT STATE SERVICE
139/tcp open netbios-ssn
MAC Address: 00:0C:29:73:FE:EC (VMware)

Host script results:
|_smb-enum-shares: ERROR: Script execution failed (use -d to debug)
|_smb-enum-users: ERROR: Script execution failed (use -d to debug)
|_smb-os-discovery: ERR
---
SMB Enum: Nmap scan report for 192.168.195.130
Host is up (0.00013s latency).

PORT STATE SERVICE
445/tcp open microsoft-ds
MAC Address: 00:0C:29:73:FE:EC (VMware)

Host script results:
|_smb-enum-shares: ERROR: Script execution failed (use -d to debug)
|_smb-enum-groups: ERROR: Script execution
```

Immediate Containment:	Restrict anonymous/null SMB sessions by setting `restrict anonymous = 2` and `map to guest = Never` in `/etc/samba/smb.conf`, then reload `smbd`.
Short-Term Hardening:	Block TCP ports 139 and 445 from untrusted internal subnets at the host firewall.
Long-Term Strategic Fix:	Audit all Samba/Windows shares to ensure explicit group ACLs are required for share enumeration.
Verification Command:	Run `rpcclient -U "" -N <target>` and `smbclient -L //<target>/ -N` and verify anonymous enumeration is rejected.

F-84: SMB NetBIOS/CIFS Service Enumeration Surface (Ports 139/445) [LOW · CVSS 3.1 · OBSERVED]

Target Asset & Port:	192.168.195.139 (Ports: N/A) | Asset Class: Multi-Role Application & Database Server (CROWN_JEWEL)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: SERVICE_BANNER_ONLY | Impact: EXPOSURE_ONLY | EvidenceConf: 42%

Access Scope Summary:	Service version or banner matched to known vulnerability; active exploitation not demonstrated in this record
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-SMB-ENUM-SURFACE | Advisory: CIS Benchmark — Restrict Anonymous SMB/NetBIOS Enumeration | CWE: CWE-287 | MITRE: T1021.002 (Lateral Movement — Remote Services: SMB/Windows Admin Shares) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-287 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 2 unique endpoint(s): 192.168.195.139:139, 192.168.195.139:445
Root Cause & Compliance:	Network Segmentation & Perimeter Filter Exposure [INFERRED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	SMB ports 139/445 were probed via Nmap NSE (smb-enum-shares, smb-enum-users). Script output did not return unauthenticated share contents or user RIDs; classified as OBSERVED service exposure.
Security Risk:	LOW (OBSERVED) — Exposed SMB enumeration surface; restrict anonymous/null session queries.
EvidenceCore V3 Record:	ID: PROV-F-84-F67B245A4B | SHA-256: f67b245a4b7138b21f25042b... | HMAC-SHA256: 09f532d7e5df01083cac7475... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-84):

```
SMB Enum: Nmap scan report for 192.168.195.139
Host is up (0.00s latency).

PORT STATE SERVICE
139/tcp open netbios-ssn
MAC Address: 00:0C:29:05:D7:6A (VMware)

Host script results:
|_smb-enum-users: ERROR: Script execution failed (use -d to debug)
|_smb-os-discovery: ERROR: Script execution failed (use -d to debug)
|_smb-enum-groups: ERR
---
SMB Enum: Nmap scan report for 192.168.195.139
Host is up (0.00s latency).

PORT STATE SERVICE
445/tcp open microsoft-ds
MAC Address: 00:0C:29:05:D7:6A (VMware)

Host script results:
|_smb-enum-users: ERROR: Script execution failed (use -d to debug)
|_smb-enum-groups: ERROR: Script execution fail
```

Immediate Containment:	Restrict anonymous/null SMB sessions by setting `restrict anonymous = 2` and `map to guest = Never` in `/etc/samba/smb.conf`, then reload `smbd`.
Short-Term Hardening:	Block TCP ports 139 and 445 from untrusted internal subnets at the host firewall.
Long-Term Strategic Fix:	Audit all Samba/Windows shares to ensure explicit group ACLs are required for share enumeration.
Verification Command:	Run `rpcclient -U "" -N <target>` and `smbclient -L //<target>/ -N` and verify anonymous enumeration is rejected.

F-85: SMB NetBIOS/CIFS Service Enumeration Surface (Ports 139/445) [LOW · CVSS 3.1 · OBSERVED]

Target Asset & Port:	192.168.195.140 (Ports: N/A) | Asset Class: Application & Web Middleware Server (HIGH_VALUE)
Proof State & Scope:	OBSERVED (Legacy Tier: OBSERVED) | Privilege: NONE | Auth: False | File: NONE | DB: NONE | Privesc Lifecycle: NOT_APPLICABLE_NON_PRIVESC
7-D Proof Vector (V3):	Execution: BANNER_ONLY | Auth: UNAUTHENTICATED_PROBE | Exploitability: UNVERIFIED_CANDIDATE | Privilege: NONE | DataAccess: SERVICE_BANNER_ONLY | Impact: EXPOSURE_ONLY | EvidenceConf: 42%
Access Scope Summary:	Service version or banner matched to known vulnerability; active exploitation not demonstrated in this record
Taxonomy & CVSS v3.1:	CVE: N/A | Custom ID: XSTRIKE-SMB-ENUM-SURFACE | Advisory: CIS Benchmark — Restrict Anonymous SMB/NetBIOS Enumeration | CWE: CWE-287 | MITRE: T1021.002 (Lateral Movement — Remote Services: SMB/Windows Admin Shares) | Taxonomy: OWASP-A05:2021 (Security Misconfiguration) | CWE-287 | Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:L/I:N/A:N
Separated Confidence (V3):	EvidenceConf=48% | ExploitabilityConf=15% | ImpactConf=22% | 5-D Vector: EvidenceConf=48% (Det=86%, Val=25%, Art=45%) | ExploitabilityConf=15% | ImpactConf=22% | FP Band: Tier 3 — Unverified Banner/Port Heuristic (Manual Verification Required)
Deduplication & Endpoints:	Consolidated 2 raw observation(s) across 2 unique endpoint(s): 192.168.195.140:139, 192.168.195.140:445
Root Cause & Compliance:	Network Segmentation & Perimeter Filter Exposure [INFERRED] | Compliance Status: POTENTIAL_CONTROL_WEAKNESS
Technical Description:	SMB ports 139/445 were probed via Nmap NSE (smb-enum-shares, smb-enum-users). Script output did not return unauthenticated share contents or user RIDs; classified as OBSERVED service exposure.
Security Risk:	LOW (OBSERVED) — Exposed SMB enumeration surface; restrict anonymous/null session queries.
EvidenceCore V3 Record:	ID: PROV-F-85-A931980144 | SHA-256: a931980144775413cd7d5848... | HMAC-SHA256: 7a4730b805f10e6398dcabee... | Module: AutoPentest-Exploit

Captured Technical Evidence (F-85):

```
SMB Enum: Nmap scan report for 192.168.195.140
Host is up (0.00s latency).

PORT STATE SERVICE
139/tcp open netbios-ssn
MAC Address: 00:0C:29:86:5E:7D (VMware)

Host script results:
|_smb-enum-shares: ERROR: Script execution failed (use -d to debug)
---
SMB Enum: Nmap scan report for 192.168.195.140
Host is up (0.00s latency).

PORT STATE SERVICE
445/tcp open microsoft-ds
MAC Address: 00:0C:29:86:5E:7D (VMware)

Host script results:
|_smb-enum-shares: ERROR: Script execution failed (use -d to debug)
```

Immediate Containment:	Restrict anonymous/null SMB sessions by setting `restrict anonymous = 2` and `map to guest = Never` in `/etc/samba/smb.conf`, then reload `smbd`.
Short-Term Hardening:	Block TCP ports 139 and 445 from untrusted internal subnets at the host firewall.
Long-Term Strategic Fix:	Audit all Samba/Windows shares to ensure explicit group ACLs are required for share enumeration.

Verification Command:	Run `rpcclient -U "" -N <target>` and `smbclient -L //<target>/ -N` and verify anonymous enumeration is rejected.

3B. Consolidated Reconnaissance & Attack Surface Inventory (R-01 through R-25)

To prevent reconnaissance telemetry from inflating vulnerability or confirmed-exploitation counts, all **25 consolidated surface discovery groups** (representing **603** raw discovery observations, including `ZAP: User Agent Fuzzer`, `ZAP: Modern Web Application`, `JavaScript Source File Discovered`, `Postfix SMTP Banner`, `Open Port`, and `Host Discovered` records) are isolated below with `Proof State = DISCOVERED` and `CVSS = 0.0`.

Recon ID	Host	Canonical Discovery Class	Normalized Title	Raw Observations Consolidated	Unique Endpoints / Ports	Provenance ID
R-01	192.168.195.1	RECON_HOST_DISCOVERED	Active Network Host Discovered (ICMP/ARP/TCP Probe)	1	1	PROV-R-01-4DF732E1A8
R-02	192.168.195.1	RECON_CRED_SURFACE	Authentication Service Surface Identified (FTP/SSH/SMB)	1	1	PROV-R-02-498CF90543
R-03	192.168.195.1	RECON_WEB_SURFACE	HTTP/HTTPS Web Application Surface & Endpoints Enumerated	3	3	PROV-R-03-9466C79F96
R-04	192.168.195.1	RECON_ZAP_MODERN_WEB_APP	OWASP ZAP Informational Telemetry: Modern Web Application Fingerprinted	75	1	PROV-R-04-2514D97FFD
R-05	192.168.195.1	RECON_ZAP_USER_AGENT_FUZZER	OWASP ZAP Informational Telemetry: User-Agent Response Differential Probe	420	1	PROV-R-05-4C2218D36A
R-06	192.168.195.1	RECON_OPEN_PORT	Open Network Service Port Enumerated	16	16	PROV-R-06-062135D8C7
R-07	192.168.195.129	RECON_HOST_DISCOVERED	Active Network Host Discovered (ICMP/ARP/TCP Probe)	1	1	PROV-R-07-332D21C307
R-08	192.168.195.129	RECON_OPEN_PORT	Open Network Service Port Enumerated	2	2	PROV-R-08-A576618718
R-09	192.168.195.130	RECON_HOST_DISCOVERED	Active Network Host Discovered (ICMP/ARP/TCP Probe)	1	1	PROV-R-09-B269E35E96
R-10	192.168.195.130	RECON_CRED_SURFACE	Authentication Service Surface Identified (FTP/SSH/SMB)	1	1	PROV-R-10-45753A0067
R-11	192.168.195.130	RECON_OPEN_PORT	Open Network Service Port Enumerated	4	4	PROV-R-11-B4D9725EAD
R-12	192.168.195.131	RECON_HOST_DISCOVERED	Active Network Host Discovered (ICMP/ARP/TCP Probe)	1	1	PROV-R-12-A33B5ED1ED
R-13	192.168.195.139	RECON_HOST_DISCOVERED	Active Network Host Discovered (ICMP/ARP/TCP Probe)	2	1	PROV-R-13-279CDB8950
R-14	192.168.195.139	RECON_CRED_SURFACE	Authentication Service Surface Identified (FTP/SSH/SMB)	1	1	PROV-R-14-DFC555AA54

Recon ID	Host	Canonical Discovery Class	Normalized Title	Raw Observations Consolidated	Unique Endpoints / Ports	Provenance ID
R-15	192.168.195.139	RECON_JS_ASSET	Client-Side JavaScript Source Asset Enumerated	5	5	PROV-R-15-3852261785
R-16	192.168.195.139	RECON_WEB_SURFACE	HTTP/HTTPS Web Application Surface & Endpoints Enumerated	2	2	PROV-R-16-F151D623E4
R-17	192.168.195.139	RECON_OPEN_PORT	Open Network Service Port Enumerated	24	24	PROV-R-17-55A5019535
R-18	192.168.195.139	RECON_SMTP_BANNER	Postfix SMTP Service Banner Enumerated (Port 25)	1	1	PROV-R-18-5C22434893
R-19	192.168.195.140	RECON_HOST_DISCOVERED	Active Network Host Discovered (ICMP/ARP/TCP Probe)	2	1	PROV-R-19-4EF5E83C8A
R-20	192.168.195.140	RECON_CRED_SURFACE	Authentication Service Surface Identified (FTP/SSH/SMB)	1	1	PROV-R-20-609DA36E96
R-21	192.168.195.140	RECON_WEB_SURFACE	HTTP/HTTPS Web Application Surface & Endpoints Enumerated	4	4	PROV-R-21-E71A4FB22A
R-22	192.168.195.140	RECON_ZAP_MODERN_WEB_APP	OWASP ZAP Informational Telemetry: Modern Web Application Fingerprinted	2	1	PROV-R-22-B0CB13465D
R-23	192.168.195.140	RECON_ZAP_USER_AGENT_FUZZER	OWASP ZAP Informational Telemetry: User-Agent Response Differential Probe	24	1	PROV-R-23-54884FB1B4
R-24	192.168.195.140	RECON_OPEN_PORT	Open Network Service Port Enumerated	8	8	PROV-R-24-39D254F68A
R-25	192.168.195.254	RECON_HOST_DISCOVERED	Active Network Host Discovered (ICMP/ARP/TCP Probe)	1	1	PROV-R-25-9EC1080559

4. Four-Stage Evidence-Backed Attack Graph V3, 5-State Taxonomy & Choke-Point Analysis

Path Count Reconciliation: The raw graph generator enumerated 98 port-level path permutations (0 high/critical permutations across open ports). AttackGraph V3 deduplicated these by underlying CVE/CWE and proof state into 6 distinct attack chains: 3 EXECUTED (2 Full Root Compromises on 192.168.195.139 + 1 DB Superuser Auth), 2 PARTIALLY_VALIDATED, and 1 INFERRED_NOT_EXECUTED, plus 2 BLOCKED_BY_CONTROLS host paths and 3 NOT_TESTED_BY_SAFETY_ROE payloads, governed by 5 remediation choke points.

Attack Path Priority Score Formula (0–100 Priority Index, Not Statistical Exploitation Probability):

```
AttackPathPriorityScore(C) = min(100, round(BaseCVSS * 6.0 * W_exec + ImpactBonus + HopBonus)) [0-100 Priority Index, not statistical probability], where W_exec = 1.00 (Confirmed Full Root Compromise), 0.90 (Confirmed DB/Service Authentication), 0.70-0.85 (Partially Validated Foothold/Precondition), 0.35 (Inferred Port-Open Candidate), or 0.00 (Blocked by Controls).
```

Chain ID	Attack Chain Name	Target Host(s)	Validation State	7-Metric Path Vector (Feas / Conf / Priv / Imp / Hops / Lev)	Attack Path Priority Score
CHAIN-01	vsFTPd 2.3.4 Backdoor -> Unauthenticated Root Shell -> /etc/shadow Dump	192.168.195.139	CONFIRMED_EXECUTED	Feas:0.99 | Conf:99.0% | Priv:ROOT (uid=0) | Imp:CROWN_JEWEL_HOST_TAKEOVER | Hops:4 | Lev:9 raw paths broken by stopping vsFTPd / blocking TCP 21 & 6200 chains	100/100 (P0_IMMEDIATE)
CHAIN-02	Ingreslock TCP/1524 Bindshell -> Unauthenticated Root Shell -> Credential Extraction	192.168.195.139	CONFIRMED_EXECUTED	Feas:0.99 | Conf:99.0% | Priv:ROOT (uid=0) | Imp:CROWN_JEWEL_HOST_TAKEOVER | Hops:4 | Lev:8 raw paths broken by killing /bin/sh on TCP 1524 & disabling inetd chains	100/100 (P0_IMMEDIATE)
CHAIN-03	SSH Default Credentials (uid=1000) -> Validated SUID Nmap / DirtyCow PrivEsc -> Local Root	192.168.195.139	PARTIALLY_VALIDATED	Feas:0.88 | Conf:88.0% | Priv:USER (uid=1000) -> Projected ROOT (uid=0) | Imp:PRIVILEGE_ESCALATION_TO_ROOT | Hops:4 | Lev:7 raw paths + FTP/SSH credential reuse broken by rotating msfadmin & removing SUID bits chains	76/100 (P1_HIGH)
CHAIN-04	PostgreSQL 5432 Default Superuser (postgres:postgres) -> DB Catalog Read Access	192.168.195.139	CONFIRMED_EXECUTED	Feas:0.95 | Conf:95.0% | Priv:DB_SUPERUSER (postgres) | Imp:RELATIONAL_DATABASE_COMPROMISE | Hops:4 | Lev:5 raw DB paths broken by rotating postgres password & binding 5432 to localhost chains	78/100 (P0_IMMEDIATE)
CHAIN-05	NFS Export Surface & Web SQLi / DVWA -> Inferred Filesystem/DB Takeover	192.168.195.139	PARTIALLY_VALIDATED	Feas:0.78 | Conf:83.0% | Priv:WEB_APP_DB_READ -> Projected FILESYSTEM_ACCESS | Imp:APPLICATION_AND_DATA_EXPOSURE | Hops:4 | Lev:5 raw paths broken by stopping nfs-kernel-server & parameterizing SQL queries chains	63/100 (P1_HIGH)

Chain ID	Attack Chain Name	Target Host(s)	Validation State	7-Metric Path Vector (Feas / Conf / Priv / Imp / Hops / Lev)	Attack Path Priority Score
CHAIN-06	Internal Subnet SMB Port 445 Exposure -> Theoretical Lateral Movement (Unverified)	192.168.195.1, 192.168.195.130, 192.168.195.140	INFERRED_NOT_EXECUTED	Feas:0.25 | Conf:30.0% | Priv:NONE (Port-Open Candidate Only) | Imp:POTENTIAL_LATERAL_MOVEMENT | Hops:2 | Lev:63 raw port permutations broken by disabling SMBv1 & enforcing SMB signing chains	19/100 (P2_MEDIUM)

4.1 Five-State Attack Path Classification Model (Executed vs Validated vs Inferred vs Blocked vs Safety-RoE Held)

Path Classification State	Count	Chain IDs & W_exec Weight	Technical Definition & Execution Boundary
EXECUTED_PATHS	3	CHAIN-01 (vsFTPd Root), CHAIN-02 (Ingreslock Root), CHAIN-04 (PostgreSQL Superuser Auth) [W_exec = 1.00 (Full OS Compromise: CHAIN-01, CHAIN-02) | W_exec = 0.90 (Confirmed DB Superuser Auth: CHAIN-04)]	End-to-end command execution or database superuser authentication verified with captured session transcript.
PARTIALLY_VALIDATED_PATHS	2	CHAIN-03 (SSH uid=1000 + SUID/DirtyCow Vector), CHAIN-05 (NFS Export + Web SQLi/DVWA) [W_exec = 0.85 (Authenticated Shell + Validated PrivEsc Precondition) | W_exec = 0.70 (Validated Surface)]	Initial foothold or vulnerability precondition validated, while destructive final escalation/write was withheld.
INFERRED_PATHS	1	CHAIN-06 (Internal Subnet SMB Port 445 Lateral Movement Candidate) [W_exec = 0.35 (Port-Open / Protocol Exposure Candidate Only)]	Theoretical multi-host pivot inferred from open SMB ports (139/445); lateral exploitation NOT verified or executed.
BLOCKED_BY_CONTROLS_PATHS	2	192.168.195.131 & 192.168.195.254 (0 Open TCP Ports / Host Firewall Filtered) [W_exec = 0.00 (Blocked at Network/Transport Layer)]	Target endpoints discovered via ARP/DHCP where host firewall or service filtering blocked all TCP connection attempts.
NOT_TESTED_BY_SAFETY_ROE_PATHS	3	DirtyCow Kernel Race Execution (CVE-2016-5195), Active MS17-010 Kernel Payload, Destructive SQL/NFS Writes [W_exec = N/A (Restricted by Mandatory Authorization & RoE Safety Policy Engine)]	Techniques intentionally withheld by the XStrike Safety Policy Engine to prevent kernel panic, data corruption, or denial of service.

4.2 Four-Stage Attack Graph Edge Provenance Table

Stage	Source -> Target Node	MITRE Technique	Finding & Provenance Ref	Validation State	Conf
S1: Initial Access	Assessor (192.168.195.130) -> 192.168.195.139:21 (vsFTPd 2.3.4)	T1190: Exploit Public-Facing Service (FTP Banner & USER Trigger)	F-02 (PROV-F-02-284F15F547)	CONFIRMED_EXECUTED	99%
S2: Credential / Vulnerability Exploitation	192.168.195.139:21 -> 192.168.195.139:6200 (Backdoor Shell)	T1059.004: CVE-2011-2523 Backdoor Listener Activation	F-02 (PROV-F-02-284F15F547)	CONFIRMED_EXECUTED	99%
S3: Privilege Escalation	192.168.195.139:6200 -> root@192.168.195.139 (uid=0)	T1068: Direct Daemon Root Execution (uid=0(root) Inherited)	F-02 (PROV-F-02-284F15F547)	CONFIRMED_EXECUTED	99%
S4: Target Impact	root@192.168.195.139 (uid=0) -> 192.168.195.139 (/etc/shadow & Full Host Control)	T1003.008: OS Credential Dumping (/etc/shadow & Crown-Jewel DB Host Takeover)	F-02 (PROV-F-02-284F15F547)	CONFIRMED_EXECUTED	99%
S1: Initial Access	Assessor (192.168.195.130) -> 192.168.195.139:1524 (ingreslock)	T1190: Connect to Exposed TCP Port 1524 Bind Shell	F-01 (PROV-F-01-E204DF2EED)	CONFIRMED_EXECUTED	99%
S2: Credential / Vulnerability Exploitation	192.168.195.139:1524 -> /bin/sh Interactive Session	T1059.004: Unauthenticated Unix Shell Execution (Consolidated with rsh/1524 probe)	F-01 (PROV-F-01-E204DF2EED)	CONFIRMED_EXECUTED	99%
S3: Privilege Escalation	/bin/sh Interactive Session -> root@192.168.195.139 (uid=0)	T1068: Root Context Pre-Established by Bindshell Daemon	F-01 (PROV-F-01-E204DF2EED)	CONFIRMED_EXECUTED	99%
S4: Target Impact	root@192.168.195.139 (uid=0) -> 192.168.195.139 (/etc/passwd, /etc/shadow & System Control)	T1003.008: Full Host Compromise & Credential Hash Extraction	F-01 (PROV-F-01-E204DF2EED)	CONFIRMED_EXECUTED	99%
S1: Initial Access	Assessor (192.168.195.130) -> 192.168.195.139:22 (OpenSSH 4.7p1)	T1078.001: Default Account Login (msfadmin:msfadmin)	F-05 (PROV-F-05-6715A75078)	CONFIRMED_EXECUTED	95%
S2: Credential / Vulnerability Exploitation	192.168.195.139:22 -> msfadmin@192.168.195.139 (uid=1000)	T1021.004: Authenticated Non-Root Shell Session Established (uid=1000)	F-05 (PROV-F-05-6715A75078)	CONFIRMED_EXECUTED	95%
S3: Privilege Escalation	msfadmin@192.168.195.139 (uid=1000) -> SUID /usr/bin/nmap & DirtyCow (CVE-2016-5195) Vector	T1548.001: Local Privilege Escalation Vector Validated (Not Triggered via SSH Session)	F-06 (PROV-F-06-E061A6DE7E)	PARTIALLY_VALIDATED	88%
S4: Target Impact	SUID /usr/bin/nmap & DirtyCow (CVE-2016-5195) Vector -> root@192.168.195.139 (Projected Escalation)	T1068: Local Root Compromise via nmap --interactive !sh	F-06 (PROV-F-06-E061A6DE7E)	INFERRED_NOT_EXECUTED	84%
S1: Initial Access	Assessor (192.168.195.130) -> 192.168.195.139:5432 (PostgreSQL 8.3.1)	T1190: Network Connection to Exposed PostgreSQL Port 5432	F-03 (PROV-F-03-5EBB5E02B0)	CONFIRMED_EXECUTED	95%
S2: Credential / Vulnerability Exploitation	192.168.195.139:5432 -> postgres@192.168.195.139:5432	T1078.001: Default Superuser Authentication (postgres:postgres)	F-03 (PROV-F-03-5EBB5E02B0)	CONFIRMED_EXECUTED	95%

Stage	Source -> Target Node	MITRE Technique	Finding & Provenance Ref	Validation State	Conf
S3: Privilege Escalation	postgres@192.168.195.139:5432 -> PostgreSQL Superuser Role (rolsuper=true)	T1078: Database Superuser Context Granted on Login	F-03 (PROV-F-03-5EBB5E02B0)	CONFIRMED_EXECUTED	95%
S4: Target Impact	PostgreSQL Superuser Role (rolsuper=true) -> pg_catalog / template1 / postgres Databases	T1005: Authenticated Database Schema & Catalog Enumeration	F-03 (PROV-F-03-5EBB5E02B0)	CONFIRMED_EXECUTED	95%
S1: Initial Access	Assessor (192.168.195.130) -> 192.168.195.139:2049 (NFS) & :80 (HTTP)	T1190: RPC/NFS Export Mount Discovery & HTTP Parameter Fuzzing	F-39 (PROV-F-39-D5CCEDCEC8)	CONFIRMED_EXECUTED	92%
S2: Credential / Vulnerability Exploitation	192.168.195.139:2049 & :80 -> Exported NFS Share & Mutillidae SQLi / DVWA Admin	T1190: Validated NFS Export Surface & Web SQLi / Default Creds	F-08 (PROV-F-08-9CC3744D82)	PARTIALLY_VALIDATED	88%
S3: Privilege Escalation	Exported NFS Share -> Root Filesystem Write / SSH authorized_keys Injection	T1098.004: Inferred NFS Mount Modification (Not Mounted Destructively)	F-39 (PROV-F-39-D5CCEDCEC8)	INFERRED_NOT_EXECUTED	78%
S4: Target Impact	Root Filesystem Write / SSH authorized_keys Injection -> 192.168.195.139 (Projected Host Takeover / DB Extraction)	T1005: Inferred Full Filesystem Access & Backend Database Extraction	F-39 (PROV-F-39-D5CCEDCEC8)	INFERRED_NOT_EXECUTED	75%
S1: Initial Access	192.168.195.139 (Compromised Pivot) -> 192.168.195.1 / .130 / .140 (TCP Port 445)	T1046: Internal Subnet SMB Port 445 Discovery	F-53 (PROV-F-53-483C65504D)	CONFIRMED_EXECUTED	90%
S2: Credential / Vulnerability Exploitation	192.168.195.1 / .130 / .140 (TCP Port 445) -> Unverified SMBv1 / Lateral Movement Candidate	T1210: Theoretical Lateral Exploitation (NOT Verified / NOT Executed)	F-53 (PROV-F-53-483C65504D)	INFERRED_NOT_EXECUTED	25%

4.3 AttackGraph V3 Remediation Choke-Point Analysis (Maximum Chain Disruption)

Choke-Point ID	Remediation Control & Target Node	Finding Ref	Chains Broken	Risk Score Reduction	SLA Priority
CP-01	Isolate 192.168.195.139 & Terminate Backdoor Daemons (192.168.195.139 (TCP/21, TCP/6200, TCP/1524)) — Stop vsFTPd 2.3.4 (:21/:6200) & kill /bin/sh on Ingreslock (:1524)	F-01, F-02	2 (CHAIN-01, CHAIN-02)	-44.0 pts	P0_IMMEDIATE (0–24h)
CP-02	Rotate Default Accounts & Enforce SSH Public-Key Auth (192.168.195.139 (TCP/22, TCP/21, TCP/5432, HTTP /dvwa/)) — Rotate postgres, msfadmin (SSH/FTP), and DVWA admin credentials; disable anonymous FTP	F-03, F-05	2 (CHAIN-03, CHAIN-04)	-20.0 pts	P0_IMMEDIATE (0–24h)
CP-03	Remove SUID Bits & Upgrade Legacy Linux 2.6.24 Kernel (192.168.195.139 (Local Privilege Escalation Surface)) — chmod u-s /usr/bin/nmap /bin/mount; upgrade kernel to remediate DirtyCow (CVE-2016-5195)	F-06	1 (CHAIN-03 (Stage 3->4 Escalation))	-9.5 pts	P1_HIGH (7 Days)

Choke-Point ID	Remediation Control & Target Node	Finding Ref	Chains Broken	Risk Score Reduction	SLA Priority
CP-04	Disable Unauthenticated NFS Exports & Parameterize SQL Queries (192.168.195.139 (TCP/111, TCP/2049, TCP/80)) — Stop nfs-kernel-server (:2049), restrict /etc/exports, and remediate Mutillidae SQLi (:80)	F-08	1 (CHAIN-05)	-8.0 pts	P1_HIGH (7 Days)
CP-05	Disable SMBv1 Dialect & Enforce Mandatory SMB Signing (192.168.195.1, .130, .139, .140 (TCP/445)) — Disable SMBv1 protocol and require SMB2/3 packet signing across all subnet hosts	F-12	1 (CHAIN-06)	-6.5 pts	P2_MEDIUM (7–30 Days)

5. Verified Exploitation & 3-State Post-Exploitation Privilege Lifecycle Matrix

The table below distinguishes confirmed root shell compromises (`CONFIRMED_COMPROMISE`), confirmed default/anonymous credential logins (`CONFIRMED_AUTH`), and post-exploitation local privilege escalation vectors using XStrike V3's **3-State Post-Exploitation Privilege Transition Model** (`PRECONDITION_DISCOVERED`, `TECHNIQUE_VALIDATED`, `PRIVILEGE_TRANSITION_EXECUTED`). Because initial access via `F-01` (`:1524` Ingreslock) and `F-02` (`:21/:6200` vsFTPd) already yielded `uid=0(root)` on `192.168.195.139`, local privilege escalation vectors discovered from the `msfadmin` (`uid=1000`) SSH session were validated non-destructively without executing kernel-modifying payloads.

Finding ID	Target Host:Port	Vulnerability / Vector	Proof State	Privilege Achieved	File / DB Access Scope	3-State Privesc Lifecycle
F-01	192.168.195.139:	Ingreslock / Bindshell Unauthenticated Root Shell on TCP Port 1524	CONFIRMED_COMPROMISE	ROOT (uid=0 / gid=0)	File: READ_WRITE (/etc/shadow verified) | DB: NONE	PRIVILEGE_TRANSITION_EXECUTED
F-02	192.168.195.139:	vsFTPd 2.3.4 Smiley-Face Backdoor Remote Root Compromise (CVE-2011-2523)	CONFIRMED_COMPROMISE	ROOT (uid=0 / gid=0)	File: READ_WRITE (/etc/shadow verified) | DB: NONE	PRIVILEGE_TRANSITION_EXECUTED
F-03	192.168.195.139:	PostgreSQL 8.3.1 Default Superuser Authentication (postgres:postgres)	CONFIRMED_AUTH	DB_SUPERUSER (postgres)	File: NONE | DB: AUTHENTICATED_SCHEMA_READ (PostgreSQL 8.3.1 databases & pg_catalog tables enumerated)	TECHNIQUE_VALIDATED
F-04	192.168.195.139:	FTP Default Credential Authentication (msfadmin:msfadmin)	CONFIRMED_AUTH	STANDARD_USER (msfadmin)	File: READ_LIST (user directory listing verified; write not executed) | DB: NONE	PRECONDITION_DISCOVERED
F-05	192.168.195.139:	SSH Default Credential Shell Access (msfadmin:msfadmin)	CONFIRMED_AUTH	STANDARD_USER (uid=1000 msfadmin, gid=1000)	File: READ_WRITE (standard user home directory) | DB: NONE	PRECONDITION_DISCOVERED
F-06	192.168.195.139:	Local Privilege Escalation Vector: SUID Binary (/usr/bin/nmap)	EXPLOITABLE	LOCAL_ESCALATION_PRECONDITION (uid=1000 -> uid=0 via nmap --interactive)	File: READ_ONLY (/usr/bin/nmap SUID-root bit verified) | DB: NONE	TECHNIQUE_VALIDATED

Finding ID	Target Host:Port	Vulnerability / Vector	Proof State	Privilege Achieved	File / DB Access Scope	3-State Privesc Lifecycle
F-09	192.168.195.139:	Legacy Linux Kernel 2.6.24-16-server Vulnerable to DirtyCow (CVE-2016-5195)	VALIDATED	LOCAL_ESCALATION_VECTOR (uid=1000 -> uid=0 potential)	File: NONE | DB: NONE	PRECONDITION_DISCOVERED
F-17	192.168.195.139:	FTP Anonymous Authentication Enabled	CONFIRMED_AUTH	ANONYMOUS (ftp)	File: READ_ONLY_EMPTY (0 files in anonymous root; no sensitive files exfiltrated) | DB: NONE	NOT_APPLICABLE_NON_PRIVESC
F-18	192.168.195.139:	Local Privilege Escalation Vector: SUID Binary (/bin/mount)	VALIDATED	LOCAL_ESCALATION_VECTOR (uid=1000 -> uid=0 potential)	File: NONE | DB: NONE	PRECONDITION_DISCOVERED

6. Four-Tier Evidence-Driven Compliance Control Mapping

Instead of marking every scanner observation as a `NON-COMPLIANT` audit failure, Engine 07 classifies each finding into one of four evidence-backed compliance states: `CONFIRMED_COMPLIANCE_DEFICIENCY` (backed by active compromise or credential auth), `EVIDENCE_OF_CONTROL_FAILURE` (backed by validated technical proof), `POTENTIAL_CONTROL_WEAKNESS` (backed by observed banner/port exposure), or `TECHNICAL_OBSERVATION` (reconnaissance telemetry).

ID	Finding Title	4-Tier Compliance Status	PCI-DSS v4.0	NIST SP 800-53 Rev. 5	ISO 27001:2022 & CIS v8	Evidence Rationale
F-01	Ingreslock / Bindshell Unauthenticated Roots	CONFIRMED_COMPLIANCE_DEFICIENCY	PCI-DSS v4.0 Req 2.2.2 & 8.2 (Default Accounts & Strong Authentication)	NIST 800-53 IA-2, IA-5 & AC-3 (Identification, Authenticator Mgmt, Access Enforcement)	ISO 27001:2022 A.5.17 & A.8.5 (Authentication Information & Secure Authentication) | CIS Control 4.7 & 5.2 (Manage Default Accounts & Use Unique Passwords)	Direct exploitation/ authentication proof (CONFIRMED_COMPROMISE) demonstrates an active failure of access control or system integrity.

ID	Finding Title	4-Tier Compliance Status	PCI-DSS v4.0	NIST SP 800-53 Rev. 5	ISO 27001:2022 & CIS v8	Evidence Rationale
F-02	vsFTPd 2.3.4 Smiley-Face Backdoor Remote Root	CONFIRMED_COMPLIANCE_DEFICIENCY	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Direct exploitation/ authentication proof (CONFIRMED_COMPROMISE) demonstrates an active failure of access control or system integrity.
F-03	PostgreSQL 8.3.1 Default Superuser Authentica	CONFIRMED_COMPLIANCE_DEFICIENCY	PCI-DSS v4.0 Req 2.2.2 & 8.2 (Default Accounts & Strong Authentication)	NIST 800-53 IA-2, IA-5 & AC-3 (Identification, Authenticator Mgmt, Access Enforcement)	ISO 27001:2022 A.5.17 & A.8.5 (Authentication Information & Secure Authentication) | CIS Control 4.7 & 5.2 (Manage Default Accounts & Use Unique Passwords)	Direct exploitation/ authentication proof (CONFIRMED_AUTH) demonstrates an active failure of access control or system integrity.
F-04	FTP Default Credential Authentication (msfadm	CONFIRMED_COMPLIANCE_DEFICIENCY	PCI-DSS v4.0 Req 2.2.2 & 8.2 (Default Accounts & Strong Authentication)	NIST 800-53 IA-2, IA-5 & AC-3 (Identification, Authenticator Mgmt, Access Enforcement)	ISO 27001:2022 A.5.17 & A.8.5 (Authentication Information & Secure Authentication) | CIS Control 4.7 & 5.2 (Manage Default Accounts & Use Unique Passwords)	Direct exploitation/ authentication proof (CONFIRMED_AUTH) demonstrates an active failure of access control or system integrity.
F-05	SSH Default Credential Shell Access (msfadmin	CONFIRMED_COMPLIANCE_DEFICIENCY	PCI-DSS v4.0 Req 2.2.2 & 8.2 (Default Accounts & Strong Authentication)	NIST 800-53 IA-2, IA-5 & AC-3 (Identification, Authenticator Mgmt, Access Enforcement)	ISO 27001:2022 A.5.17 & A.8.5 (Authentication Information & Secure Authentication) | CIS Control 4.7 & 5.2 (Manage Default Accounts & Use Unique Passwords)	Direct exploitation/ authentication proof (CONFIRMED_AUTH) demonstrates an active failure of access control or system integrity.
F-06	Local Privilege Escalation Vector: SUID Binar	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 7.2.2 (Assign Privileges Based on Job Function / Least Privilege)	NIST 800-53 AC-6 (Least Privilege)	ISO 27001:2022 A.8.2 (Privileged Access Rights) | CIS Control 5.4 & 6.8 (Restrict Administrator Privileges)	Deterministic HTTP/protocol validation (EXPLOITABLE) confirms a technical control gap requiring remediation.

ID	Finding Title	4-Tier Compliance Status	PCI-DSS v4.0	NIST SP 800-53 Rev. 5	ISO 27001:2022 & CIS v8	Evidence Rationale
F-07	DVWA Web Application Default Administrative C	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 2.2.2 & 8.2 (Default Accounts & Strong Authentication)	NIST 800-53 IA-2, IA-5 & AC-3 (Identification, Authenticator Mgmt, Access Enforcement)	ISO 27001:2022 A.5.17 & A.8.5 (Authentication Information & Secure Authentication) | CIS Control 4.7 & 5.2 (Manage Default Accounts & Use Unique Passwords)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-08	Web Application SQL Injection Indicator (Muti	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 6.2.4 & 6.4.1 (Prevent Injection, SSRF & Insecure Deserialization)	NIST 800-53 SI-10 & SI-2 (Information Input Validation & Flaw Remediation)	ISO 27001:2022 A.8.26 & A.8.28 (Application Security Requirements & Secure Coding) | CIS Control 16.12 (Implement Code-Level Security Checks)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-09	Legacy Linux Kernel 2.6.24-16-server Vulnerab	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-10	Deprecated SMBv1 Dialect Enabled on SMB Servi	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 2.2.7 & 4.2.1 (Encrypt Non-Console Admin Access & Strong Cryptography)	NIST 800-53 SC-8 & SC-13 (Transmission Confidentiality & Cryptographic Protection)	ISO 27001:2022 A.8.24 (Use of Cryptography) | CIS Control 3.10 (Encrypt Sensitive Data in Transit)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-11	Web Application Server-Side Request Forgery (	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 6.2.4 & 6.4.1 (Prevent Injection, SSRF & Insecure Deserialization)	NIST 800-53 SI-10 & SI-2 (Information Input Validation & Flaw Remediation)	ISO 27001:2022 A.8.26 & A.8.28 (Application Security Requirements & Secure Coding) | CIS Control 16.12 (Implement Code-Level Security Checks)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.

ID	Finding Title	4-Tier Compliance Status	PCI-DSS v4.0	NIST SP 800-53 Rev. 5	ISO 27001:2022 & CIS v8	Evidence Rationale
F-12	Deprecated SMBv1 Dialect Enabled on SMB Servi	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 2.2.7 & 4.2.1 (Encrypt Non-Console Admin Access & Strong Cryptography)	NIST 800-53 SC-8 & SC-13 (Transmission Confidentiality & Cryptographic Protection)	ISO 27001:2022 A.8.24 (Use of Cryptography) | CIS Control 3.10 (Encrypt Sensitive Data in Transit)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-13	SMB Message Signing Not Required (NTLM Relay	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 2.2.7 & 4.2.1 (Encrypt Non-Console Admin Access & Strong Cryptography)	NIST 800-53 SC-8 & SC-13 (Transmission Confidentiality & Cryptographic Protection)	ISO 27001:2022 A.8.24 (Use of Cryptography) | CIS Control 3.10 (Encrypt Sensitive Data in Transit)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-14	Cleartext Telnet Service Enabled on TCP Port	POTENTIAL_CONTROL_WEAKNESS	PCI-DSS v4.0 Req 2.2.7 & 4.2.1 (Encrypt Non-Console Admin Access & Strong Cryptography)	NIST 800-53 SC-8 & SC-13 (Transmission Confidentiality & Cryptographic Protection)	ISO 27001:2022 A.8.24 (Use of Cryptography) | CIS Control 3.10 (Encrypt Sensitive Data in Transit)	Service port or version banner exposure observed; requires host configuration verification to confirm compliance failure.
F-15	End-of-Life Apache HTTP Server 2.2.8 Deployed	POTENTIAL_CONTROL_WEAKNESS	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Service port or version banner exposure observed; requires host configuration verification to confirm compliance failure.
F-16	PHP-CGI Argument Injection Candidate (CVE-201	POTENTIAL_CONTROL_WEAKNESS	PCI-DSS v4.0 Req 6.2.4 & 6.4.1 (Prevent Injection, SSRF & Insecure Deserialization)	NIST 800-53 SI-10 & SI-2 (Information Input Validation & Flaw Remediation)	ISO 27001:2022 A.8.26 & A.8.28 (Application Security Requirements & Secure Coding) | CIS Control 16.12 (Implement Code-Level Security Checks)	Service port or version banner exposure observed; requires host configuration verification to confirm compliance failure.

ID	Finding Title	4-Tier Compliance Status	PCI-DSS v4.0	NIST SP 800-53 Rev. 5	ISO 27001:2022 & CIS v8	Evidence Rationale
F-17	FTP Anonymous Authentication Enabled	CONFIRMED_COMPLIANCE_DEFICIENCY	PCI-DSS v4.0 Req 2.2.2 & 8.2 (Default Accounts & Strong Authentication)	NIST 800-53 IA-2, IA-5 & AC-3 (Identification, Authenticator Mgmt, Access Enforcement)	ISO 27001:2022 A.5.17 & A.8.5 (Authentication Information & Secure Authentication) | CIS Control 4.7 & 5.2 (Manage Default Accounts & Use Unique Passwords)	Direct exploitation/ authentication proof (CONFIRMED_AUTH) demonstrates an active failure of access control or system integrity.
F-18	Local Privilege Escalation Vector: SUID Binar	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 7.2.2 (Assign Privileges Based on Job Function / Least Privilege)	NIST 800-53 AC-6 (Least Privilege)	ISO 27001:2022 A.8.2 (Privileged Access Rights) | CIS Control 5.4 & 6.8 (Restrict Administrator Privileges)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-19	Client-Manipulable Session/Tracking Cookie Pa	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-20	Content-Security-Policy (CSP) Header Not Set	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.

ID	Finding Title	4-Tier Compliance Status	PCI-DSS v4.0	NIST SP 800-53 Rev. 5	ISO 27001:2022 & CIS v8	Evidence Rationale
F-21	Missing Anti-Clickjacking Protection (X-Frame	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-22	Session Cookie Missing 'Secure' Flag	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 2.2.7 & 4.2.1 (Encrypt Non-Console Admin Access & Strong Cryptography)	NIST 800-53 SC-8 & SC-13 (Transmission Confidentiality & Cryptographic Protection)	ISO 27001:2022 A.8.24 (Use of Cryptography) | CIS Control 3.10 (Encrypt Sensitive Data in Transit)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-23	Verbose Application / Database Error Message	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-24	Web Application Missing Anti-CSRF Tokens & Ha	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.

ID	Finding Title	4-Tier Compliance Status	PCI-DSS v4.0	NIST SP 800-53 Rev. 5	ISO 27001:2022 & CIS v8	Evidence Rationale
F-25	Content-Security-Policy (CSP) Header Not Set	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-26	Web Application Missing Anti-CSRF Tokens & Ha	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-27	Content-Security-Policy (CSP) Header Not Set	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-28	Web Application Missing Anti-CSRF Tokens & Ha	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.

ID	Finding Title	4-Tier Compliance Status	PCI-DSS v4.0	NIST SP 800-53 Rev. 5	ISO 27001:2022 & CIS v8	Evidence Rationale
F-29	Content-Security-Policy (CSP) Header Not Set	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-30	DNS Open Recursion Enabled on ISC BIND 9.4.2	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-31	Missing Anti-Clickjacking Protection (X-Frame	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-32	Web Application Missing Anti-CSRF Tokens & Ha	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.

ID	Finding Title	4-Tier Compliance Status	PCI-DSS v4.0	NIST SP 800-53 Rev. 5	ISO 27001:2022 & CIS v8	Evidence Rationale
F-33	Content-Security-Policy (CSP) Header Not Set	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-34	Missing Anti-Clickjacking Protection (X-Frame	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.
F-35	Web Application Missing Anti-CSRF Tokens & Ha	EVIDENCE_OF_CONTROL_FAILURE	PCI-DSS v4.0 Req 1.2.1 & 2.2.1 (Configuration Standards & Network Service Restrictions)	NIST 800-53 CM-6, CM-7 & RA-5 (Configuration Settings, Least Functionality, Vuln Monitoring)	ISO 27001:2022 A.8.8 & A.8.9 (Management of Technical Vulnerabilities & Configuration Mgmt) | CIS Control 4.1 & 7.1 (Establish Secure Configurations & Vulnerability Management)	Deterministic HTTP/protocol validation (VALIDATED) confirms a technical control gap requiring remediation.

7. Deduplicated Root-Cause Intelligence & Attribution Certainty

Root-cause distribution is computed strictly on the **85 deduplicated security vulnerabilities** (never on the 1,559 raw scanner alerts, which previously skewed 'Configuration Error' to 88.3%). Each root cause is paired with a 4-tier attribution certainty (`DIRECTLY_ESTABLISHED`, `PROBABLE`, `INFERRED`, `UNKNOWN`).

Root-Cause Category	Deduplicated Findings	Share (%)	Directly Established	Probable / Inferred	Primary Remediation Owner
Web Server & HTTP Security Header Hardening Gap	50	58.8%	50	0	Infrastructure / AppSec Team
Network Segmentation & Perimeter Filter Exposure	13	15.3%	0	13	Infrastructure / AppSec Team
Identity & Credential Hygiene Deficiency	5	5.9%	5	0	SecOps / Platform Engineering
Service Hardening & Configuration Baseline Gap	5	5.9%	0	5	Infrastructure / AppSec Team
Patch & Lifecycle Management Deficit (EOL Software)	4	4.7%	4	0	Infrastructure / AppSec Team
Insecure Application Coding (Input Validation)	3	3.5%	2	1	Infrastructure / AppSec Team
Malicious Backdoor / Intentional Vulnerable Build	2	2.4%	2	0	SecOps / Platform Engineering
Excessive Local Privilege / File Permission Misconfiguration	2	2.4%	2	0	Infrastructure / AppSec Team
Access Control & Network Share Misconfiguration	1	1.2%	1	0	Infrastructure / AppSec Team

8. Prioritized Remediation Roadmap & Reproducible SLA Risk-Reduction Trajectory

The SLA Risk-Reduction Trajectory below is computed by re-running the 4-component Evidence-Weighted Risk Formula (`R = min(100, S_compromise + S_exposure + S_asset + S_reachability)`) after removing the findings remediated in each SLA window:

Residual Architectural Risk Floor (`1.5/100` at Day 30 when `0` Active Findings Remain): When all 85 active security findings are remediated (30-Day Forecast: 0 remaining findings), Active Vulnerability Risk drops to 0.0/100 (S_compromise=0.0, S_exposure=0.0). The 1.5/100 Projected Score represents the explicit Residual Architectural Risk Floor: S_asset=1.0 pt (intrinsic operational criticality of Crown-Jewel database/application host 192.168.195.139) + S_reachability=0.5 pt (baseline internal /24 LAN reachability prior to zero-trust micro-segmentation).

Remediation Phase	SLA Window	Remaining Findings	S_comp (45)	S_exp (25)	S_asset (15)	S_reach (15)	Active Vuln Risk	Total Projected Risk (incl. Floor)
Current Baseline (T0 — Measured)	Immediate	85	45.0	25.0	15.0	13.5	98.5/100	98.5/100 (-0.0%)
Phase 1 — Emergency Containment (Forecast)	24 Hours	82	15.0	25.0	15.0	9.0	64.0/100	64.0/100 (-35.0%)

Remediation Phase	SLA Window	Remaining Findings	S_comp (45)	S_exp (25)	S_asset (15)	S_reach (15)	Active Vuln Risk	Total Projected Risk (incl. Floor)
Phase 2 — High-Priority Hardening (Forecast)	7 Days	69	3.0	25.0	5.0	9.0	42.0/100	42.0/100 (-57.4%)
Phase 3 — Full Closure & Residual Floor (Forecast)	30 Days	0	0.0	0.0	1.0	0.5	0.0/100	1.5/100 (-98.5%)

9. EvidenceCore V3 Cryptographic Manifest, SQLite Verifier & 18-Rule Quality Gate Attestation

To enable independent offline verification of every finding, command transcript, and provenance hash, **EvidenceCore V3** generates an HMAC-SHA256 signed evidence manifest, an OASIS SARIF v2.1.0 machine-verifiable bundle, and an independently verified 5-table SQLite database snapshot:

Manifest Version:	EvidenceCore-V3.0
Assessment ID:	XSTRIKE-V3-20261031
Total Signed Records:	110 (85 Security + 25 Recon)
Manifest Root SHA-256:	2c953f61914c3b0df4bb82d65d5dce7742b65f3890b73c06df97003b2cb021af
Manifest HMAC-SHA256:	be6c55ab5af98256a437af1105415dd23b5a80786af11cdc93cbcb75f17c3718
Signing Key ID & Policy:	XSTRIKE-V3-DEFAULT-KEY-01 | 90-Day Automated Key Rotation with Per-Assessment Key ID Binding & Non-Repudiation Audit Log
Independent Hash & HMAC Check:	110/110 SHA-256 Verified | 110/110 HMAC-SHA256 Verified
Tamper-Detection Self-Test:	PASSED (1-byte mutation immediately rejected by HMAC-SHA256 verifier)
SQLite 5-Table Snapshot Audit:	VERIFIED — 110/110 DB record hashes matched (v2_intelligence.db)
OASIS SARIF v2.1.0 Bundle:	85 machine-verifiable results exported with 7-D proof vectors & provenance hashes

Prior to generating this deliverable, **Engine 08 (ReportQualityEngine V3)** executed all **18 mandatory acceptance checks** (`AC-01` through `AC-18`) against the enriched finding dataset, cross-section transition table, attack graph, risk model, and SQLite snapshot. Overall Gate Status: **18 automated acceptance criteria passed out of 18 (18/18 PASSED)**.

Rule ID	Quality Gate Verification Rule	Status	Verification Telemetry
AC-01	Zero Unverified / Port-Open Findings Marked CONFIRMED	PASS	0 violations across 110 findings
AC-02	Repeated Web/Network Observations Consolidated with Endpoint Counts Preserved	PASS	1552 target observations consolidated into 85 security findings + 25 recon items
AC-03	Zero Title / Description / CWE / CVE / MITRE Semantic Mismatches	PASS	100% semantic alignment across all findings
AC-04	7-Dimensional Proof Vector & Granular Access Scope Separation	PASS	All findings carry 7-D proof vectors (Execution, Auth, Exploitability, Privilege, DataAccess, Impact, EvidenceConf)
AC-05	Per-Host Asset Classification & Classification Source/Confidence	PASS	0 misclassified non-DB hosts; 7 hosts reconciled with explicit classification_source & confidence
AC-06	Reconnaissance & Telemetry Separated from Security Vulnerabilities	PASS	25 recon items isolated in Section 3B with CVSS 0.0 and DISCOVERED state
AC-07	Separated Confidence Model (EvidenceConf vs ExploitabilityConf vs ImpactConf)	PASS	All findings separate Evidence Confidence from Exploitability Confidence and Impact Confidence
AC-08	AttackGraph V3: 4-Stage Provenance, 7 Path Metrics & Choke-Point Analysis	PASS	6 chains, 22 edges, and 5 remediation choke points
AC-09	RiskEngine V3: 4-Component Score, Calibrated Exposure & SLA Trajectory	PASS	Baseline 98.5 -> 24h (64.0) -> 7d (42.0) -> 30d (1.5)
AC-10	Root-Cause Analysis Computed on Deduplicated Findings with 4-Tier Certainty	PASS	Root cause computed on 85 deduplicated findings (not raw scanner volume)

Rule ID	Quality Gate Verification Rule	Status	Verification Telemetry
AC-11	100% Vulnerability-Specific Remediation Guidance (Zero Generic Fallbacks)	PASS	0 generic fallback remediations across 110 findings
AC-12	4-Tier Evidence-Driven Compliance Mapping with Rationale	PASS	All findings mapped to PCI-DSS v4.0, NIST 800-53, ISO 27001, and CIS Controls with 4-tier status
AC-13	EvidenceCore V3: 10-Control Provenance & HMAC-SHA256 Integrity Signatures	PASS	100% of findings (110) bound to SHA-256 + HMAC-SHA256 signed provenance records
AC-14	Strict CVE Syntax (^CVE-YYYY-NNNN+$) & Specific CWE Mapping (Zero CWE-16)	PASS	0 custom IDs in CVE field; 0 generic CWE-16 placeholders across all security findings
AC-15	Strict NVD CVSS v3.1 Score-to-Severity Band Alignment	PASS	100% CVSS-to-Severity alignment across 85 security findings
AC-16	Endpoint Hygiene (Zero w3.org/nmap.org URLs) & Zero Generic OWASP Rollups	PASS	0 external reference URLs in affected_endpoints; 0 generic OWASP A01..A10 rollup rows
AC-17	Cross-Section Finding ID Reconciliation & 3-State Privesc Lifecycle Model	PASS	100% cross-section ID alignment (Phase 3A=['F-03', 'F-04', 'F-05', 'F-17'], Phase 3B=['F-01', 'F-02'], Phase 4=['F-06', 'F-09', 'F-18'])
AC-18	Independent Cryptographic (SHA-256 / HMAC / Merkle) & SQLite Snapshot Verification	PASS	Verified 110/110 SHA-256 & HMAC signatures, Merkle root, tamper self-test, and 110 SQLite records

10. XStrike V3 Autonomous Architecture & Unified TrePath v7.0 Integration

The tables below summarize the five core XStrike V3 autonomous capabilities operational in this assessment and how **TrePath v7.0** (Web/API DAST) and **XStrike V3** (Network, Host, Exploitation & Post-Exploitation) operate together on the shared EvidenceCore V3 intelligence layer:

Capability ID	XStrike V3 Autonomous Capability	Status	Technical Summary
CAP-01	Rules-of-Engagement (RoE) & Safety Policy Engine	OPERATIONAL (v3.0)	Enforces CIDR scope boundaries, per-technique exploit allowlists, DoS/crash prevention gates (holding MS17-010 and Dirty COW in non-destructive states), and read-only command whitelists.
CAP-02	Session-Aware Post-Exploitation Collector	OPERATIONAL (v3.0)	Collects structured OS, kernel, interface, SUID, NFS export, and DB catalog telemetry from established sessions using a 3-State Privilege Escalation Lifecycle (PRECONDITION_DISCOVERED, TECHNIQUE_VALIDATED, PRIVILEGE_TRANSITION_EXECUTED).
CAP-03	Cross-Host Pivot & Trust-Relationship Validator	OPERATIONAL (v3.0)	Maps SSH key reuse, NFS no_root_squash exports, database credential reuse, and SMB lateral paths across the 5-State Attack Path Taxonomy (EXECUTED, PARTIALLY_VALIDATED, INFERRED, BLOCKED_BY_CONTROLS, NOT_TESTED_BY_SAFETY_ROE).
CAP-04	Cryptographic Evidence Bundle & SQLite Verifier	OPERATIONAL (v3.0)	Generates tamper-evident SHA-256 + HMAC-SHA256 provenance records, Merkle root manifests, OASIS SARIF v2.1.0 bundles, and independently verified 5-table SQLite snapshots.

Capability ID	XStrike V3 Autonomous Capability	Status	Technical Summary
CAP-05	Automated Remediation Re-Test & Verification Loop	READY FOR RE-SCAN	Re-executes the exact deterministic proof command for each remediated finding (e.g., Ingreslock :1524, vsFTPd :21, PostgreSQL :5432) and transitions verified fixes to REMEDIATION_VERIFIED.

Architectural Dimension	TrePath v7.0 (Web & API DAST)	XStrike V3 (Network, Host & Post-Exploit)	Unified Platform Value
Primary Assessment Domain	Web Application & API DAST (HTTP/HTTPS, REST/GraphQL, OWASP Top 10:2025)	Autonomous Network, Host, Database, Exploitation & Post-Exploitation Engine	Full-stack L3–L7 coverage from network perimeter to web app & database internals
Validation & Proof Semantics	HTTP Request/Response Replay, DOM/Reflection Proof & Status Diffing	7-State Proof Model + 3-State Privesc Lifecycle + Interactive Shell/Auth Transcripts	Zero unverified 'Confirmed' inflation across both web and infrastructure findings
Attack Graph & Choke Points	Application Endpoint & Auth-Bypass Dependency Chains	5-State Multi-Hop Network/Host Attack Graph with Priority Index & Choke-Point ROI	End-to-end kill-chain modeling from web/network entry to Crown-Jewel data stores
Evidence Integrity & GRC Export	SHA-256 Evidence Hashing & OWASP/PCI/NIST Mapping	HMAC-SHA256 Signed Merkle Manifest + Independent SQLite Verifier + SARIF v2.1.0	Cryptographically verifiable audit artifacts ready for CI/CD and enterprise GRC

11. Sign-Off & Deliverables Attestation

Lead Penetration Tester: _____ Date: 2026-10-31

Principal Security Architect (Quality Gate Reviewer): _____ Date: 2026-10-31

Client System Owner: _____ Date: 2026-10-31